Benchmarking alignment penalties for single-cell batch correction

Kendall Reid[1,*], Genevieve Stein-O'Brien[2,†], Erhan Guven[3,†]

[1]Department of Computer Science, Johns Hopkins University; [2]Department of Neuroscience, Johns Hopkins University; [3]Johns Hopkins University Applied Physics Laboratory. *Corresponding author: kreid20@jh.edu. †These authors contributed equally.

Writer's outline. Target: *BMC Bioinformatics*, Research article; structured abstract at most 350 words; 3 to 10 keywords; no stated word, figure or reference limit; Declarations required; large language model use documented in Methods. Main text about 5,000 words plus 400 words of captions; paragraphs 100–150 words. Brackets: fill from results or still open. Background and Methods are the first author's prose with tracked corrections (`changes` package; change logs in `paper/drafts_20261022/`, `paper/drafts_20261023_pass3/` and `paper/pass4_changelog.md`. Pass 4 reframes the draft around the finished penalty-strength sweep: the benchmark at fixed strengths is not run, by the first author's decision of 2 November 2026; Pass 5 adds the replication across training seeds that replaced that benchmark, decided by the first author on the same day (change log `paper/pass5_changelog.md`)). Figures 1–6 are drawn from the sweep analysis (Figures 2–6 in Pass 10, Figure 1 in Pass 11, `paper/figure_scripts`, change logs `paper/pass10_changelog.md` and `paper/pass11_changelog.md`); Tables 2 and 3 hold sweep values (Pass 9, change log `paper/pass9_changelog.md`, number ledger `paper/pass9_numbers.csv`); Table 1 holds real design facts; the strength-range table of the planned fixed-strength benchmark, which is not run, was removed in Pass 4. Pass 13 numbers the main-text tables 1 to 3 (Table 1 the methods, Table 2 the ranking and Table 3 the placement table) and every reference to them is made by label. Pass 12 (change log `paper/pass12_changelog.md`) fixes the points of the Pass 11 rigour review; every abstract number is listed with its source in `paper/pass12_numbers.csv` and re-read by `paper/tools/check_ledger.py`. Pass 14 fills the replication across training seeds from its result (change log `paper/pass14_changelog.md`, numbers in `paper/pass14_numbers.csv`, Supplementary Table S9 generated by `paper/tools/make_replication.py`). Pass 15 fixes the points of the independent re-review and cuts the abstract (change log `paper/pass15_changelog.md`); the clean copy `paper/outline_clean.tex` is written by `paper/tools/accept_changes.py`. Pass 16 adds the family-level sign-flip sensitivity analysis to the replication result (change log `paper/pass16_changelog.md`). Pass 17 places the finished supplement (Supplementary Notes 2–12, Tables S1, S2, S4, S5 and S8) in the supplementary list and corrects the main text where the supplement's sources showed it wrong (change log `paper/pass17_changelog.md`).

Abstract

at most 350 words; Background, Results, Conclusions

1. **Background:** Single-cell batch correction often adds an alignment penalty (a discriminator, a Wasserstein-1 critic or a distance between batches) to a generative model, but penalties have rarely been compared on one backbone.
2. Question: which penalty keeps the most biology at equal batch removal in one scVI (single-cell variational inference) model, and does any beat plain scVI or standard tools?
3. Exploratory comparison on eight tasks of the single-cell integration benchmark: scVI plus the Jensen–Shannon discriminator, three Wasserstein-1 critics (reference, pooled, barycenter), maximum mean discrepancy (MMD) or the Sinkhorn divergence, each fitted at ten strengths from 0.1 to 3,000, with the decoder given the batch label or not and three seeds (100–102), beside plain scVI and standard tools. Instead of a planned fixed-strength run, the two sweep strengths bracketing matched batch removal were refitted with five new seeds (0–4).
4. **Results:** The pooled critic had the best mean rank with the batch-conditioned decoder (2.20 over five real tasks), and MMD with the unconditioned decoder (2.17 over six real tasks).
5. With the batch-conditioned decoder, the largest collapse rate on a real task ranged from 27% (12%–46%) for MMD to 73% (54%–88%) for the discriminator (nominal 95% intervals).
6. With this decoder, a penalty at one strength drawn at random exceeded plain scVI's selection score on at most 3 of six real tasks, and the best of six strengths on 4 to 6; at six tries every penalty exceeded Harmony and sysVI on 5 of six real tasks.
7. A replication across training seeds kept the sign of all four large within-pair differences (mean rank correlation 0.456, $p = 0.00049$).
8. **Conclusions:** Within these tasks, strength mattered more than penalty choice: penalties were near-tied at matched batch removal (160 of 195 within-pair differences under 0.02); a penalty exceeded batch-conditioned plain scVI mainly when several strengths were tried and scored on the same data, an optimistic gain; scANVI, trained with the cell-type labels that several biology metrics also use, exceeded every penalty on most tasks. Code and checksums: github.com/kreid415/wasserstein-critic-deconfounding.

Keywords: single-cell RNA sequencing; batch correction; adversarial learning; optimal transport; benchmarking

1 Background

Single-cell RNA-seq experiments provide a snapshot of a cell's transcriptional activity at a point in time; however, the biological differences in a dataset can be overshadowed by technical differences between experimental samples. Laboratory conditions, reagent lots, personnel and instruments introduce technical variation, which becomes a confounder

when it is correlated with the biological outcome (Leek et al., 2010; Yu et al., 2024). In single-cell RNA-seq, separately processed technical replicates can differ substantially (Tung et al., 2017), and single-cell data bring new challenges for modelling batch effects (Goh et al., 2022). This has led to a variety of batch correction methods, among them Harmony (Korsunsky et al., 2019), Scanorama (Hie et al., 2019), scVI (single-cell variational inference) (Lopez et al., 2018), its label-aware extension scANVI (Xu et al., 2021) and sysVI (Hrovatin et al., 2025). A central challenge of these methods is to balance the alignment of batches with the preservation of biological signal (Luecken et al., 2022). Under-correction results in datasets that retain substantial batch effects, while over-correction discards key biological information (Yu et al., 2024).

With the adoption of deep learning-based methods in single-cell analysis, techniques for trainable end-to-end batch correction are necessary. A notable class of deep-learning approaches, latent-variable models, aim to learn a low-dimensional representation of the RNA-seq readout that captures salient signals corresponding to biological programs. To align these latent spaces across batches, two primary methods have been adopted: decoder conditioning, which passes the batch label as an input variable to the decoder responsible for reconstructing the input RNA-seq vector from the latent variable (Lopez et al., 2018), and adversarial batch, which trains a network to predict the batch, or another confounder such as age, from each cell's latent code and penalizes the encoder whenever that prediction succeeds (Ganin et al., 2016; Dincer et al., 2020).

Adversarial methods are split between discriminator approaches, which minimize the Jensen–Shannon (JS) divergence (Goodfellow et al., 2014) between samples from different batches or confounding variables (Ganin et al., 2016; Dincer et al., 2020), and Wasserstein critic-based methods, which minimize the Wasserstein-1 (W1) distance (Arjovsky et al., 2017; Gulrajani et al., 2017) between such samples. These techniques have been applied extensively to single-cell batch correction.

iMAP trained a critic with the gradient penalty of the Wasserstein generative adversarial network (WGAN-GP), correcting every dataset towards one anchor dataset that is left unchanged (Wang et al., 2021). iDLC trains a WGAN-GP critic, with a Sinkhorn transport term added to the generator loss, to map each source batch onto a reference batch (Jiang et al., 2026); scStarCorrect also aligns every batch to one fixed reference batch, but through a domain-aware discriminator whose loss form its abstract does not state (Gu et al., 2025); Beaconet trains one Wasserstein critic per batch against the mixture of the other batches and needs no reference (Xu et al., 2024). IMGG trains a WGAN-GP generator towards a synthetic intermediate batch built from cells matched across batches (Wang et al., 2022a), ResPAN combines a Wasserstein GAN with random-walk mutual-nearest-neighbour pairing (Wang et al., 2022b), and IMAAE uses a loss of WGAN-GP form, an anchor batch and known cell labels (Wang et al., 2023). Methods built on a standard discriminator are numerous as well. Portal trains two logistic discriminators and clamps their logits so that cells scored as clearly dataset-specific give no gradient and are not forced to mix (Zhao et al., 2022); AIF pairs a real-versus-reconstructed discriminator with a batch classifier that the encoder is trained against (Monnier and Cournède, 2024); and ABC trains an autoencoder against a multi-class batch discriminator alongside cell-type classifier branches that use labels (Danino et al., 2024).

A third route avoids training an adversary network altogether by measuring the distance between the latent batches explicitly. These include the maximum mean discrepancy (MMD) (Gretton et al., 2012), and optimal transport (OT) measures (Cuturi, 2013; Genevay et al., 2018; Feydy et al., 2019). A full mathematical formulation of these approaches is available in Supplementary Note 1. Both measures have been used to align single-cell batches without an adversary. MMD-ResNet trains a residual network to match the distribution of one batch to that of another (Shaham et al., 2017), SAUCIE applies an MMD penalty against a reference batch on an internal layer (Amodio et al., 2019), trVAE places an MMD term across conditions in the first decoder layer (Lotfollahi et al., 2020), uniPort couples a variational autoencoder with minibatch unbalanced OT (Cao et al., 2022), and Li et al. added an MMD term between batches to the loss of an early scVI version (Li et al., 2022).

Despite their promise, alignment penalties have limitations that challenge their utility in practice. When batches contain different cell-type proportions, a lower bound on classification error grows as the latent distributions of the two batches are aligned more tightly (Zhao et al., 2019), and exact alignment under any divergence that is zero only for identical distributions leaves an error at least as large as the proportion gap (Wu et al., 2019). The sysVI study found that an adversarial conditional variational autoencoder and an adversarial multi-omics integration method mixed unrelated cell types when proportions were unbalanced across batches, and more so at higher adversarial strength (Hrovatin et al., 2025); dataset imbalance also changed clustering, annotation, differential expression, query mapping and trajectories across five integration methods (Maan et al., 2024).

Batch-correction methods have been compared in several large benchmarks (Tran et al., 2020; Luecken et al., 2022, 2025). These benchmarks compare complete methods, each with its own architecture, preprocessing and training settings, so they cannot show which alignment objective accounts for a method's performance. Controlled comparisons of alignment losses inside one framework are few. The extended single-cell integration benchmark (scIB-E) compared six batch losses, one of them a GAN discriminator, inside one scVI framework, with no Wasserstein, MMD or OT term (Yi et al., 2025); Li et al. compared MMD with an adversarially trained mutual-information penalty on an early scVI

version and traced the Pareto front between batch removal and biology (Li et al., 2022); and the iDLC ablation removes its Sinkhorn term but has no JS or MMD arm (Jiang et al., 2026). In searches of OpenAlex, arXiv and Europe PMC run in October 2026 and repeated in OpenAlex and Europe PMC on 5 November 2026 for works published since 1 October 2026, we found no study that compares JS, W1 and MMD or OT objectives on one stock scVI backbone. This absence is search-based, not exhaustive.

In this study we benchmark discriminator-, critic-, kernel- and optimal transport-based alignment penalties for batch correction. We compare the JS discriminator, three variants of the Wasserstein critic (reference, pooled and barycenter), MMD and the Sinkhorn divergence with standard baselines. Each penalty is added to the same scVI model (Lopez et al., 2018), with the decoder either given the batch label or not, and evaluated on eight tasks of the single-cell integration benchmark scIB (Luecken et al., 2022). The backbone and tasks are fixed, and the training settings are fixed except the adversary update steps and optimiser, which follow each method's usual setting, so differences between methods come from the alignment penalty as it is usually run. The comparison is exploratory. Every penalty is fitted across one strength range on every task and the analysis reads all of those fits. A benchmark at fixed strengths, planned to follow the sweep, was replaced by a smaller replication across training seeds: at the two sweep strengths on either side of the batch removal at which the penalties are compared (Methods), each penalty is refitted with five new seeds (0–4) on the same eight tasks, and the new fits are compared with the sweep's readings by a rule fixed before any of them was scored. The replication uses new seeds, not new data. It shows whether the sweep's readings hold when the training seed changes, and its rule tests only the sign of the within-pair differences of 0.04 or more (no precision calculation was made), so it does not test the order of penalties that sit close together.

2 Methods

To establish a fair comparison between methods, we use a fixed backbone from the well-established scVI (single-cell variational inference) architecture (Lopez et al., 2018), implemented in the scvi-tools python package (Gayoso et al., 2022). Every method uses scvi-tools' default settings and likelihood with epochs, minibatch size and latent size equal within a task, so that the alignment method and its strength are the only intended differences, apart from the adversary update steps and their optimiser, which follow each method's usual setting (Supplementary Note 2). Each adversary is trained with its standard settings: the reference, pooled and barycenter critics take five update steps per encoder step with a gradient penalty, as is usual for Wasserstein critics (Gulrajani et al., 2017), and the discriminator takes one update step, as is usual for discriminators, so the penalties are compared as they are normally run (Supplementary Note 2). We apply this across two scenarios: a conditioned setting in which the batch label is fed to the decoder, as scVI is usually run, and an unconditioned setting in which no batch label reaches the encoder or decoder, so the alignment penalty is the only correction (Figure 1a). The full hyperparameter and training details are available in Supplementary Table S2 and Supplementary Note 2.

Each method adds a fixed penalty, weighted by a fixed strength, to the normal scVI loss (Table 1). The discriminator predicts each cell's batch and the encoder minimizes scvi-tools' bounded fool loss, a Jensen–Shannon (JS) type objective (Goodfellow et al., 2014). We test three Wasserstein-1 (W1) critic-based methods(Arjovsky et al., 2017), (Gulrajani et al., 2017): reference based, pooled, and barycenter. The reference method aims to align all batches with a single reference batch, which we assign based on the highest cell-type entropy (the Shannon entropy of a batch's cell-type proportions, computed from the cell-type labels; ties go to the larger batch). The entropy rule prefers the batch whose cell types are most evenly represented, so that the target does not favour a few cell types, which a merely large batch, the other obvious choice, may do because a few cell types can dominate it. The chosen reference holds every cell type in five of the eight tasks, and 10 of 17 (lung), 14 of 19 (human–mouse immune) and 3 of 4 (simulation 2) in the others, so in those three tasks some cells of other batches have no cell of their type in the reference. The pooled critic compares each batch against the cells of all other batches pooled together. The Wasserstein barycenter approach (Cuturi and Doucet, 2014) estimates an average distribution of the batches present in each minibatch, recomputed at every step, and aligns every batch to it. The network-free approaches are the maximum mean discrepancy (MMD) of each batch to the others, with a fixed multi-scale Gaussian kernel (Gretton et al., 2012), and the debiased Sinkhorn divergence, an entropy-regularised optimal transport (OT) distance (Cuturi, 2013; Genevay et al., 2018; Feydy et al., 2019), which at the smoothing used is not an estimate of W1. For the full details of the loss equations and the training settings, see Supplementary Notes 1 and 2.

We benchmark these methods on eight integration tasks of the single-cell integration benchmark (scIB) (Luecken et al., 2022), used as distributed with their batch and cell-type labels. These consist of six real datasets from different biological settings, four of single-cell RNA sequencing (human pancreas, lung, immune cells, and joint human–mouse immune cells) and two of chromatin accessibility (assay for transposase-accessible chromatin, ATAC; mouse brain, a small and a large dataset, as gene-activity matrices), and two simulated datasets. RNA tasks use 2,000 highly variable genes and the ATAC tasks all features; counts are used as distributed. The full details of the datasets are available

in Supplementary Table S1. We score each method using the scIB batch, biology, and aggregate selection scores. The batch score is the mean of five metrics: principal-component regression on batch, batch average silhouette width (ASW), graph integration local inverse Simpson's index (iLISI), graph connectivity and the k-nearest-neighbor batch-effect test (kBET). The biology score is the mean of the task's biology metrics: normalized mutual information and adjusted Rand index between clusters and cell-type labels, cell-type ASW, isolated-label F1, isolated-label silhouette, graph cell-type LISI (cLISI), plus cell-cycle conservation on the four real RNA tasks and trajectory conservation on the two immune tasks, which have pseudotime. The mathematical formulation of these metrics is available in Supplementary Note 3. The scIB combined selection score is the weighted average $0.4\,\text{batch} + 0.6\,\text{biology}$.

Penalty-strength sweep. We fit plain scVI and the six penalties on all eight tasks at ten strengths in half-decade steps from 0.1 to 3,000 (0.1, 0.3, 1, 3, 10, 30, 100, 300, 1,000 and 3,000), with both decoders and three seeds (100–102). Two edge strengths were added in two places, 0.01 and 0.03 for the discriminator with the unconditioned decoder and 10,000 and 30,000 for the Sinkhorn divergence with the conditioned decoder, where a strength-choice rule (below) reached the end of the grid on the first three tasks fitted (the small chromatin-accessibility, immune and simulation 1 tasks). Each task therefore has 378 fits: 6 of plain scVI, 360 of the six penalties at ten strengths with three seeds and two decoders, and 12 at edge strengths; 3,024 fits in all. Nine further fits from an earlier strength-extension round, at strength 0.03 for the discriminator, the reference critic and the pooled critic on the immune task with the conditioned decoder, are also read by the analysis, which therefore uses 3,033 fits; that pair is low-signal and not ranked (Supplementary Note 4). A fit fails if its training loss or its latent becomes non-finite; failed fits are counted and left out of seed means (Supplementary Table S5). The sweep was planned as a pilot. Rules fixed before it ran were to use its seed means to choose six strengths per penalty family and decoder for a planned fixed-strength benchmark at five new seeds; that benchmark was not run, so the rules select nothing and every analysis below reads all strengths of the sweep. The comparison is therefore exploratory: it describes how each penalty behaves over its whole strength range on these tasks and tests no hypothesis fixed in advance.

Replication across training seeds. The planned fixed-strength benchmark was replaced by a smaller replication across training seeds, which asks whether the sweep's readings at matched batch removal (the next paragraph defines the matched level b^*) hold on new fits; the sweep's readings rest on three seeds per strength. For every task and decoder pair that is ranked (13 of 16; the three low-signal pairs, immune, simulation 1 and simulation 2 with the batch-conditioned decoder, are skipped) and every penalty, we took the two sweep strengths that bracket the first crossing of b^* on the sweep's seed-mean curve, with b^* fixed at its sweep value, and refitted them with five new seeds (0–4); 46 of these 78 curves cross b^* more than once, and the first crossing sets the bracket, as in the sweep reading. In one cell the first crossing lies between plain scVI and the lowest sweep strength, so plain scVI at the new seeds is its lower point. Plain scVI was refitted at the same five seeds on every task and decoder. This gives 855 fits, 775 with a penalty and 80 of plain scVI (Supplementary Note 4). The strengths were chosen from the sweep's batch scores alone, never from its biology scores. Each task's replication fits are made on the same kind of machine as its sweep fits and scored with the same scoring code; for three tasks the scoring site differs (the sweep fits of lung, simulation 2 and pancreas were scored on the workstation and their replication fits on the cluster, under the sweep's rule for combining the two sites; Supplementary Note 11). The reading is the sweep's: on the means over the new seeds, biology at b^* by linear interpolation in batch score between the two bracketing strengths, walking up from plain scVI at the new seeds and taking the first crossing. If the new means do not bracket b^*, the next sweep strength outward is fitted once with five seeds; if b^* is still not bracketed, the penalty takes the bottom rank in that pair and is marked as not replicating the crossing. The sweep's readings and ranks were written down as the prediction, and the agreement rule was fixed, before any replication fit was scored (Supplementary Notes 4 and 10). The replication agrees with the sweep if two conditions hold: each of the four within-pair differences of 0.04 or more in the sweep keeps its sign on the new seeds, and the mean over the 13 ranked pairs of Kendall's rank correlation (τ_b) between the sweep ranks and the replication ranks of the six penalties is above zero by a one-sided sign-flip permutation test over pairs (all 2^{13} sign patterns, $p < 0.05$). The sign-flip test assumes that the 13 pairs' correlations are independent and symmetric about zero when the two orders are unrelated; this holds only roughly, because the two decoders of a task use the same task data and some tasks are related (two nested pairs), so its p-value is approximate. A sensitivity analysis that flips the signs per task, both decoders of a task together (all 2^8 sign patterns over the eight tasks), was added after the rule was fixed; it is reported beside the rule and is not part of it, and because it still flips the two tasks of a nested pair separately, its p-value is approximate too; a second sensitivity analysis, also added after the rule was fixed, flips the signs per task family, so that the two tasks of each nested pair and both decoders flip together (all 2^6 sign patterns over six families: the two nested pairs, pancreas, lung, simulation 1 and simulation 2, which share no data); it handles the nesting but not other dependence between tasks, so it is approximate as well. Beside the rule, and not part of it, we report the share of sweep near-ties (differences under 0.02) whose replication difference stays within ± 0.04, the rank correlation per pair, and the batch score that every replication point reached. Sweep (seeds 100–102) and replication (seeds 0–4) readings are reported separately. The rule tests only the sign of the within-pair differences of 0.04 or more, and no precision calculation was made, so the replication is not set up to separate penalties whose readings differ by less, and because it refits the same

eight tasks it says nothing about other data.

Because strength has a different scale for each penalty and a stronger penalty trades biology for batch removal along a curve, as Li et al. showed for scVI (Li et al., 2022), we do not compare methods at a fixed strength. Instead we identify, for each task and decoder, a batch score, b^*, halfway between plain scVI's mean batch score and the highest mean batch score that every one of the six penalties reaches at a sweep strength with no failed fit (Supplementary Note 5); we therefore read each penalty's biology score at b^*, a batch removal that every penalty reaches (Figure 1b). This matched level only works if the penalties remove clearly more batch effect than plain scVI. We call a task and decoder pair low-signal when that gap is under 0.02, so that b^* sits less than 0.01 above plain scVI and cannot be placed reliably; low-signal pairs are left out of the ranking, shown in a separate table. The sweep identifies which pairs are low-signal: the immune task and both simulations with the batch-conditioned decoder, where plain scVI's batch score and the highest batch score every penalty reaches were 0.638 and 0.657 for immune (gap 0.019), 0.812 and 0.819 for simulation 1 (gap 0.007) and 0.656 and 0.665 for simulation 2 (gap 0.009); no pair is low-signal with the unconditioned decoder.

The main analysis is a descriptive ranking. For each task and decoder, a penalty's curve joins plain scVI and its sweep strengths (ten, or twelve where edge strengths were fitted) by seed-mean scores over the three seeds (100–102), and its biology score is read at b^* by linear interpolation where the curve first reaches b^* (Figure 1b). For each penalty, task and decoder the uncertainty is the standard error of that reading over repeats with each of the three seeds left out in turn, holding b^* fixed and leaving the same seed out of plain scVI, and the range (lowest to highest) of the penalty's rank over the three repeats. These are reported only if at least two of the three leave-one-seed-out readings exist, and otherwise as not available; the penalty is ranked either way (Supplementary Note 6). A task and decoder pair is ranked unless it is low-signal. Within each ranked pair the six penalties are ordered by biology at b^* (rank 1 is best); tied penalties get the average of their ranks. A penalty whose curve does not reach b^* takes the bottom rank in that pair, penalties that all fail sharing the last places, following scIB, where methods that could not be run on a task were given the same rank as unintegrated data (Luecken et al., 2022). As in scIB, the mean rank over all real tasks that are not low-signal is reported for each decoder; simulations are ranked and reported separately, and low-signal pairs are shown separately and left out of the ranking (Supplementary Note 6). We report no significance tests: ranks and their uncertainty describe the data. Each penalty's selection score is also compared with plain scVI's, at one strength not tuned on the task and as the expected best of k tries (below). The ranking rules were written for the planned fixed-strength benchmark and are applied to the sweep with the changes listed in Supplementary Note 10: the three tasks fitted first had already been read by the strength-choice rules, whereas no sweep score of the other five tasks existed when the ranking rules were written (dated 22 and 23 October 2026; both commits were made on 22 October), although score files of some baseline fits on four of them are older, from 6 to 12 October (Supplementary Notes 10 and 11).

We place the penalties against plain scVI and against standard tools: scvi-tools' own adversarial classifier (its built-in batch classifier, run with the conditioned decoder only), Harmony (Korsunsky et al., 2019), Scanorama (Hie et al., 2019), sysVI (Hrovatin et al., 2025), scANVI, which uses the cell-type labels (Xu et al., 2021), and principal-component analysis (PCA) as the uncorrected reference. We compare all of these approaches at equal numbers of tries, allowing for the outcome that no alignment penalty improves on plain scVI. Harmony, Scanorama and sysVI run at six settings each, scANVI and PCA at one (settings, seeds and fit counts in Supplementary Table S4 and Supplementary Note 8). A try is one setting of a method, fitted and scored. The scvi-tools adversary, scANVI and sysVI are fitted with five seeds (0–4), and Harmony, Scanorama and PCA once per setting, all on every task: 448 planned fits (447 fitted and scored; Results), including three sensitivity settings (Harmony at 50 principal components, Scanorama at 100 dimensions and PCA at 50 components). Because the penalties and the tools have different numbers of settings, placement compares methods at equal numbers of tries, by the expected best selection score among k settings drawn at random, and is descriptive. Each penalty is also scored at one strength not tuned on that task, the strength with the best mean selection score on the real tasks of the other task families, shown beside the task's own best strength as an optimistic bound (Supplementary Note 7). Harmony, Scanorama and sysVI get their setting by the same rule: the setting with the best mean selection score on the real tasks of the other families, with the task's own best setting in parentheses as the optimistic bound (Supplementary Note 8). scANVI and PCA run at one setting, so they involve no choice.

For each penalty we also report how often it collapses and how narrow its range of good strengths is, together with its hypervolume, its share of non-dominated strengths, its local versus global batch removal and label transfer; all six views are descriptive, and each rule is given in full in Supplementary Notes 7 and 9. A fit collapses if it fails or its biology score is at least 0.10 below the same seed's plain scVI; the collapse rate is the collapsed share of a penalty's fits in a task and decoder, with exact binomial (Clopper–Pearson) 95% intervals (Clopper and Pearson, 1934), which are nominal because the fits of one seed are judged against one plain-scVI threshold and are not independent (Discussion). The good-strength width is the span, in decades, of the strengths whose mean selection score is at least the lower 95% confidence bound of the best strength's mean. The hypervolume (Zitzler and Thiele, 1998), which Li et al. used to

compare trade-off fronts (Li et al., 2022), is the part of the batch-score by biology-score plane that a penalty's sweep points reach beyond a reference point at plain scVI's batch score and 0.10 below its biology score, divided by the batch gap that sets b^* so that tasks share one scale. The share of non-dominated strengths, also from Li et al., is the fraction of a penalty's strengths that none of its other strengths beats on both the batch and the biology score.

Local removal is the mean of the three neighbourhood batch metrics (iLISI, kBET and graph connectivity) and global removal the mean of the two whole-embedding metrics (principal-component regression and batch ASW), so that at b^* more local removal means less global; we report each penalty's local minus global reading at its crossing of b^*. Label transfer holds each batch out in turn and predicts its cell types from the other batches with a 15-nearest-neighbour classifier, on each penalty's embedding at the strength whose batch score is nearest b^*, never chosen by accuracy (Supplementary Note 9).

A direct biology check reads four per-cell-type quantities on the saved embeddings, with no new fits: neighbourhood purity, the share of a cell's 15 nearest neighbours in the latent space that carry its own cell-type label, averaged within each type; the same purity and the label-transfer recall for rare types, those with under 1% of a task's cells; marker-gene preservation, the Jaccard overlap between each type's top 20 markers from the expression as distributed and from the expression averaged over the 15 neighbours, on the four RNA tasks; and the label-transfer confusion table. Penalties are read at the label-transfer strength and every baseline at each of its settings, and each value is reported as a difference from plain scVI. The check is exploratory: its rules were written after the sweep scores had been read and before any of its values was computed (Supplementary Note 12).

Use of large language models. Large language model tools were used in this work: Claude (Anthropic), through the Claude Science and Claude Code applications, and Codex (OpenAI). They wrote and checked analysis, scoring and figure code; prepared, ran and monitored computational jobs on the workstation and the cluster; assisted with literature searches; checked the numbers in the text against their source files; reviewed the text independently; and drafted and edited text. The study design and the decisions about what was run, measured and reported were made or approved by the authors. All figures are drawn from the data by plotting scripts in the repository (no image was made by a generative model), and every number in the paper is re-read from its source file by a script (`paper/tools/check_ledger.py`). The authors checked all outputs and take responsibility for the content.

Results

1,500 words; 8 paragraphs plus one for the replication across training seeds and one for the direct biology check

Res-1. We first report which fits completed and where batch removal could be matched.
All 3,033 sweep fits completed, with 0 failed fits across the penalties and decoders (Supplementary Table S5). The analysis includes 387 fits on the immune task and 378 on each other task; the immune total includes additional strengths for the discriminator, the reference critic and the pooled critic with the batch-conditioned decoder. Of the 448 baseline fits, 447 were scored and 0 failed: the scvi-tools adversary 40 of 40, scANVI 40 of 40, sysVI 240 of 240, Harmony 56 of 56, Scanorama 55 of 56 and PCA 16 of 16. The remaining fit, Scanorama with 80 neighbours on the large chromatin-accessibility task, could not be fitted (it ran out of memory, and a retry stopped at a 32-bit sparse-matrix index limit); it is reported as an incomplete setting, not as a failed fit, so on that task Scanorama is placed with five of its six settings (Supplementary Table S5). The low-signal pairs were the immune task and both simulations with the batch-conditioned decoder: plain scVI and the highest common batch scores were 0.638 and 0.657 (gap 0.019) for the immune task; 0.812 and 0.819 (gap 0.007) for simulation 1; 0.656 and 0.665 (gap 0.009) for simulation 2. Every penalty reached b^* in every task and decoder pair, including the low-signal pairs; none received a bottom rank for a missing full-seed reading. Low-signal pairs are reported separately under the Methods rule (Supplementary Note 5).

Res-2. Penalties are ranked by the biology they keep at matched batch removal (Figure 2, Table 2).
With the batch-conditioned decoder, the ranking rested on five real tasks. Mean ranks (range across those tasks) were the pooled critic 2.20 (1–3); the Sinkhorn divergence 2.40 (1–5); the barycenter critic 3.00 (1–5); the reference critic 3.80 (1–5); MMD 4.60 (2–6); the discriminator 5.00 (3–6). With the unconditioned decoder, all six real tasks contributed: MMD 2.17 (1–6); the reference critic 3.33 (1–6); the pooled critic 3.33 (2–5); the discriminator 3.50 (1–5); the barycenter critic 4.00 (1–6); the Sinkhorn divergence 4.67 (3–6). The Sinkhorn divergence ranked first most often with the batch-conditioned decoder, on the human–mouse immune and lung tasks; the discriminator ranked last on the small chromatin-accessibility, lung and pancreas tasks. With the unconditioned decoder, MMD ranked first most often, on the large chromatin-accessibility, human–mouse immune and pancreas tasks; the reference critic ranked last on the small chromatin-accessibility and human–mouse immune tasks, and the Sinkhorn divergence on the large chromatin-accessibility and lung tasks.
Neighbouring biology readings often differed by less than at least one of their reported leave-one-seed-out standard errors (Methods; Supplementary Table S6). For example, with the batch-conditioned decoder on the small chromatin-accessibility task, the pooled critic and the Sinkhorn divergence read 0.632 and 0.631, with standard errors 0.007 and 0.004, respectively. With the unconditioned decoder on the large chromatin-accessibility task, MMD and the reference

critic read 0.622 and 0.621, with standard errors 0.010 and 0.006. Some gaps exceeded both neighbouring standard errors: on the small chromatin-accessibility task with the batch-conditioned decoder, MMD exceeded the reference critic by 0.017 (standard errors 0.007 and 0.003). The simulations were reported separately (Supplementary Figure S1): with the unconditioned decoder, the order from highest to lowest biology was the discriminator, the reference critic, MMD, the Sinkhorn divergence, the pooled critic and the barycenter critic in simulation 1, and the discriminator, the pooled critic, MMD, the reference critic, the barycenter critic and the Sinkhorn divergence in simulation 2. Neither simulation was ranked with the batch-conditioned decoder because both pairs were low-signal.

Res-2b. The replication across training seeds reports whether the sweep's readings at b^* hold on new fits (Supplementary Table S9).

- Rule: the sweep's readings and ranks at b^* in the 13 ranked pairs are the prediction; the replication agrees with the sweep if each of the four large sweep gaps (0.04 or more within a pair) keeps its sign and the mean Kendall's τ_b over the 13 pairs is above zero by the sign-flip permutation test; everything else in this paragraph is descriptive (Methods; Supplementary Notes 4 and 10).
- All 895 replication fits were scored and 0 failed: the 855 planned fits and 40 fits of the extension step, which refitted the next sweep strength once, with seeds 0–4, for the 8 readings whose new-seed means did not bracket b^* (Supplementary Note 11). Plain scVI at seeds 0–4 reached batch scores within -0.006 to 0.005 of the sweep's plain scVI in the 13 ranked pairs. Before the extension step 70 of the 78 readings bracketed b^*; after it 77 did. One reading, MMD on the pancreas task with the batch-conditioned decoder, did not replicate the crossing (its highest new-seed mean batch score was 0.674, against $b^* = 0.684$) and takes the bottom place in its pair.
- Biology at b^* on the new seeds differed from the sweep reading by a median of 0.004 and at most 0.024 (absolute differences over the 77 readings that reach b^*); the mean absolute difference per pair ranged from 0.002 to 0.011. In 7 readings the new-seed mean at the lower refitted strength had already reached b^*, so the reading interpolates from plain scVI at seeds 0–4. The first-ranked penalty was the same as in the sweep in 6 of 13 pairs; 29 of the 78 penalties kept their sweep rank and 56 moved by at most one place. Supplementary Table S9 gives every reading with the batch and biology scores of its two bracketing points.
- Each of the four large sweep gaps kept its sign on the new seeds (4 of 4); the closest to flipping was the barycenter critic minus the reference critic on the small chromatin-accessibility task with the unconditioned decoder (sweep $+0.041$, replication $+0.013$). Kendall's τ_b ranged from -0.067 to 0.867 over the 13 pairs (11 positive), with mean 0.456 and one-sided sign-flip $p = 0.00049$ (4 of the 2^{13} sign patterns reach the observed mean). The agreement rule is met. Before the extension step that gap could not be read, because the barycenter critic's new-seed mean had not reached b^*; the other 3 gaps kept their sign and the mean τ_b was 0.426 ($p = 0.00049$), so the rule was met only once the step it prescribes had been run.
- Sensitivity analysis, not part of the rule: with the signs flipped per task, both decoders of a task together (all 2^8 patterns over the eight tasks), $p = 0.0039$. Only the observed pattern reaches the observed mean (1 of 256), because the summed τ_b is positive in all 8 tasks, so this is the smallest p that eight tasks allow. It keeps the two decoders of a task together but still flips the two tasks of each nested pair separately, so it is approximate as well (Methods). With the signs flipped per task family (the two tasks of each nested pair and both decoders together; all 2^6 patterns over six families), $p = 0.0156$. Only the observed pattern reaches the observed mean (1 of 64), because the summed τ_b is positive in all 6 families, so this is the smallest p that six families allow; it was defined after the rule and is approximate as well (Methods).
- Of the 160 sweep near-ties (differences under 0.02), 155 kept a replication difference within ± 0.04 (127 before the extension step). Near-tied penalties stayed close, but the rule tests only the sign of the large gaps, so their order is not tested by it (Methods). The slope-adjusted reading, a sensitivity analysis whose formula was defined after the rule was fixed (the rule names it without one), kept the sign of 4 of the 4 large gaps, with mean τ_b 0.374 and $p = 0.00073$.

Res-3. Figure 3 shows each method's trade-off between batch removal and biology in every real task.
The real-task curves overlapped near plain scVI and around many matched readings, but differed more at stronger penalties (Figure 3; curve construction in Supplementary Note 5). With the batch-conditioned decoder, the lung and pancreas curves were close near b^*, whereas the discriminator lost more biology at stronger settings. On the large chromatin-accessibility task, the barycenter critic reached a lower maximum batch score, 0.674, than the reference critic, 0.815. This shorter batch-removal range did not reflect failed fits or failure to reach b^*. At the strongest fitted setting on lung with the batch-conditioned decoder, the barycenter critic ended at batch and biology scores of 0.466 and 0.397, below plain scVI on both scores. With the unconditioned decoder, the chromatin-accessibility curves lost biology as batch removal increased, whereas on pancreas several penalties increased both scores over part of their strength range. Thus, a curve could turn back or lose biology after reaching the matched level; no real-task curve ended early because of failed fits.

Res-4. Hypervolume, share of non-dominated strengths and collapse rate give three further views of each penalty

(Figure 4a,b, Supplementary Table S8).
The secondary summaries varied across real tasks (Figure 4; Supplementary Table S8; definitions in Supplementary Note 7). Hypervolume ranges with the batch-conditioned and unconditioned decoders, respectively, were the discriminator 0.029–0.255 and 0.060–0.386; the reference critic 0.022–0.269 and 0.038–0.383; the pooled critic 0.047–0.344 and 0.074–0.440; the barycenter critic 0.055–0.108 and 0.072–0.128; MMD 0.039–0.178 and 0.089–0.390; the Sinkhorn divergence 0.060–0.155 and 0.085–0.192. The corresponding ranges of the share of non-dominated strengths were the discriminator 0.400–0.545 and 0.083–0.583; the reference critic 0.364–0.600 and 0.100–0.900; the pooled critic 0.364–0.600 and 0.200–0.700; the barycenter critic 0.200–0.600 and 0.200–0.600; MMD 0.300–0.600 and 0.300–0.700; the Sinkhorn divergence 0.250–0.500 and 0.300–0.600. These shares and collapse summaries include the low-signal immune pair. The largest real-task collapse rate for each penalty, with its exact nominal 95% interval, was as follows with the batch-conditioned decoder: the discriminator 73% (54%–88%; the human–mouse immune task); the reference critic 57% (37%–75%; the small chromatin-accessibility task); the pooled critic 47% (28%–66%; the small chromatin-accessibility task); the barycenter critic 33% (17%–53%; the small chromatin-accessibility task); MMD 27% (12%–46%; the small chromatin-accessibility task); the Sinkhorn divergence 42% (26%–59%; the small chromatin-accessibility task). With the unconditioned decoder, the corresponding maxima were the discriminator 61% (43%–77%; the small chromatin-accessibility task); the reference critic 57% (37%–75%; the small chromatin-accessibility task); the pooled critic 57% (37%–75%; the large chromatin-accessibility task); the barycenter critic 30% (15%–49%; the large chromatin-accessibility task); MMD 13% (4%–31%; the large chromatin-accessibility task); the Sinkhorn divergence 30% (15%–49%; the small chromatin-accessibility task). All recorded collapses reflected loss of biology rather than failed fits. MMD had lower collapse rates than the discriminator on every real task with the batch-conditioned decoder, although it ranked above the discriminator on only some of the ranked tasks.

Res-5. We then report whether each penalty removes batch effects more locally or more globally, and the label-transfer check (Figure 4c; label transfer in Supplementary Figure S3).
At b^*, every penalty had more global than local batch removal on the ranked real tasks except the large chromatin-accessibility task, where the local-minus-global reading was positive (Figure 4c; Methods and Supplementary Note 7). On that task, readings for the discriminator, the reference critic, the pooled critic, the barycenter critic, MMD and the Sinkhorn divergence were 0.043, 0.041, 0.043, 0.039, 0.036, 0.039 with the batch-conditioned decoder and 0.105, 0.109, 0.111, 0.113, 0.106, 0.091 with the unconditioned decoder. The ordering was task-dependent: the discriminator had the highest local-minus-global reading on lung with the batch-conditioned decoder (-0.356), whereas the Sinkhorn divergence had the highest on lung with the unconditioned decoder (-0.289). In the same penalty order, mean label-transfer macro F1 over the ranked real tasks was 0.752, 0.790, 0.771, 0.796, 0.799, 0.769 with the batch-conditioned decoder and 0.726, 0.801, 0.818, 0.815, 0.811, 0.783 with the unconditioned decoder (Supplementary Figure S3; Supplementary Note 9). These orders differed from the biology ranking at b^*. Label-transfer embeddings were not exactly matched: for example, on the small chromatin-accessibility task with the batch-conditioned decoder, the pooled critic and the Sinkhorn divergence had macro F1 scores of 0.840 and 0.851, at batch-score distances of 0.024 and 0.010 from b^*, respectively. Along the full strength curves, label transfer also deteriorated at some strong settings: the barycenter critic ended at 0.360 on lung with the batch-conditioned decoder.

Res-5b. A direct biology check reads purity, rare types, marker genes and label-transfer confusion per cell type (Supplementary Table S10).
This check is exploratory and descriptive: it was defined after the sweep scores had been read, uses 783 saved fits (penalties at the label-transfer strength, seeds 100–102; plain scVI; every scored baseline setting) and runs no test (Methods; Supplementary Note 12). Over the six real tasks, neighbourhood purity per cell type differed from plain scVI with the same decoder and seed by -0.052 for the discriminator with the batch-conditioned decoder and -0.077 with the unconditioned decoder, and by -0.022 and -0.033 for the Sinkhorn divergence; the reference, pooled and barycenter critics and MMD stayed between -0.019 and $+0.001$ with either decoder. Harmony, averaged over its 7 settings, differed from batch-conditioned plain scVI by -0.141, and scANVI, which is trained with the labels, by $+0.115$. Rare types (present on the pancreas, lung, immune and human–mouse immune tasks) lost purity in 23 of 48 task, penalty and decoder cells, most with the discriminator (-0.107 with the batch-conditioned decoder). The largest drop, -0.406, was for the 46 ionocytes of the lung task with the batch-conditioned discriminator at strength 3,000: their label-transfer recall was 0.326 (seed SD 0.407) against 0.797 for plain scVI, and a share of 0.232 of them was predicted as dendritic cells. Over all cell types, label-transfer recall differed from plain scVI by -0.040 and -0.085 for the discriminator with the two decoders and by -0.024 to $+0.003$ for the reference, pooled and barycenter critics and MMD. Marker genes were kept alike: the Jaccard overlap of the top 20 markers was 0.390 to 0.399 for every penalty and plain scVI, with differences from plain scVI of at most 0.006, whereas Harmony (-0.031) and PCA (-0.028) were lower. The direct readouts single out the discriminator, which also had the worst mean rank with the batch-conditioned decoder and the lowest mean label-transfer F1 with both decoders (Res-5); with the batch-conditioned decoder its loss of purity was larger for rare types than over all types.

Res-6. The range of good penalty strengths is reported for every method (Figure 5).
Across the real tasks, including the low-signal immune pair, good-strength widths with the batch-conditioned and unconditioned decoders, respectively, ranged as follows: 0.0–4.5 and 0.0–3.5 decades for the discriminator; 0.0–3.5 and 1.0–4.0 decades for the reference critic; 0.0–2.0 and 0.5–4.5 decades for the pooled critic; 0.0–3.5 and 0.0–2.5 decades for the barycenter critic; 0.0–4.5 and 0.0–3.5 decades for MMD; 2.0–5.5 and 1.0–4.5 decades for the Sinkhorn divergence (Figure 5; definition in Supplementary Note 7). The Sinkhorn divergence had a nonzero width on every real task with either decoder. Each other penalty had a zero-width window on at least one real task and decoder pair; a zero width indicates a single qualifying strength, not a missing result. For example, MMD had widths of 4.5 decades on each chromatin-accessibility task with the batch-conditioned decoder but 0.0 on the human–mouse immune task. Every real-task pair had a defined window. The best strength lay at a grid edge for some penalties: on pancreas with the batch-conditioned decoder, MMD was best at the largest fitted strength, 3,000; on the large chromatin-accessibility task with the unconditioned decoder, the Sinkhorn divergence was best at the smallest, 0.1. Some windows contained gaps between qualifying strengths, so their reported span need not represent a continuous range of good strengths.

Res-7. Single-strength tables place each method against plain scVI and the standard tools (Table 3, Figure 6b).

- Rule: each penalty, and each of Harmony, Scanorama and sysVI at one of its settings, is shown at a strength or setting not tuned on that task, the one with the best mean selection score on real tasks of other families (related tasks never choose for each other), beside the task's own best strength as an optimistic bound; descriptive (Supplementary Notes 7 and 8, Table S4).
- At the strength chosen on real tasks of other families, the penalties with the batch-conditioned decoder exceeded plain scVI's selection score on 3 (discriminator), 4 (reference critic), 5 (pooled critic), 5 (barycenter critic), 4 (MMD), 4 (Sinkhorn divergence) of the six real tasks, with differences from plain scVI from -0.023 to $+0.045$ (median $+0.006$; Table 3); 11 of these 36 differences were smaller than 0.005 in size (for example -0.0004 for the Sinkhorn divergence on pancreas, which Table 3 prints as a tie), against seed standard deviations of the penalty's selection score of 0.001 to 0.016 in those cells, so the counts mix small gains, small losses and near-ties; at each task's own best strength, the optimistic bound, they did so on 5 to 6 of six. At their transferred settings, Harmony, Scanorama and sysVI exceeded batch-conditioned plain scVI on 1, 0 and 1 of six real tasks. With the unconditioned decoder, whose plain scVI receives no batch information, the penalties at the transferred strength exceeded plain scVI on 2 to 6 of six (Supplementary Table S7). The transferred strengths with the batch-conditioned decoder were discriminator 0.1–0.3; reference critic 0.3–3; pooled critic 1; barycenter critic 0.3–3; MMD 30; Sinkhorn divergence 1–100.

Res-8. The comparison at equal numbers of tries reports whether any penalty improves on plain scVI (Figure 6a; per-task curves in Supplementary Figure S6).

- Rule: each method gets the expected best selection score among k of its settings drawn at random, $k = 1$–6, the number of settings of Harmony, Scanorama and sysVI (comparisons at equal tries stop at six, although the penalties have ten or twelve sweep strengths), so methods with more settings gain no advantage; optimistic because the best setting is chosen and scored on the same data; scANVI has its own stratum (Supplementary Note 7).
- With the batch-conditioned decoder, a penalty's expected best selection score at one try exceeded plain scVI's on 0, 2, 2, 0, 3, 1 of the six real tasks (in the order discriminator, reference, pooled and barycenter critics, MMD, Sinkhorn divergence), 8 of 36 task–penalty cells, and at six tries on 4, 5, 5, 5, 5, 6 (30 of 36). Averaged over the six real tasks, plain scVI scored 0.648; the penalties ranged from 0.609 to 0.653 at one try and from 0.661 to 0.666 at six (Figure 6a), and their real-task means first exceeded plain scVI's at 4, 3, 2, 2, 1, 2 tries in the same order. Against the standard tools at six tries, every penalty exceeded Harmony on 5 of six real tasks, sysVI on 5 of six and Scanorama on all 5 tasks where its six settings exist; at one try the counts were Harmony 4 to 5, sysVI 4 to 5 and Scanorama 6 of six. The scvi-tools adversary (mean 0.651) was exceeded at one try on 0 to 4 of six tasks and at six tries on 5 to 6. scANVI, which uses the cell-type labels (mean 0.700), was exceeded by no penalty at one try with either decoder; at six tries with the batch-conditioned decoder every penalty exceeded it on pancreas only, by 0.002 to 0.010, and with the unconditioned decoder on no task. With the unconditioned decoder the reference is plain scVI without batch information (mean 0.578), which the penalties exceeded at one try on 1 to 5 of six real tasks and at six tries on 5 to 6. Every best-of-k value is optimistic for methods with more than one setting, because the best setting is picked and scored on the same data (Supplementary Note 7).
- Descriptive; best-of-k labelled optimistic.

Discussion

700 words; 5 paragraphs

Disc-1. The penalty with the best mean rank depended on the decoder: the pooled critic ranked highest with the batch-conditioned decoder, followed closely by the Sinkhorn divergence, whereas MMD ranked highest with the unconditioned decoder. The task-level orders varied, and many neighbouring biology readings were within a leave-one-seed-out standard error. These descriptive rankings therefore do not establish a consistent ordering across tasks. The secondary

summaries also gave different orders: MMD had relatively low collapse rates despite ranking near the bottom with the batch-conditioned decoder, and the local-versus-global readings did not give a consistent advantage to one penalty. This exploratory comparison on eight tasks with three seeds (100–102) supports a decoder-specific description of the sweep; the replication across training seeds, on new seeds of the same tasks, kept the four large within-pair gaps and agreed with the pair orders in part (Res-2b).

- The replication across training seeds met its agreement rule: the four large within-pair gaps kept their sign, and the sweep and replication orders were positively correlated (mean τ_b 0.456, $p = 0.00049$; $p = 0.0039$ with signs flipped per task and $p = 0.0156$ per task family; all three values are approximate, because related tasks are not independent). The orders did not hold in detail: the first-ranked penalty changed in 7 of 13 pairs, τ_b was slightly negative in 2 pairs (pancreas and the human–mouse immune task, both with the batch-conditioned decoder; $\tau_b = -0.067$), and MMD on pancreas with the batch-conditioned decoder did not replicate its crossing. Because the replication refits the same tasks with new seeds, not new data, and its rule tests only the sign of gaps of 0.04 or more, it supports the large gaps and the broad order, not the order of near-tied penalties.

Disc-2. The secondary analyses suggest that the penalties differ partly in how much biology they lose as strength increases and in how broadly useful strengths are distributed within the fitted range. The discriminator often lost biology beyond the matched level, while MMD generally had lower collapse rates; nevertheless, lower collapse did not ensure a higher biology rank at b^*. Hypervolume and the share of non-dominated strengths describe different aspects of the curves, so their orders need not agree with a ranking at a single batch score. Likewise, label-transfer scores did not follow the matched-biology ranking, and their batch-score distances from b^* limit direct comparisons. Local-versus-global differences varied by task, suggesting no common local-removal explanation for the ranking. These patterns describe each penalty as it is usually trained, with each adversary's update steps and optimiser at its standard setting (Methods).

Disc-3. For users of scVI, the sweep points to the penalty strength, more than the choice among penalties, as the setting that matters. At matched batch removal the six penalties were near-tied: 160 of the 195 within-pair differences in biology over the 13 ranked pairs were under 0.02, and 4 were 0.04 or more. With the batch-conditioned decoder, going from one to six tries raised each penalty's mean expected best selection score over the real tasks by 0.012 to 0.053, whereas at six tries the six penalties' means spanned only 0.661 to 0.666. Added to scVI as it is usually run, with the batch-conditioned decoder, a penalty at one strength drawn at random rarely beat plain scVI (8 of 36 task–penalty cells); the best of six strengths did in 30 of 36, by small margins (real-task means 0.661–0.666 against 0.648).

- When to add a penalty: when several strengths can be fitted and scored on the task, because the real-task mean of the best of k first exceeded batch-conditioned plain scVI at 1 (MMD) to 4 (discriminator) tries. Strength range: the strengths chosen on other task families with the batch-conditioned decoder lay between 0.1 and 100. Standard tools: at equal tries the penalties exceeded Harmony, Scanorama and sysVI on most real tasks, but not scANVI, which uses cell-type labels and was exceeded only on pancreas at six tries. These are optimistic best-of-k readings on eight scIB tasks.

Disc-4. The design limits what the data can show.

- Six real tasks in four families, two of them nested pairs (human–mouse immune contains all immune cells; large ATAC contains small); related tasks count as separate tasks in the mean rank, and the ranking is descriptive. A penalty whose curve does not reach b^* is ranked last in that pair, so a missing reading counts as the worst result.
- The comparison is exploratory. The strength grid was fixed in advance and the sweep reports all ten strengths, none left out (the strengths of the single-strength tables and of the replication were then chosen from sweep scores, as Methods describes); the planned fixed-strength benchmark that was to follow the sweep (six strengths per family, five new seeds) was replaced by the smaller replication across training seeds. The sweep results describe the sweep and test no hypothesis fixed in advance, and the ranking rules were applied to the same fits that the strength-choice rules had already read. Three seeds per strength: the standard errors in Supplementary Table S6 show how far apart neighbouring ranks are. The replication checks the sweep's readings at b^* on new fits against a prediction and an agreement rule fixed before any replication score was read, but it refits the same eight tasks with new training seeds, not new data, so it says nothing about other datasets, batch structures or preprocessing. Its rule tests only the sign of the within-pair differences of 0.04 or more, and no precision calculation was made, so it does not separate penalties whose readings differ by less. Its strengths and b^* were set from the sweep's batch scores after the sweep had been read, so it is a replication of the sweep's readings, not a test fixed before the sweep.
- All fits use one backbone, scVI, with one fixed architecture and latent size (10 latent dimensions, one hidden layer of 128 units), so the ranking may not hold for other backbones, architectures or latent sizes. The eight tasks are of the size of the single-cell integration benchmark, not of an atlas, so the ranking may not hold for integrations of that scale. The replication across training seeds refits the same eight tasks, so it tests the sweep's readings against the training seed and not against new datasets. The biology checks, including label transfer, are exploratory: they were defined after the sweep scores had been read.

- The strength sweep is what Li et al. call scalarization: one penalty is run at a series of fixed strengths and both scores are read at each (Li et al., 2022). The ranking therefore describes the penalties as users run them, not the best trade-off front that a multi-objective solver could reach for each penalty. Li et al. also used an adversarially trained mutual-information penalty (MINE); it is not among the six penalties and was not tested here.
- One batch variable per task; ATAC as gene-activity matrices; non-integer counts as distributed; scoring includes training cells; Sinkhorn not W1; reference choice uses labels.
- The collapse-rate intervals are nominal exact binomial 95% intervals. They treat a penalty's fits in a task and decoder (30; 33 for the discriminator, the reference critic and the pooled critic on the immune task with the batch-conditioned decoder, which have additional strengths; 36 where edge strengths were fitted) as independent, although each seed's fits are judged against the same plain-scVI threshold and can collapse together, so their actual coverage may differ from 95%. An illustrative simulation of this dependence is in Supplementary Note 7; it does not bound the coverage of the intervals reported here.

Disc-5. Several factors are outside the scope of this comparison.

- Not varied or tested: architecture, Kullback–Leibler (KL) warm-up (incomplete on shorter tasks, penalty from start), reference batch, batch number; composition shift read from strength curves, not a dose-response test.

Conclusions

about 150 words; 1 paragraph

- On eight tasks of the single-cell integration benchmark, six alignment penalties added to one scVI model kept similar biology at matched batch removal: the best mean rank was the pooled critic's with the batch-conditioned decoder (2.20) and MMD's with the unconditioned decoder (2.17), but 160 of the 195 within-pair differences were under 0.02, so the ranking does not separate most penalties. At equal numbers of tries with the batch-conditioned decoder, the penalties exceeded Harmony, Scanorama and sysVI on most real tasks and plain scVI mainly when several strengths were tried and scored on the same data; scANVI, which uses cell-type labels, was exceeded only on pancreas.
- Observations for scVI users, from the scores on these eight tasks and not tested on independent data: (i) the strength mattered more than the choice of penalty, and plain scVI with the batch-conditioned decoder remained a strong candidate, since a penalty at one strength drawn at random beat it in 8 of 36 task–penalty cells and the best of six strengths in 30, a count that is optimistic because the strengths were chosen and scored on the same data; (ii) the strengths chosen on other task families with that decoder lay between 0.1 and 100 across penalties (Results); (iii) the biology was worth checking directly at the chosen strength, because the discriminator lost the most neighbourhood purity in the direct biology check (-0.052 against plain scVI, -0.107 for rare types, batch-conditioned decoder); (iv) when cell-type labels were available, scANVI scored higher than every penalty on most tasks, but it is trained with the labels that several biology metrics also use, so its biology scores are not label-free.

Declarations

eight BMC subheadings, each filled or marked not applicable

Ethics approval and consent to participate. Not applicable. This study analyses only publicly available, previously published data: the integration tasks of the single-cell integration benchmark (figshare article 12420968, version 8), which hold human and mouse single-cell data from earlier studies, and two simulated datasets. No new data were collected from human participants or animals.

Consent for publication. Not applicable.

Availability of data and materials. Code and data access.

- Repository github.com/kreid415/wasserstein-critic-deconfounding; fitting and scoring code at commit 10a1ca5, sweep analysis at commit 8595a16, biology check at commit c4a793f, figure scripts and their inputs in `paper/figure_scripts` and `paper/figure_data`; [*to be decided: release tag and commit of the submitted version*]; checksums in supplement.
- The eight task files are the integration-task datasets of the single-cell integration benchmark, figshare article 12420968, version 8 (doi:10.6084/m9.figshare.12420968.v8); file checksums in Supplementary Table S1.

Competing interests. The authors declare that they have no competing interests.

Funding. [*to be decided: funding statement*]

Authors' contributions. [*to be decided: authors' contributions statement*]

Acknowledgements. [*to be decided: acknowledgements*]

Authors' information (optional). Not applicable.

Supplementary material

Supplementary Notes 1–12, Tables S1, S2 and S4–S11, Figures S1–S6

Every supplementary note and table listed here is in this document: Supplementary Notes 2–12 and Tables S1, S2, S4, S5 and S8 are printed below in the place of their entries (Table S11 within Note 9); Supplementary Note 1, Tables S6, S7 and S9 and Figures S1–S6 follow the main figures and tables, and Table S10 follows Table 3. Table S3 (strength ranges of the planned fixed-strength benchmark) was removed with that benchmark, and its number is not used.

Supplementary Notes

- Supplementary Note 1, model and loss equations: notation, with the reconstruction term written per gene g as $p_{\mathrm{ZINB}}(x_g \mid \mu_g, \theta_g, \pi_g)$ (mean, dispersion, dropout) and the discriminator loss as $\min_h \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \mathcal{L}_{\mathrm{CE}}(h(u_i), s_i)$ on the posterior mean $u_i = \mu_i$ of cell i in minibatch $\mathcal{B}$, for batch label s_i; scVI loss (zero-inflated negative binomial reconstruction, Kullback–Leibler (KL) term and KL weight schedule); discriminator fool loss as in scvi-tools (Jensen–Shannon type); the three Wasserstein-1 critics with gradient penalty weight 10: reference (each batch against the reference batch, chosen as the batch with the highest cell-type entropy, which uses labels), pooled (each batch against all others), barycenter (each batch against the minibatch Wasserstein barycenter, an average distribution); barycenter target construction and solver (Wasserstein-2, at most 10 iterations, threshold 10^{-7}); maximum mean discrepancy (MMD) with multi-scale Gaussian kernel (the squared MMD of each batch against the cells of all other batches in the minibatch, as a biased V-statistic, averaged over the batches present; the kernel is the mean of five Gaussian kernels $\exp(-\|z - z'\|^2/2\sigma^2)$ with $\sigma = \sqrt{d}\,m$ for $m \in \{0.25, 0.5, 1, 2, 4\}$, so for $d = 10$ latent dimensions the bandwidths are $\sigma = 0.79$, 1.58, 3.16, 6.32 and 12.65; they are fixed in advance and not tuned, and the MMD results hold for this kernel); debiased Sinkhorn divergence (entropy-regularised optimal transport; not W1 at the smoothing used; Euclidean cost, entropic smoothing $0.05\sqrt{2d}$ for latent dimension d, float64, marginal error 10^{-4}, cap of 500,000 iterations); scvi-tools adversarial classifier (conditioned decoder only; 32 hidden units, weight one minus the KL weight); plain scVI as strength zero; each critic head has two hidden layers of 128 units with one output per batch.

Supplementary Note 2. Training settings.

One model; differences by design. Plain scVI, the six penalties and the scvi-tools adversary are the same scVI model of scvi-tools 1.4.2 with the same settings within a task (Supplementary Table S2, Part A): 10 latent dimensions, one hidden layer of 128 units in the encoder and in the decoder, the zero-inflated negative binomial likelihood, 90% of the cells for training and the rest for validation, minibatches of 128 cells, and scvi-tools' own optimiser for the encoder and decoder (Adam, learning rate 10^{-3}, epsilon 0.01, weight decay 10^{-6}). Plain scVI is this model at strength $\lambda = 0$: the same training plan runs scvi-tools' own training step with no penalty. The values in Supplementary Table S2 are those recorded in the fits' own latent files, and every fit of a method has the same value. The settings that differ between methods by design are the adversary update steps and the adversary optimiser, and each is the method's standard setting (Supplementary Table S2, Part B). Adversary update steps per encoder step: the Jensen–Shannon discriminator takes one, as discriminators are trained, and the reference, pooled and barycenter critics take five, with a gradient penalty of weight 10, as Wasserstein critics are trained (Gulrajani et al.; main-text reference). Adversary optimiser: the discriminator uses the encoder's optimiser settings (Adam, learning rate 10^{-3}, epsilon 0.01, weight decay 10^{-6}, as scvi-tools trains its adversarial classifier), and the critics use Adam with learning rate 10^{-4} and betas 0 and 0.9, the settings of the same reference. MMD and the Sinkhorn divergence have no adversary network, so they take no adversary step. The adversary steps of an encoder step reuse that step's minibatch: its posterior means are computed once and detached, every adversary step is taken on them, and the encoder step that follows uses the same minibatch (Supplementary Note 1).

Adversary input and the two input pilots. Every penalty reads the encoder's posterior mean, without per-dimension standardisation (Supplementary Note 1). Two pilots tested the alternatives, with switching criteria fixed before they were run: a posterior sample instead of the mean, for the three adversarial methods (the discriminator and the reference and pooled critics), and standardising each latent dimension by the minibatch mean and standard deviation of the posterior means, for five methods (those three, MMD and the Sinkhorn divergence). Both ran on the immune and small chromatin-accessibility tasks with both decoders and seeds 100–102, at three strengths per task–decoder–method combination around its matched batch removal, against the sweep fits at the same settings as the default. A combination is evaluable when both its default and its alternative curve reach the matched batch removal b^*, and D is its biology at b^* with the alternative minus that with the default. The rule switched only if four criteria held together: at least two-thirds of the combinations evaluable, with an evaluable combination in every task–decoder pair; mean D above 0.01; mean D positive within every task–decoder pair; and no more failed fits with the alternative. The posterior sample lowered biology at b^* by mean -0.004 (9 of 12 combinations evaluable) and standardisation by -0.003 (14 of 20 evaluable), so neither met the criteria and both defaults were kept. These values are from the decision record recomputed after the Sinkhorn fits were redone (commit 2168d75); the first record (commit cea50ca, the version at commit 10a1ca5) gave -0.001 (12 of 12) and $+0.001$ (17 of 20) and the same decision.

Epochs, KL warm-up and failed fits. Every fit trains for the number of epochs given by scvi-tools' rule, $\min(400, \mathrm{round}(20000 \times 400/n))$ epochs for a task of n cells, with no early stopping, so epochs, minibatch and latent size are equal for every method within a task: 400 epochs on the four tasks with fewer than 20,000 cells and 82 to 246 on the other four (Supplementary Table S2, Part C). The weight of the latent KL term rises linearly over 400 epochs (scvi-tools' default warm-up), with epochs counted from zero, so in the last of E epochs it is $(E-1)/400$: 0.9975 on the four tasks with 400 epochs, and 0.2025 to 0.6125 on the four

shorter tasks, on which the warm-up is not complete. The penalty acts at its full strength from the first step and is not scaled by the warm-up. Every fit is scored on the posterior mean of every cell (Supplementary Note 3). A fit fails if a loss term that a training step computes or back-propagates becomes non-finite, or if its posterior-mean latent holds a non-finite value; failed fits are counted in Supplementary Table S5.

Supplementary Note 3. scIB metrics and scores.

Scoring. Every fit is scored once, on its posterior-mean embedding $e_i \in \mathbb{R}^d$ of every cell, training cells included, against the prepared task file, with the metrics function of scib 1.1.7 for embedding outputs (Luecken et al.; main-text reference). Scoring all cells is the scIB convention; Li et al. instead report held-out cells (main-text reference). Every metric is used as scib returns it, a raw value where higher is better; nothing is rescaled across fits. Every value lies between 0 and 1 except the adjusted Rand index, which is corrected for chance and can fall slightly below 0: its lowest value, -0.00003, is that of two fits on simulation 2. Below, s_i is the batch and c_i the cell type of cell i, K the number of batches and L the number of cell types of the task, and G_{15} the graph of the 15 nearest neighbours of every cell in the embedding (scanpy's default), built once per fit.

Batch metrics. (1) *Principal-component regression* (PCR). For a matrix with principal components j explaining variance V_j, $\mathrm{PCR} = \sum_j R_j^2 V_j / \sum_{j'} V_{j'}$, with R_j^2 the coefficient of determination of a linear regression of component j on the one-hot batch. It is computed for the unintegrated data (50 components of scIB's normalised values on all genes) and for the embedding (its d components), and the score is $\max\{0, (\mathrm{PCR}_{\mathrm{before}} - \mathrm{PCR}_{\mathrm{after}})/\mathrm{PCR}_{\mathrm{before}}\}$. (2) *Batch average silhouette width* (ASW). Within each cell type that holds more than one batch, a_i is the silhouette of cell i with the batches as clusters (Euclidean distance); the score is the mean over cell types of the mean of $1-|a_i|$ over their cells. (3) *Graph integration local inverse Simpson's index* (iLISI; Korsunsky et al., main-text reference). On a graph of the 15 nearest neighbours, each cell's neighbourhood is its 90 nearest cells by shortest path, weighted with perplexity 30; the inverse Simpson's index of the batch labels in that neighbourhood is computed for about half of the cells, drawn at random, and with its median m the score is $(m-1)/(K-1)$. (4) *Graph connectivity.* For each cell type c, the fraction of its cells in the largest connected component of G_{15} restricted to its cells; the score is the mean over cell types. (5) *k-nearest-neighbour batch-effect test* (kBET). On a graph of the 50 nearest neighbours, for each cell type with at least 10 cells and more than one batch, kBET's χ^2 test compares the batch composition of each cell's k_0 diffusion neighbours with that of the cell type, with $k_0 = \min\{70, \max(10, \lfloor \bar{n}_c/4 \rfloor)\}$ and $\bar{n}_c$ the mean number of cells per batch in the cell type; r_c is the rejection rate. If the cell type's graph is not connected, components smaller than $3k_0$ are dropped when at least 75% of its cells remain, and otherwise $r_c = 1$. The score is $1 - \frac{1}{L'}\sum_c r_c$ over the L' cell types tested. R's random seed is set to 0 before every kBET call, so the score is reproducible.

Biology metrics. (6) *Normalised mutual information* (NMI). G_{15} is clustered with Leiden at the ten resolutions $0.2, 0.4, \ldots, 2.0$, and the clustering κ with the highest NMI to the cell types is kept; $\mathrm{NMI} = 2I(\kappa; c)/(H(\kappa) + H(c))$, with I the mutual information and H the entropy. (7) *Adjusted Rand index* (ARI) of the same clustering, with n_{uv} the cells in cluster u and cell type v, a_u and b_v the margins, $\Sigma_{ab} = \sum_u \binom{a_u}{2} \sum_v \binom{b_v}{2} / \binom{n}{2}$: $\mathrm{ARI} = \left(\sum_{uv} \binom{n_{uv}}{2} - \Sigma_{ab}\right) / \left(\frac{1}{2}\left[\sum_u \binom{a_u}{2} + \sum_v \binom{b_v}{2}\right] - \Sigma_{ab}\right)$. (8) *Cell-type ASW*: $(\bar{a}+1)/2$, with $\bar{a}$ the mean silhouette of all cells with the cell types as clusters. (9) *Isolated-label F1.* The isolated labels are the cell types present in the fewest batches (none when every cell type is in every batch). For each, the Leiden resolution (of the same ten) is chosen whose best-matching cluster has the highest F_1 score for that label, and that F_1 is the label's value; the score is the mean over isolated labels. (10) *Isolated-label ASW*: for each isolated label, $(\bar{a}_\ell + 1)/2$ with $\bar{a}_\ell$ the mean cell-type silhouette of its cells; the score is the mean. (11) *Graph cLISI*: as iLISI with the cell types as labels, scored $(L-m)/(L-1)$. (12) *Cell-cycle conservation* (pancreas, lung and both immune tasks). In each batch, S and G2M scores are computed on the unintegrated data from human cell-cycle genes; their variance contribution v is the PCR of the two scores, on the unintegrated data of the batch (50 components) and on its embedding, and the batch scores $\max\{0, 1 - |v_{\mathrm{after}} - v_{\mathrm{before}}|/v_{\mathrm{before}}\}$; the score is the mean over batches. (13) *Trajectory conservation* (both immune tasks, the tasks whose file holds scIB's diffusion pseudotime). For the cells with a pseudotime, diffusion pseudotime is recomputed on the embedding, within the largest connected component of its neighbourhood graph, from a root cell that scib chooses from the file's pseudotime and cell types; values above 1, and the values of cells outside that component, are set to 0, and with ρ the Spearman correlation of the two pseudotimes the score is $(\rho+1)/2$ (0 if scib finds no root cell, which each score file records). Highly variable gene conservation compares gene selections and is not defined for an embedding, so it is not computed for any task.

Biology metrics per task and blank metrics. Each task's biology score uses every biology metric that applies to it: metrics (6)–(11) for all eight tasks; with (12) for pancreas and lung (seven metrics); with (12) and (13) for the immune and human–mouse immune tasks (eight metrics). For the two chromatin-accessibility tasks, cell-cycle conservation (scIB did not compute it for gene-activity data), trajectory conservation (their files hold no pseudotime) and highly variable gene conservation are blank in every score file, so their biology score is the mean of the six metrics (6)–(11); the two simulations have the same six, as simulated genes have no cell-cycle genes and the files no pseudotime. A blank metric is left out of the mean. Every batch metric is present for every task.

Scores. The batch score B is the mean of the five batch metrics, the biology score C the mean of the task's biology metrics, and the selection score $S = 0.4\,B + 0.6\,C$, the weights of the scIB overall score. scIB's own overall score first scales each metric from 0 to 1 across the methods compared on a task; here the raw values are used, so a fit's scores do not depend on which other fits are scored. For the local and global split, B_{local} is the mean of the three neighbourhood metrics (iLISI, kBET and graph connectivity) and B_{global} the mean of the two whole-embedding metrics (PCR and batch ASW), so that $B = (3B_{\mathrm{local}} + 2B_{\mathrm{global}})/5$.

Input preparation. Each task file of scIB (figshare article 12420968, version 8) is prepared once (`scripts/prep_scib_task.py`), keeping the cells as distributed with no further filtering, with one batch variable and one cell-type variable per task: those of scIB's pipeline configuration for the RNA tasks and the files' own batch and cell-label variables for the chromatin-accessibility

tasks (Supplementary Table S1). The training features are 2,000 highly variable genes chosen by scIB's batch-aware selection (cell-ranger flavour, 20 bins, on scIB's normalised values) for the RNA tasks, and all 3,429 and 3,580 gene-activity features of the small and large chromatin-accessibility tasks, the features on which scIB scored them; the chromatin-accessibility tasks enter as these gene-activity matrices. The models read the counts layer as scIB distributes it, as scIB itself fed it to scVI. In 18 batches of four tasks these values are not integers (Supplementary Table S1): per the preparation script's record, transcripts per million (the Villani batch of both immune tasks), reads per kilobase per million (pancreas, smarter), expected counts of a transcript quantifier (pancreas, fluidigmc1), counts corrected for unique-molecular-identifier collisions (pancreas, celseq and celseq2) and normalised values (the twelve 10x batches of the lung task). The scorer's unintegrated reference is scIB's normalised data on all genes.

Supplementary Note 4. Penalty-strength sweep: design and coverage.

Design. Plain scVI and the six penalties (Supplementary Note 1) were fitted on all eight tasks at the ten half-decade strengths 0.1, 0.3, 1, 3, 10, 30, 100, 300, 1,000 and 3,000, with the batch-conditioned and the unconditioned decoder and with seeds 100, 101 and 102. Edge strengths were added in two places: the discriminator with the unconditioned decoder was also fitted at 0.01 and 0.03, and the Sinkhorn divergence with the batch-conditioned decoder at 10,000 and 30,000. Each task therefore has 378 fits: 6 of plain scVI (three seeds, two decoders), 360 of the penalties (six penalties, ten strengths, three seeds, two decoders) and 12 at edge strengths (two places, two strengths, three seeds), so the sweep has 3,024 fits on the eight tasks, 96 of them at edge strengths. Nine further fits, at strength 0.03 for the discriminator, the reference critic and the pooled critic on the immune task with the batch-conditioned decoder (seeds 100–102), come from an earlier strength-extension round (below). The analysis reads every fitted strength, so it reads 3,033 fits, 387 of them on the immune task. A penalty's curve therefore has ten strengths, twelve for the discriminator with the unconditioned decoder and for the Sinkhorn divergence with the batch-conditioned decoder on every task, and 11 for the discriminator, the reference critic and the pooled critic on the immune task with the batch-conditioned decoder. None of the 3,033 fits failed (Supplementary Table S5).

Why the edge strengths exist. The sweep was first run as a pilot on the three tasks fitted first (the small chromatin-accessibility, immune and simulation 1 tasks). A strength-choice rule, written before the pilot ran, was to read the pilot's seed means and set, for each penalty family and decoder, a range of six strengths for a planned fixed-strength benchmark; where a range reached the end of the grid, the rule asked for two more half-decade strengths beyond that end. On the three tasks this happened in two places, the discriminator with the unconditioned decoder at the low end and the Sinkhorn divergence with the batch-conditioned decoder at the high end, and the two extra strengths were fitted there (36 fits on the three tasks). When the sweep was extended to the other five tasks, the same 378 fits per task were made, the edge strengths included, so every task has the same strengths. The nine further fits come from a second rule, which matched batch removal at a strength nearest a target batch score and, where that strength was the lowest of its grid, fitted the next lower half-decade value; this happened for three penalties on the immune task with the batch-conditioned decoder. They are not strengths of the strength-choice rule, but they are fitted points of these penalties' curves and are read like every other strength. That pair is low-signal and is not ranked (Supplementary Note 5).

The planned fixed-strength benchmark was not run. The strength-choice rules were written to set up a benchmark at six strengths per penalty family and decoder with five new seeds. That benchmark was not run, so the strength-choice rules choose no strength: every analysis reads every sweep strength, and the comparison is exploratory. The replication across training seeds (below) refits strengths chosen from the sweep's batch scores, not from its biology scores.

Machines. All fits of a task were made on one kind of machine, so that the hardware never differs between penalties, strengths or decoders within a task: the small chromatin-accessibility, immune and simulation 1 tasks on the workstation's NVIDIA RTX 3080 graphics card, and each of the other five tasks in one cluster job on NVIDIA L40S graphics cards (Supplementary Note 11).

Sinkhorn solver. All 528 Sinkhorn fits of the sweep (66 per task) were made with the corrected Sinkhorn solver, at commit 10a1ca5 (Supplementary Note 1 gives the solver). On the three workstation tasks, the fits made with the earlier solver were replaced: the 180 fits at the ten grid strengths were refitted on 17–18 October 2026 and the 18 fits at the edge strengths 10,000 and 30,000 on 19 October 2026, 198 in all. The five cluster tasks were fitted from 22 October 2026 at the same commit.

Replication across training seeds: design. For each of the 13 ranked task and decoder pairs and each of the six penalties, we took the two sweep strengths that bracket the first crossing of b^* on the sweep's seed-mean curve, with b^* fixed at its sweep value (Supplementary Note 5), and refitted them with five new seeds (0–4). 46 of the 78 curves cross b^* more than once; the first crossing sets the bracket, as in the sweep reading. In one cell, the discriminator with the batch-conditioned decoder on the human–mouse immune task, the curve already lies above b^* at the lowest sweep strength, 0.1, so the bracket runs from plain scVI to that strength and plain scVI at the new seeds is its lower point. Plain scVI was refitted at seeds 0–4 on all eight tasks with both decoders. This gives 855 fits, 775 with a penalty and 80 of plain scVI. The three low-signal pairs (the immune task and both simulations with the batch-conditioned decoder) were skipped. If the new seed means at the two strengths do not bracket b^*, the next sweep strength outward is fitted once with seeds 0–4; if b^* is still not bracketed, the penalty takes the bottom rank in that pair and is marked as not replicating the crossing. The fits of this extension step are counted in Supplementary Note 11, and the readings that needed it are marked in Supplementary Table S9.

Replication: provenance. The design script `experiments/sweep_replication/replication_design.py` wrote the run list (`manifests/sweep_replication_x1.tsv`, 855 rows) and its summary in `experiments/sweep_replication/design_20261102/` (commit 2113ec7). The prediction file `PREDICTION.json` (the sweep readings and ranks at b^* of the 13 ranked pairs and the 4 within-pair differences of 0.04 or more) and the agreement rule `AGREEMENT.md` were committed under the tag

`replication-prediction-v1` (commit 3a90d0f) before any replication fit was scored. The prediction file records the table it was read from, the ranking table of the sweep analysis at commit a361ba2, with its SHA-256 (786cb8c8…); that table is byte-identical to the ranking table of the final sweep analysis used in this paper (`paper/figure_data/ranking_pairs.csv`), so the prediction holds the readings and ranks of Supplementary Table S6.

Supplementary Note 5. Matched batch removal b^* and the low-signal rule.

Why equal strength is not equal correction. The strength λ multiplies a different quantity for each penalty (a cross-entropy for the discriminator, a Wasserstein-1 estimate for the critics, a squared MMD and a Sinkhorn divergence; Supplementary Note 1), and these quantities have different scales, so one strength removes different amounts of batch effect. For example, on the small chromatin-accessibility task with the unconditioned decoder, the seed-mean batch scores at strength 1 range from 0.401 (MMD) to 0.714 (the discriminator), against 0.402 for plain scVI. A stronger penalty also trades biology for batch removal along a curve. Penalties are therefore compared at one batch score, b^*, and their biology is read there.

Definition of b^*. For each task and decoder, b_0 is plain scVI's batch score averaged over its three fits (seeds 100–102). For each penalty, M is the highest seed-mean batch score over its strengths with no failed fit (edge and extension strengths included), and b_{common} is the lowest M of the six penalties, the highest batch score that every penalty reaches. Then

$$b^* = b_0 + \tfrac{1}{2}\,(b_{\text{common}} - b_0). \tag{1}$$

Batch and biology scores are the unscaled scIB means of Supplementary Note 3.

Curves and the reading. A penalty's curve starts at plain scVI ($\lambda = 0$; seed means of batch and biology score) and continues through its fitted strengths in increasing order, each point the mean batch and biology score over that strength's fits that did not fail; a strength whose fits all failed is not a point. The reading is the biology score where the curve first reaches b^*: walking up from plain scVI, the first point whose batch score equals b^* (within 10^{-9}), or else the first pair of neighbouring points whose batch scores lie on either side of b^*, with the biology score interpolated linearly in batch score between them. A curve can fall back below b^* and cross it again at higher strengths; later crossings are counted (Supplementary Table S6) and not used. If plain scVI's batch score is at or above b^*, the curve gives no reading. Supplementary Note 6 gives the leave-one-seed-out uncertainty of the reading and the ranking.

Every full-seed curve reaches b^*. Every curve starts at b_0 and contains a point whose batch score is at least b_{common}; when $b_{\text{common}} > b_0$, b^* lies strictly between them, so the curve crosses b^*. Because M is taken over strengths with no failed fit, it is a point of the curve, so a full-seed curve can fail to reach b^* only when $b_{\text{common}} \le b_0$, which makes the pair low-signal (below); a curve read with one seed left out, or on new seeds, is read against the fixed b^* and need not reach it. In the sweep 0 of the 3,033 fits failed and every one of the 96 task, decoder and penalty curves reached b^*, so the bottom-rank rule for a penalty that does not reach b^* (Supplementary Note 6) was never used.

Low-signal rule. A task and decoder pair is low-signal when $b_{\text{common}} - b_0 < 0.02$, twice the smallest effect (0.01) of the strength-choice rules; b^* then lies less than 0.01 above plain scVI and cannot be placed reliably. Three pairs are low-signal, all with the batch-conditioned decoder: the immune task (plain scVI 0.638, highest common batch score 0.657, gap 0.019), simulation 1 (0.812 and 0.819, gap 0.007) and simulation 2 (0.656 and 0.665, gap 0.009). No pair is low-signal with the unconditioned decoder. Low-signal pairs are left out of every result that uses b^*: the ranking (ranks, standard errors, rank ranges and mean ranks; Supplementary Note 6), the hypervolume, whose divisor is the gap (Supplementary Note 7), the summaries of the local versus global reading and the cross-task label-transfer means (Supplementary Note 9). Their readings are reported in separate tables: Part B of Supplementary Tables S6 and S8 and the low-signal blocks of the label-transfer table. Analyses that do not use b^* keep them: the share of non-dominated strengths, the collapse rate, the good-strength width, the single-strength placement and the best of k settings (Supplementary Note 7).

Matched strength. Label transfer (Supplementary Note 9) and the direct biology check (Supplementary Note 12) read each penalty at one fitted strength, the matched strength: among the strengths with no failed fit, the one whose seed-mean batch score is nearest b^*, ties to the smaller strength, edge and extension strengths included. It is never chosen by label-transfer accuracy or by any biology score, and the distance of its batch score from b^* is reported beside each value.

Failed fits. A fit fails when its training loss or its latent becomes non-finite (recorded as diverged or non-finite latent). Failed fits are left out of the seed means of their strength, and a strength with a failed fit cannot set M or be the matched strength; every failed fit counts as a collapse (Supplementary Note 7). No sweep fit failed.

Supplementary Note 6. Benchmark ranking at matched batch removal.

Reading. For each penalty, task and decoder, biology at b^* is read as in Supplementary Note 5: on the curve of seed means over the three seeds (100–102) through plain scVI and the fitted sweep strengths, at the first crossing of b^*, by linear interpolation in batch score. 46 of the 78 ranked curves cross b^* more than once (32 once, 35 twice, 6 three times and 5 four times); only the first crossing is read, and Supplementary Table S6 gives the number of crossings of each curve.

Leave-one-seed-out standard error. The reading is repeated three times, each time without one seed: the same seed is left out of the penalty's fits and of plain scVI's fits, so the anchor is the mean of the two remaining plain scVI fits, and b^* stays fixed at its full value. A repeat gives a reading when its curve reaches b^*. From the m readings $x_{(i)}$ that exist, with mean $\bar{x}$,

$$\text{SE} = \Big(\frac{m-1}{m}\sum_{i=1}^{m}\big(x_{(i)} - \bar{x}\big)^2\Big)^{1/2}, \tag{2}$$

the jackknife standard error. It is reported when $m \geq 2$ of the three repeats give a reading and otherwise as not available, and m is reported beside it (column m of Supplementary Table S6). The rule of the planned five-seed benchmark, at least four of five readings, also allowed at most one repeat without a reading; with three seeds it becomes at least two of three. In the sweep $m = 3$ for all 78 ranked readings, so every reading has a standard error and a rank range.

Rank range. In each of the three repeats all six penalties of the pair are read again and ranked with the rules below, a penalty that does not reach b^* in a repeat taking the bottom places of that repeat. The rank range is the lowest to the highest rank a penalty gets over the three repeats; the rank of the full reading is not part of it. It is reported under the same condition as the standard error ($m \geq 2$), and the penalty is ranked either way. The standard error and the rank range describe the spread of the readings over three repeats of the same data; no interval, test or probability is built from them.

Which pairs are ranked. Every task and decoder pair that is not low-signal is ranked: 13 of the 16 pairs, the six real tasks and both simulations with the unconditioned decoder and five real tasks with the batch-conditioned decoder (the immune task and both simulations are low-signal with that decoder; Supplementary Note 5). A pair stays ranked whether or not every penalty reaches b^* in it.

Ranking rules. Within each ranked pair the six penalties are ranked by biology at b^*, rank 1 being the highest. Readings within 10^{-9} are tied, and tied penalties get the average of the places they occupy, so two penalties tied for second both get 2.5. A penalty whose curve does not reach b^* takes the bottom rank of its pair; with f such penalties each gets the mean of places $7 - f$ to 6, so they share the last places, and the six ranks of a pair always sum to 21. This follows scIB (main-text reference), where methods that could not be run on a task were given the same rank as unintegrated data. In the sweep no two readings of a pair were tied and every curve reached b^*, so every rank is a whole number and no penalty took a bottom rank by this rule (Supplementary Note 5).

Mean rank. As in scIB, each penalty's ranks are averaged over the real tasks that are ranked, separately for each decoder (five tasks with the batch-conditioned decoder and six with the unconditioned decoder), with the number of tasks shown beside the mean and the range over tasks (Table 2). Decoders are never combined. The simulations are ranked with the same rules and reported per task, never averaged with each other or with the real tasks (Supplementary Figure S1). Low-signal pairs are not ranked; their readings are in Part B of Supplementary Table S6.

Related tasks. Related tasks are counted as separate tasks in the mean rank: the human–mouse immune task contains all 33,506 cells of the immune task, and the large chromatin-accessibility task contains all 11,270 cells of the small one (matched by batch and barcode, with the same cell-type labels; 3,618 of them, the 10x Genomics batch, under identical cell names; Supplementary Table S1). The six real tasks are therefore four task families, and two families weigh twice in the mean rank.

Sensitivity of the ranking. Three variants re-read biology at b^* from the same fits and re-rank (Supplementary Figure S2): the per-run reading, in which each seed is read on its own curve with its own plain scVI fit as the anchor, the six penalties are ranked within each seed and the three per-seed ranks are averaged; biology without trajectory conservation on the two immune tasks (b^* is unchanged, because it depends on batch scores only); and the reading with every point that contains a failed fit removed, which equals the primary reading here because no fit failed. The scaling checks (scIB min-max scaling, with each method family left out in turn, z-scores and a composite without the silhouette metrics) recompute the scores, b^*, the readings and the ranks (Supplementary Figure S5); they are exploratory and used for no conclusion.

No tests. The ranking is descriptive. No significance test, probability, margin or multiplicity correction is computed. Neighbouring readings often differ by less than their standard errors, and a reading is a linear interpolation between half-decade strengths, so it depends on the spacing of the strengths; penalties whose readings differ by little can change places on new fits or another spacing (Supplementary Table S9 reports how the ranks moved on new seeds).

Supplementary Note 7. Rules for the secondary analyses.

All analyses in this note are descriptive: no difference between penalties is tested, and no value is pooled over tasks except where a mean over tasks is stated. Results per penalty, task and decoder are in Supplementary Table S8 and Figures 4–6. A point is a strength's seed means of batch score B and biology score C over its fits that did not fail, as in Supplementary Note 5; b_0 and c_0 are plain scVI's means, and $S = 0.4B + 0.6C$ is the selection score (Supplementary Note 3).

Hypervolume. The hypervolume of a penalty in a task and decoder pair is the area of the batch-score by biology-score plane reached by its points beyond a reference point $(b_{\mathrm{ref}}, c_{\mathrm{ref}}) = (b_0,\, c_0 - 0.10)$, plain scVI's batch score and 0.10 below its biology score (the collapse bound below): the union of the rectangles $[b_{\mathrm{ref}}, B_i] \times [c_{\mathrm{ref}}, C_i]$ over the points with $B_i > b_{\mathrm{ref}}$ and $C_i > c_{\mathrm{ref}}$, divided by $b_{\mathrm{common}} - b_0$, the batch gap that sets b^* (Supplementary Note 5), so that it is in biology-score units and on one scale across tasks. Plain scVI's own point is the reference, not a point of the penalty. A pair with no point beyond the reference point gets 0. Low-signal pairs get no hypervolume, because their gap is under 0.02. The hypervolume is reported per penalty, task and decoder; the analysis record also holds it with the reference point at biology 0, a sensitivity analysis not used in the paper.

Share of non-dominated strengths. A point is dominated when another point of the same penalty is at least as good on both the batch and the biology score (within 10^{-9}) and better on one of them by more than 10^{-9}; identical points do not dominate each other. The share is the number of points that no other point dominates divided by the number of points, reported with both counts. A strength whose fits all failed is not a point. The share needs neither b^* nor the reference point, so it is computed in every task and decoder pair, the low-signal pairs included. Li et al.'s number of distinct choices (main-text reference) is not used, because it needs a grid width that would have to be chosen arbitrarily.

Collapse rate. A fit collapses when it failed or when its biology score is at least 0.10 below that of plain scVI with the same

decoder and the same seed (within 10^{-9}). The collapse rate is the collapsed share of a penalty's fits in a task and decoder pair, over all its fitted strengths and the three seeds: 30 fits (ten strengths), 33 for the discriminator, the reference critic and the pooled critic on the immune task with the batch-conditioned decoder (eleven strengths, the additional strength of Supplementary Note 4), and 36 where edge strengths were fitted (twelve strengths). Each rate has an exact binomial (Clopper–Pearson) 95% interval. The interval is nominal, because the fits of one seed are judged against one plain scVI threshold and are not independent. Failure types (diverged, non-finite latent) are reported separately; no sweep fit failed, so every collapse is a loss of biology. All task and decoder pairs are used, the low-signal pairs included, because the collapse rate does not use b^*.

Coverage of the collapse-rate interval: an illustrative simulation. The script `paper/analysis_pass10/coverage_sim_sweep.py` (random seed 7, 20,000 replicates per scenario) estimates the probability that the exact 95% interval computed from 30 fits (ten strengths, three seeds) or 36 fits (twelve strengths) contains the mean collapse probability over the strength grid. In scenario A the fits are independent and the collapse probability rises with strength along a profile set by hand (0.0, 0.0, 0.05, 0.2, 0.6, 0.95 at six points, interpolated linearly to ten or twelve strengths). In scenario B each seed has one collapse threshold shared by its strengths, so a seed that collapses at one strength collapses at every higher one; the threshold position is drawn from a distribution set by hand (0.0, 0.0, 0.0, 0.1, 0.15, 0.2, 0.55 over seven positions, the last meaning no collapse, interpolated and renormalised). A control with independent fits at a constant probability of 0.3 must cover at least 0.95. Coverage was at least 0.999 in scenario A, 0.88 (30 fits) and 0.83 (36 fits) in scenario B, and 0.97 and 0.95 in the control, with Monte Carlo standard errors of at most 0.003. Both profiles are illustrative choices, not estimated from the sweep; other dependence structures give other coverage, and the 33-fit design is not simulated, so the simulation shows that dependence can lower coverage below the nominal level but does not bound the coverage of the reported intervals.

Good-strength width. For each penalty, task and decoder, the best strength is the one with the highest mean selection score over its fits (ties to the smaller strength). The window holds every strength whose mean selection score is at least the lower end of the two-sided 95% t interval of the best strength's mean (t quantile 0.975 with $n-1$ degrees of freedom, $n=3$ fits). The width is $\log_{10}$ of the largest strength in the window minus $\log_{10}$ of the smallest, in decades (0 for a single strength), reported with the number of strengths in the window. The strengths of a window need not be neighbours: a window is flagged when a strength between its smallest and largest falls below the bound (19 of the 96 windows), and its width then spans that strength. No width is computed when the best strength has fewer than two fits that did not fail. All pairs are used, the low-signal pairs included, because the selection score does not use b^*. Every fitted strength counts, so a window can span more of the strength range than a six-strength grid would allow; the widest window spans 5.5 decades.

Local versus global batch removal. Local removal L is the mean of the three neighbourhood batch metrics (graph iLISI, kBET and graph connectivity) and global removal G the mean of the two whole-embedding metrics (principal-component regression on batch and batch ASW), so that the batch score is exactly $B = (3L + 2G)/5$. For each penalty, task and decoder, L and G are read on their curves of seed means at the same crossing as the biology reading, with the same interpolation weights, and the value reported is $L - G$. At the crossing $B = b^*$, so $L - G = 2.5\,(L - b^*)$: at matched batch removal, more local removal means less global removal. A penalty that does not reach b^* has no value. Low-signal pairs are shown separately (Part B of Supplementary Table S8) and left out of the means of Figure 4c. Supplementary Table S8 also gives L and G, recomputed from the per-fit metrics on the same curves and equal to the analysis record's $L - G$ within 10^{-9}.

Single-strength placement. For each real task, penalty and decoder, the transferred strength is the fitted strength with the best mean selection score over the training tasks, the real tasks of the other task families (pancreas; lung; the immune and human–mouse immune tasks; the small and large chromatin-accessibility tasks), so that related tasks never choose for each other. A strength is eligible when it was fitted and had no failed fit in every training task and was fitted in the task it is applied to; ties go to the smaller strength. The simulations take the strength chosen on all six real tasks. The task's own best strength, chosen by the same failure-free rule within the task, is shown in parentheses as an optimistic bound, and each score is compared with plain scVI's mean selection score in the same task and decoder. Harmony, Scanorama and sysVI get their setting by the same rule (Supplementary Note 8). Results are in Table 3 and Supplementary Table S7.

Best of k settings. A setting's score is the mean selection score over its fits (one fit for Harmony, Scanorama and PCA). For a method with N settings whose scores sorted in increasing order are $S_{(1)} \le \cdots \le S_{(N)}$, the expected best score among k settings drawn at random without replacement is

$$E_k = \sum_{i=k}^{N} S_{(i)} \binom{i-1}{k-1} \Big/ \binom{N}{k}, \qquad k = 1, \ldots, N. \tag{3}$$

Each penalty has its 10 to 12 fitted strengths, Harmony, Scanorama and sysVI six settings each (Scanorama five on the large chromatin-accessibility task, where its 80-neighbour setting could not be fitted), and plain scVI, the scvi-tools adversary and scANVI one setting each; scANVI, which uses the cell-type labels, is shown in its own stratum, and PCA is the uncorrected reference, not a method family. Methods are compared at equal numbers of tries: the per-task tables continue each penalty's curve to its 10 to 12 strengths, but comparisons stop at $k = 6$, the number of settings of Harmony, Scanorama and sysVI. Plain scVI has one setting, so its selection score is the reference at every k and E_k minus plain scVI's score is reported. The mean over the six real tasks at each k is taken over the tasks where the method has at least k settings, with the number of tasks shown (Figure 6a; per-task curves in Supplementary Figure S6). Every value is optimistic, because the best setting is chosen and scored on the same data. All six real tasks enter, the low-signal pair included, because the selection score does not use b^*; the simulations are reported per task.

Supplementary Note 8. Baselines.

Methods and settings. Six baselines were fitted on each of the eight tasks, from the same prepared files and training features as the penalties (Supplementary Table S4). The *scvi-tools adversary* is scvi-tools' own adversarial batch classifier on scVI (Supplementary Note 1): it runs with the batch-conditioned decoder only, and its weight is fixed at one minus the KL weight, so it has no strength to vary. *scANVI* (Xu et al.; main-text reference) follows scIB's protocol: scVI is trained with the task's settings, and scANVI, initialised from it, is trained for $\min\{10, \max(2, \text{round}(E/3))\}$ further epochs, 10 on every task, with the cell-type labels of all cells; it is the only method here that uses the labels. *sysVI* (Hrovatin et al.; main-text reference) is the scvi-tools 1.4.2 implementation with its default prior and a Gaussian likelihood on scIB's normalised values, with no KL warm-up; its strength is the weight of the cycle-consistency loss, at 1, 2, 5, 10, 20 and 50 (scvi-tools' default is 2). The three neural baselines use the scVI-backbone settings of each task (latent size, hidden units, epochs, minibatch and training split; Supplementary Table S2) and five seeds, 0–4. *Harmony* (Korsunsky et al.; main-text reference) runs as in scIB: principal-component analysis of scIB's normalised values on the training features, then harmony-pytorch 0.1.7 on 10 principal components, with the diversity penalty θ (the weight of the penalty on batch–cluster dependence; 0 is no penalty, 2 the tool's default) at 0, 0.5, 1, 2, 4 and 8. *Scanorama* (Hie et al.; main-text reference) runs as in scIB, with Scanorama 1.7.4 at 10 output dimensions and the number of neighbours used for mutual matching at 2, 5, 10, 20, 40 and 80 (default 20). *Principal-component analysis* at 10 components is the uncorrected reference. Harmony, Scanorama and principal-component analysis run once per setting (seed 0); every Harmony fit runs in a fresh process with one thread, because Harmony was not reproducible from run to run within a shared process. Three further settings, at each tool's default size, are sensitivity checks and choose nothing: Harmony at $\theta = 2$ on 50 principal components, Scanorama with 20 neighbours at 100 dimensions and principal-component analysis with 50 components. scANVI and principal-component analysis have one setting each, so neither involves a choice.

Fits. The plan has 448 baseline fits on the eight tasks: 40 of the scvi-tools adversary, 40 of scANVI, 240 of sysVI, 56 of Harmony, 56 of Scanorama and 16 of principal-component analysis, the three sensitivity settings included. 447 were fitted and scored and none failed; Scanorama with 80 neighbours on the large chromatin-accessibility task stopped with an infrastructure error (memory or a 32-bit index limit, per the run record of the sweep analysis), which is not a fit outcome, and is reported as missing (Supplementary Table S4).

Grid amendment. The Scanorama grid was first set at 5 to 160 neighbours. A timing run on the human–mouse immune task then found that 160 neighbours did not finish within 8 hours and reached 43.9 GB of memory, while 80 neighbours took 2.1 hours and 23.2 GB. On 4 October 2026 the grid was changed to 2 to 80 neighbours, as the rule for this check, written the day before, required: an infeasible setting is replaced before the baseline grid is run, and the grid is not chosen from results.

Settings in the single-strength placement. When each method is placed at a single strength (Table 3 and Supplementary Table S7), Harmony, Scanorama and sysVI get their setting by the same rule as the penalties: on each real task, the setting with the best mean selection score on the real tasks of the other task families, so that related tasks never choose for each other (Supplementary Table S1). A setting is a candidate only if it was fitted on each of those tasks with no failed fit, and the sensitivity settings are not candidates. The task's own best setting, given in parentheses, is the optimistic bound.

Supplementary Note 9. Label transfer.

Embeddings. Label transfer asks whether a batch's cell types can be predicted from the other batches in the latent space. For each penalty, task and decoder it is read on the embeddings (posterior means) of the matched strength, the strength with no failed fit whose seed-mean batch score is nearest b^*, ties to the smaller strength, edge and extension strengths included (Supplementary Note 5), with its three sweep seeds (100–102). The strength is chosen from batch scores only, never by label-transfer accuracy. Plain scVI (three seeds, each decoder) and the scvi-tools adversarial classifier (five seeds, 0–4, batch-conditioned decoder only) have no strength to choose and are shown unmatched. Harmony, Scanorama, sysVI, scANVI and PCA are not label-transferred.

Classifier and score. Cell types present in only one batch of a task are dropped, a choice made from the labels alone. Each remaining batch is held out in turn, and its cells' types are predicted from the cells of the other batches by a 15-nearest-neighbour classifier (scikit-learn's k-nearest-neighbour classifier with Euclidean distance and equal vote weights). The score of a held-out batch is the macro F1, the F1 per cell type averaged over the cell types present in that batch; the score of an embedding is the mean over its held-out batches, and Supplementary Table S11 gives the mean and standard deviation over seeds. Every one of the 120 method, task and decoder entries could be scored (no task lost all its cell types).

Distance to b^*. Matched strengths are fitted strengths, so their batch scores are near b^* but not on it. Beside each value the table gives the seed-mean batch score of the embeddings and its distance from b^*; for plain scVI and the scvi-tools adversary, the distance of their own batch score. Distances differ between penalties, and a penalty read further above b^* has removed more batch effect, which can lower its label-transfer score; values should be compared with the distances in view.

Summary and scope. For each method and decoder, the cross-task summary is the mean macro F1 over the real tasks that are not low-signal (five with the batch-conditioned decoder, six with the unconditioned decoder). Low-signal pairs are tabulated and marked but not summarised; simulations are tabulated and not averaged with the real tasks. No test or contrast between methods is computed. The analysis also scored label transfer on every sweep embedding (3,073 embeddings, the penalties at every fitted strength, plain scVI and the scvi-tools adversary), giving each penalty a label-transfer curve over strength (`label_transfer_curve.csv` of the sweep analysis); the matched-strength values below are read from those results and are the ones reported. F1 per cell type is in Supplementary Figure S3, and the per-type recall and confusion of the direct biology check in Supplementary Note 12.

Supplementary Table S11: Label transfer at the matched strength, per method, task and decoder (definitions in Supplementary Note 9; descriptive). Strength: the matched strength (– for plain scVI and the scvi-tools adversary, which have none). Macro F1 and SD: mean and standard deviation over the seeds of the held-out-batch macro F1. Batch score B: seed-mean batch score of the embeddings; $|B - b^*|$: its distance from b^*. Last block: mean macro F1 over the real tasks that are not low-signal, per decoder.

Method	Strength	Seeds	Macro F1	SD	Batch score	$\lvert B - b^* \rvert$
Pancreas, batch-conditioned decoder; $b^* = 0.684$						
Plain scVI	–	3	0.874	0.010	0.672	0.012
scvi-tools adversary	–	5	0.890	0.013	0.679	0.005
JS discriminator	0.3	3	0.880	0.016	0.681	0.003
Reference critic	0.3	3	0.876	0.016	0.684	0.000
Pooled critic	0.3	3	0.880	0.012	0.679	0.005
Barycenter critic	1	3	0.875	0.017	0.683	0.001
MMD	10	3	0.878	0.018	0.685	0.001
Sinkhorn	10	3	0.872	0.007	0.684	0.000
Lung, batch-conditioned decoder; $b^* = 0.648$						
Plain scVI	–	3	0.742	0.006	0.633	0.015
scvi-tools adversary	–	5	0.719	0.006	0.645	0.003
JS discriminator	3,000	3	0.588	0.072	0.648	0.000
Reference critic	0.3	3	0.726	0.013	0.647	0.001
Pooled critic	0.3	3	0.725	0.007	0.647	0.001
Barycenter critic	1	3	0.722	0.009	0.648	0.000
MMD	10	3	0.725	0.010	0.650	0.002
Sinkhorn	3	3	0.727	0.008	0.646	0.002
Immune, batch-conditioned decoder (low-signal); $b^* = 0.648$						
Plain scVI	–	3	0.847	0.002	0.638	0.009
scvi-tools adversary	–	5	0.847	0.002	0.647	0.001
JS discriminator	0.1	3	0.848	0.002	0.648	0.000
Reference critic	0.1	3	0.845	0.004	0.647	0.000
Pooled critic	0.1	3	0.843	0.004	0.644	0.003
Barycenter critic	30	3	0.800	0.004	0.647	0.001
MMD	1,000	3	0.788	0.002	0.648	0.001
Sinkhorn	3	3	0.838	0.004	0.646	0.001
ATAC small, batch-conditioned decoder; $b^* = 0.746$						
Plain scVI	–	3	0.843	0.012	0.694	0.052
scvi-tools adversary	–	5	0.848	0.006	0.708	0.038
JS discriminator	1	3	0.847	0.003	0.751	0.005
Reference critic	100	3	0.813	0.012	0.738	0.008
Pooled critic	0.3	3	0.840	0.007	0.722	0.024
Barycenter critic	1	3	0.849	0.008	0.741	0.005
MMD	30	3	0.841	0.020	0.756	0.010
Sinkhorn	3	3	0.851	0.007	0.736	0.010
ATAC large, batch-conditioned decoder; $b^* = 0.632$						
Plain scVI	–	3	0.798	0.006	0.589	0.042
scvi-tools adversary	–	5	0.802	0.016	0.623	0.009
JS discriminator	30	3	0.700	0.023	0.636	0.004
Reference critic	300	3	0.787	0.004	0.647	0.016
Pooled critic	300	3	0.762	0.012	0.627	0.004
Barycenter critic	10	3	0.789	0.005	0.633	0.002
MMD	3	3	0.803	0.012	0.612	0.020
Sinkhorn	30,000	3	0.650	0.037	0.630	0.002
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$						
Plain scVI	–	3	0.750	0.002	0.540	0.014
scvi-tools adversary	–	5	0.730	0.004	0.560	0.006
JS discriminator	0.1	3	0.744	0.004	0.555	0.001
Reference critic	0.3	3	0.747	0.002	0.552	0.002
Pooled critic	300	3	0.650	0.027	0.553	0.001
Barycenter critic	1	3	0.746	0.003	0.554	0.000
MMD	3	3	0.751	0.005	0.548	0.005
Sinkhorn	3	3	0.743	0.001	0.553	0.001
Simulation 1, batch-conditioned decoder (low-signal); $b^* = 0.815$						
Plain scVI	–	3	1.000	0.000	0.812	0.003
scvi-tools adversary	–	5	1.000	0.000	0.836	0.021
JS discriminator	300	3	0.812	0.021	0.837	0.022
Reference critic	0.1	3	1.000	0.000	0.824	0.009
Pooled critic	0.1	3	1.000	0.000	0.826	0.011
Barycenter critic	0.1	3	0.999	0.001	0.817	0.002
MMD	1	3	1.000	0.000	0.814	0.001
Sinkhorn	0.3	3	1.000	0.000	0.818	0.003
Simulation 2, batch-conditioned decoder (low-signal); $b^* = 0.660$						
Plain scVI	–	3	0.752	0.020	0.656	0.004
scvi-tools adversary	–	5	0.737	0.019	0.659	0.001
JS discriminator	0.1	3	0.762	0.048	0.672	0.012
Reference critic	0.1	3	0.788	0.032	0.655	0.005
Pooled critic	0.3	3	0.731	0.037	0.659	0.001
Barycenter critic	0.1	3	0.756	0.023	0.657	0.003
MMD	1	3	0.765	0.035	0.657	0.004
Sinkhorn	0.3	3	0.766	0.032	0.658	0.002
Pancreas, unconditioned decoder; $b^* = 0.492$						
Plain scVI	–	3	0.838	0.018	0.446	0.046
JS discriminator	0.1	3	0.838	0.005	0.488	0.005

Supplementary Table S11, continued

Method	Strength	Seeds	Macro F1	SD	Batch score	$\lvert B - b^* \rvert$
Reference critic	1	3	0.845	0.008	0.490	0.002
Pooled critic	1	3	0.849	0.005	0.496	0.004
Barycenter critic	3	3	0.847	0.007	0.498	0.006
MMD	10	3	0.850	0.016	0.477	0.016
Sinkhorn	3	3	0.851	0.014	0.489	0.004
Lung, unconditioned decoder; $b^* = 0.535$						
Plain scVI	–	3	0.792	0.007	0.481	0.054
JS discriminator	0.1	3	0.753	0.006	0.529	0.006
Reference critic	3	3	0.764	0.008	0.535	0.000
Pooled critic	1	3	0.787	0.007	0.515	0.020
Barycenter critic	10	3	0.772	0.010	0.527	0.008
MMD	30	3	0.766	0.004	0.543	0.008
Sinkhorn	30	3	0.785	0.003	0.521	0.014
Immune, unconditioned decoder; $b^* = 0.531$						
Plain scVI	–	3	0.839	0.009	0.499	0.032
JS discriminator	0.1	3	0.835	0.007	0.532	0.002
Reference critic	1	3	0.843	0.007	0.533	0.003
Pooled critic	1	3	0.838	0.007	0.539	0.008
Barycenter critic	3	3	0.840	0.006	0.526	0.004
MMD	10	3	0.838	0.008	0.522	0.009
Sinkhorn	10	3	0.840	0.005	0.525	0.005
ATAC small, unconditioned decoder; $b^* = 0.522$						
Plain scVI	–	3	0.853	0.008	0.402	0.120
JS discriminator	1,000	3	0.657	0.101	0.496	0.026
Reference critic	1	3	0.845	0.012	0.496	0.026
Pooled critic	1	3	0.860	0.008	0.509	0.013
Barycenter critic	100	3	0.864	0.004	0.540	0.018
MMD	30	3	0.847	0.004	0.539	0.017
Sinkhorn	10	3	0.844	0.010	0.528	0.006
ATAC large, unconditioned decoder; $b^* = 0.550$						
Plain scVI	–	3	0.823	0.007	0.538	0.012
JS discriminator	10	3	0.741	0.049	0.557	0.008
Reference critic	1,000	3	0.756	0.028	0.551	0.002
Pooled critic	0.3	3	0.825	0.005	0.553	0.004
Barycenter critic	1	3	0.822	0.012	0.548	0.002
MMD	10	3	0.821	0.020	0.556	0.006
Sinkhorn	10	3	0.825	0.006	0.545	0.004
Human–mouse immune, unconditioned decoder; $b^* = 0.450$						
Plain scVI	–	3	0.750	0.001	0.432	0.019
JS discriminator	1,000	3	0.528	0.003	0.451	0.001
Reference critic	0.3	3	0.752	0.003	0.447	0.003
Pooled critic	0.3	3	0.745	0.002	0.448	0.002
Barycenter critic	3	3	0.747	0.006	0.449	0.001
MMD	3	3	0.746	0.006	0.448	0.003
Sinkhorn	3,000	3	0.552	0.026	0.450	0.000
Simulation 1, unconditioned decoder; $b^* = 0.369$						
Plain scVI	–	3	0.999	0.001	0.232	0.137
JS discriminator	0.1	3	1.000	0.000	0.340	0.030
Reference critic	100	3	0.943	0.055	0.366	0.003
Pooled critic	1	3	1.000	0.000	0.369	0.000
Barycenter critic	3	3	1.000	0.000	0.417	0.048
MMD	10	3	1.000	0.000	0.321	0.049
Sinkhorn	3	3	1.000	0.000	0.360	0.009
Simulation 2, unconditioned decoder; $b^* = 0.318$						
Plain scVI	–	3	0.928	0.042	0.209	0.110
JS discriminator	10	3	0.727	0.062	0.303	0.016
Reference critic	300	3	0.913	0.026	0.314	0.004
Pooled critic	3	3	0.946	0.019	0.310	0.008
Barycenter critic	1,000	3	0.741	0.249	0.312	0.006
MMD	1,000	3	0.802	0.056	0.331	0.013
Sinkhorn	100	3	0.753	0.135	0.384	0.066

Mean macro F1 over the real tasks that are not low-signal

Decoder	conditioned (5)	uncond. (6)
Plain scVI	0.801	0.816
scvi-tools adversary	0.798	–
JS discriminator	0.752	0.726
Reference critic	0.790	0.801
Pooled critic	0.771	0.818
Barycenter critic	0.796	0.815
MMD	0.799	0.811
Sinkhorn	0.769	0.783

Supplementary Note 10. Rules fixed before scoring, and deviations.

Status of the study. The study is exploratory. It was planned in two stages: a penalty-strength pilot, whose seed means were to choose six strengths per penalty family and decoder by rules fixed before it ran, followed by a benchmark at those fixed strengths with five new seeds, to be analysed by rules fixed before any of its scores was read. That benchmark was stopped and not run. The pilot, extended to all eight tasks, is the sweep reported here, and it was analysed with the ranking rules written for the benchmark, with the changes listed below. This note gives, for each set of rules, when it was fixed and which scores had been read by then, and it lists every deviation. The rule documents are kept in the repository as a record: the decision rules with their dated amendments (`docs/PREREG.md`), the ranking-analysis rules (`docs/ANALYSIS_SPEC.md`), how these were applied to the sweep (`docs/SWEEP_ANALYSIS.md` and `docs/RESCOPE_20261102.md`, branch `sweep-paper`), the replication's prediction and agreement rule (`experiments/sweep_replication/PREDICTION.json` and `AGREEMENT.md`) and the rules of the direct biology check (`docs/BIOLOGY_CHECK.md`, branch `biology-check`). The table lists the commits that fixed or changed these rules, with dates in New York time (the commit date, or the date of the git tag for a tagged commit).

Date (New York)	Commit (tag)	Rule record or change
3 Oct 2026, 15:07	129142b (git tag)	Decision rules of the sweep (then a pilot) and of the planned fixed-strength benchmark first fixed, before the first sweep fit
4 Oct 2026, 17:42	c88cce3 (git tag)	Amendments of 4 October; the sweep restarted from zero at this tag (fits made at the first tag unused)
16 Oct 2026, 17:48	10a1ca5 (git tag)	Amendments of 13 to 16 October (Sinkhorn correction and redo, checks before the tag, low-signal rule, first analysis specification); code of every fit and score
18 Oct 2026, 10:52	57241a2	Five follow-up experiments dropped before any was run (decision of 17 October)
21 Oct 2026, 12:03	635be7f (git tag)	Rule of the follow-up on regularisation, optimiser and update steps fixed
22 Oct 2026, 08:24	98732ff	Amendment of 22 October: planned fixed-strength benchmark stopped (21 October); sweep extended to the five further tasks with unchanged code and rules
22 Oct 2026, 22:25	065a486	Ranking-analysis rules: descriptive ranking design replaces the contrast design; the follow-up on regularisation, optimiser and update steps and the premise check dropped
22 Oct 2026, 22:34	5d01da0	Ranking-analysis rules: answers to the nine open points written in
1 Nov 2026, 20:16	2bc5b00	Rescope: the planned fixed-strength benchmark is not run; the sweep is the result; all baselines of the run list added
1 Nov 2026, 21:39	8e32bb2	Sweep analysis code for all eight tasks (first run on all eight tasks the same day, for debugging)
1 Nov 2026, 21:45	2113ec7	Replication design: 855-row run list from the sweep's batch scores
2 Nov 2026, 11:03	3a90d0f (tag `replication-prediction-v1`)	Replication prediction and agreement rule fixed, before any replication fit was scored
2 Nov 2026, 15:15	8595a16	Hypervolume and share of non-dominated strengths added; final sweep analysis run (5 November) at this commit
5 Nov 2026, 09:57	639c302	Direct biology check: rules written, before any of its values was computed
5 Nov 2026, 13:31	803ed80	Replication result before the extension step (first replication scores read)
5 Nov 2026, 17:00	648f7e8	Replication extension run list: 40 fits required by the agreement rule
5 Nov 2026, 18:50	f618484	Scaling checks of Supplementary Figure S5 computed (paper branch, supplement figures)
5 Nov 2026, 19:41	9faac6e	Per-task sign-flip sensitivity analysis of the replication defined (paper branch, text)
5 Nov 2026, 20:56	a475ea3	Replication result with the extension step

Sweep rules. The fitting and scoring rules of the sweep (the strengths 0.1 to 3,000 in half-decade steps, both decoders, seeds 100–102, the scores and the definition of a failed fit) and the rules that were to read it (the strength-grid rule, the matched-strength rule and the switching rules of the two input pilots) were first fixed on 3 October 2026, before the first sweep fit. They were amended on 4 October and the sweep restarted from zero at that commit; the 202 fits made under the first version are not used. A third version, with the amendments decided from 13 to 16 October, was fixed on 16 October at commit 10a1ca5, whose fitting and scoring code made every fit and score in this paper (fits and scores made at c88cce3, the commit of the second version, were declared result-identical to it for every method except Sinkhorn, whose implementation was corrected and whose pilot fits were redone). The amendments of the third version were decided after the sweep on the small chromatin-accessibility, immune and simulation 1 tasks and both input pilots had been fitted and scored, and after the strength-grid, matched-strength and input-pilot rules had read their seed means; the amendment table in `docs/PREREG.md` gives, for each, the data that existed when it was decided. The extension of the sweep to the five further tasks (human–mouse immune, large chromatin-accessibility, lung, simulation 2 and pancreas) was decided on 21 and 22 October and written on 22 October (commit 98732ff); it added 1,890 fits with the same code, rules, seeds and strengths, the edge strengths included, and no score of these fits existed when it was decided.

Ranking-analysis rules. The descriptive ranking (biology at the matched batch removal b^* per task and decoder, ranks with their leave-one-seed-out spread, the secondary per-penalty analyses, label transfer, single-strength placement, best-of-k and the sensitivity analyses) replaced a contrast design (the reference critic's biology at b^* minus the discriminator's, pooled over tasks, with a margin and a correction for several tests) by the first author's decisions of 22 and 23 October 2026. They were written into `docs/ANALYSIS_SPEC.md` by commits 065a486 and 5d01da0, both made on the evening of 22 October, New York time. The rules were written for the fixed-strength benchmark and were never fixed by a tag, because that benchmark was not run. When they were written, sweep scores had been read on the three small tasks only (by the strength-grid, matched-strength and input-pilot rules) and no score of the benchmark had been read; the sweep fits of the other five tasks were being made (from 22 October) and their score files are dated 28 to 30 October. Score files of some baseline fits on four of these tasks are older: those of the scvi-tools

adversary, scANVI and sysVI on lung, pancreas and simulation 2 are dated 6 and 7 October, and at least one of the scvi-tools adversary on the human–mouse immune task 12 October (Supplementary Note 11, Table A). The rescope that made the sweep the result kept these rules (`docs/RESCOPE_20261102.md`, commit 2bc5b00; the document dates the decision 2 November, the commit is dated 1 November at 20:16). The sweep analysis code was written on 1 November (commit 8e32bb2) and run on all eight tasks that day for debugging, so the sweep scores of all eight tasks had been read before the last two changes to the analysis, made on 2 November (commit 8595a16): the trade-off area was renamed the hypervolume, with its corner unchanged, and the share of non-dominated strengths was added. The final run of the sweep analysis was made on 5 November at that commit.

Changes from the ranking specification. Applied to the sweep, the specification could not be followed as written at these points (`docs/SWEEP_ANALYSIS.md`, section 3): (i) three seeds (100–102) instead of five, so the leave-one-seed-out analysis has three repeats and its standard error is reported where at least 2 of the 3 repeats give a reading (the specification asked for 4 of 5); (ii) every fitted strength is a point of a penalty's curve (10 to 12 strengths) instead of the six strengths of the strength-grid record; (iii) b^*, the highest batch score reached by every penalty and the low-signal flag are computed from the sweep's own rows with the specification's formula, because no matched-strength record exists for the sweep; (iv) the label-transfer strength is chosen among every fitted strength by the same rule (the failure-free strength whose seed-mean batch score is nearest b^*, ties to the smaller strength), and label transfer is computed on every penalty fit, the matched-strength table being read from those results; (v) a strength is eligible as the transfer strength across tasks when it was fitted and is failure-free in every training task and was also fitted in the held-out task; (vi) the baselines enter as scored, with a coverage table, instead of the analysis refusing an incomplete set; (vii) baseline settings keep their own seeds (five for the scvi-tools adversary, scANVI and sysVI; one run for Harmony, Scanorama and PCA); (viii) the collapse rate counts every fitted strength, 30 to 36 fits per penalty, task and decoder, instead of 30; (ix) a curve whose plain-scVI anchor already lies at or above b^* gives no reading, as the specification states and the earlier code did not; (x) the analysis was not run once after a freezing tag: development runs on three and then on eight tasks came before the final run, and the final run was made without the check that every baseline row is scored or failed, because one baseline setting could not be fitted (deviation 6).

Replication rule. The replication's run list (855 fits: for each penalty in each of the 13 ranked pairs, the two sweep strengths that bracket the first crossing of b^*, and plain scVI, with seeds 0–4) was built from the sweep's batch scores on 1 November (commit 2113ec7). The prediction (the sweep's readings and ranks at b^* in the 13 ranked pairs and its four within-pair differences of 0.04 or more, taken from the sweep analysis at commit a361ba2) and the agreement rule were committed under the tag `replication-prediction-v1` (commit 3a90d0f) on 2 November 2026, before any replication fit was scored. By then every sweep score, b^* and the sweep readings had been read. The rule makes the raw reading on the new seeds primary and names a slope-adjusted reading as a sensitivity analysis only, without giving its formula; it prescribes one further sweep strength outward, with five seeds, for a reading whose new-seed means do not bracket b^*, and the bottom place for a penalty that still does not reach it. The share of sweep near-ties that stay close, the rank correlation per pair and the batch score reached by every replication point are reported beside the rule and are not part of it (Supplementary Note 4).

Deviations.

1. *The planned fixed-strength benchmark was not run.* It was stopped on 21 October 2026 and, by the rescope, not run; the replication across training seeds took its place (its run list was committed on 1 November and its rule on 2 November). The strength-grid and matched-strength rules, whose purpose was to choose the benchmark's strengths, therefore select nothing in this paper. About 572 fits had been made for the benchmark: 167 on the workstation and 405 on the cluster (the cluster count is the record's and was not re-counted). Of the workstation fits, all on the immune task with the batch-conditioned decoder, 160 had been scored, 19 on 12 October and 141 on 20 and 21 October; the cluster fits were not scored. None of these fits or scores is used, and by the first author's statement none of the scores was read.
2. *Follow-up experiments were dropped.* Five follow-ups (composition dose-response, reference choice, batch-count scaling, architecture variants and KL warm-up sensitivity) were dropped on 17 October 2026, before any of them was run. The follow-up on regularisation, optimiser and update steps, whose rule was fixed on 21 October, and a premise check (batch separability, adversary gradient norms, discriminator accuracy) were dropped with the ranking design on 22 and 23 October. No follow-up fit is used.
3. *The sweep was extended to eight tasks* (above). The strength-grid rule was to be re-run on all eight tasks and a keep-or-refit step for the benchmark's fits was to follow; neither was done, because the benchmark was not run.
4. *Edge-strength and matching fits.* The first extension round of the strength-grid rule added two strengths at each of two grid edges (0.01 and 0.03 for the discriminator with the unconditioned decoder, 10,000 and 30,000 for Sinkhorn with the batch-conditioned decoder), 12 fits per task and 96 in all; the matched-strength rule added strength 0.03 for the discriminator and the reference and pooled critics with the batch-conditioned decoder on the immune task (9 fits). All are analysed as sweep points: 3,024 fits with the edge strengths, and 3,033 with the 9 matching fits (Supplementary Note 11).
5. *The replication's extension step.* Before it, 70 of the 78 replication readings bracketed b^*. For the 8 that did not, the next sweep strength outward was fitted with seeds 0–4, as the agreement rule prescribes: 40 fits, 15 on the workstation and 25 on the cluster, whose run list was committed before any of them was fitted (commit 648f7e8). After it 77 readings bracketed b^*; MMD on the pancreas task with the batch-conditioned decoder did not replicate the crossing and takes the bottom place in its pair.
6. *An unfittable Scanorama setting.* Scanorama with 80 neighbours could not be fitted on the large chromatin-accessibility task (the fit ran out of memory or overflowed a 32-bit index). A failed fit is defined by a non-finite training loss or latent, so this setting is neither scored nor failed; it is reported as missing, and 447 of the 448 baseline rows were scored. Scanorama's best-of-k curve on that task stops at $k = 5$ (Supplementary Figure S6).

7. *Baseline settings.* On 4 October 2026 the Scanorama neighbour grid became 2, 5, 10, 20, 40 and 80: 160 neighbours had not finished within 8 hours on the human–mouse immune task. The tool's default (100 dimensions at 20 neighbours) is kept as a seventh setting. At the rescope every row of the committed baseline run list was added (448 rows).
8. *Design choices after the pilots.* The adversary reads the posterior mean and its input is not standardised per dimension: the defaults, which the two input pilots' switching rules kept (Methods). The input pilots were run on the small chromatin-accessibility and immune tasks only and were not extended to the other tasks. A task and decoder pair in which the highest batch score reached by every penalty lies less than 0.02 above plain scVI's is flagged as low-signal and is not ranked (amendment of 13 October).
9. *Seed pairing.* A seed fixes the scVI initialisation and the split into training and validation cells for every method. The order of minibatches is the same only within a method and between plain scVI and the methods without an adversary network (MMD, Sinkhorn), because building an adversary draws from the random-number generator before the first epoch (amendment of 4 October). Differences from plain scVI at the same seed are therefore less tightly paired than first stated.
10. *Analyses added after the rules.* These are exploratory and are reported beside the results the rules define, never in their place: (a) the per-task sign-flip analysis of the replication, which flips the signs of both decoders of a task together (2^8 patterns), defined on 5 November after the replication result before the extension step had been read; (b) a sign-flip analysis by task family, which flips related tasks and both decoders together, added after the replication result (one-sided $p = 0.0156$, the smallest value six families allow: 1 of $2^6 = 64$ patterns); (c) the formula of the slope-adjusted reading, which the rule names without a formula, written with the agreement test on 5 November; (d) the direct biology check (Supplementary Note 12), whose rules were written on 5 November after all sweep scores had been read; (e) the scaling checks of Supplementary Figure S5: the ranking specification names them as exploratory checks used for no conclusion but does not define how they are computed, and they were computed on 5 November, after the sweep scores had been read; (f) the hypervolume's name and the share of non-dominated strengths (2 November, above).
11. *Limitations.* Ranks and mean ranks are descriptive and carry no test. A reading's leave-one-seed-out standard error comes from 2 or 3 repeats of a statistic that is not smooth (a curve crossing), and no coverage check exists for it. A reading interpolates a piecewise-linear curve, so it depends on the spacing of the strengths, and penalties whose readings differ by less than that dependence can change places. The collapse intervals treat the fits of a penalty in a task as independent, although the fits of one seed share their plain-scVI reference. b^* is set by the penalty with the lowest best batch score, so one weak penalty lowers the target for all six, and a penalty that does not reach b^* would share the last places whatever its biology. Related tasks count separately in the mean rank: all cells of the immune task are in the human–mouse immune task, and the two chromatin-accessibility tasks share cells. Low-signal pairs are not ranked. The replication refits the same tasks with new seeds and says nothing about new data. Scores made at the two scoring sites were shown identical on the latents of one identity check only (Supplementary Note 11).

Supplementary Note 11. Software, hardware, run counts and checksums.

Code. Every fit and score in this paper was made with the fitting and scoring code of commit 10a1ca5 of the repository, or of commit c88cce3, whose fits and scores were declared result-identical to it for every method except Sinkhorn (Supplementary Note 10). Each latent file records the commit, the scvi-tools and PyTorch versions and the graphics card of its fit (or, for Harmony, Scanorama and PCA, the processor and the package versions), and each score file records the scorer's commit, the processor and its instruction level, the package versions, the kBET seed and the scoring site. The commits and machines in Table A are read from these records; the analyses that followed (the sweep analysis, the replication's agreement test, the biology check and the paper's tables and figures) are at the commits listed in `paper/REPRODUCE.md`.

Software. The models are scvi-tools 1.4.2 models trained with PyTorch 2.13.0 on the graphics card; scores are the scIB metrics of scib 1.1.7, with kBET 0.99.6 run in R 4.5.2. Table B gives the versions of the main packages in each environment, from the environment exports committed with the code (`cluster/envspec/` at commit 10a1ca5: the conda and pip exports of the fitting and scoring environments on the workstation and the pip pins of the cluster environments), from the version fields of the latent and score files, from the environment export of the sweep analysis and from the record of the biology check. The environments of the replication's agreement test, the paper's tables and figures and the PDF build are exported in the repository and named in `paper/REPRODUCE.md`.

Hardware. The workstation has one NVIDIA GeForce RTX 3080 graphics card (10 GB) and an AMD Ryzen 5 3600 processor (6 cores, 12 threads, AVX2 instructions). On the JHPCE cluster, fits ran on nodes with NVIDIA L40S graphics cards (46 GB) and scores on nodes with AMD EPYC 7702 processors (Zen 2, the workstation's AVX2 instruction level). Each task's sweep was fitted on one kind of machine: the small chromatin-accessibility, immune and simulation 1 tasks on the workstation, the other five tasks on the L40S cards. Each task's replication fits were made on the same kind of machine as its sweep fits (the replication's latent files of the cluster tasks stayed on the cluster; the replication record names the L40S). The workstation's graphics driver was updated on 2 November 2026, between the sweep and the replication fits. Scoring sites differ in places (Table A): the sweep fits of the lung, simulation 2 and pancreas tasks were scored on the workstation and their replication fits on the cluster, and the human–mouse immune and large chromatin-accessibility tasks were scored on the cluster in both (one sweep fit of the human–mouse immune task on the workstation). Scores from the two sites are combined under a rule that admits only workstation rows and cluster rows made with the random seed of the local inverse Simpson's index fixed to the workstation's value (the workstation's processor returns a constant from its hardware random-number instruction, so its subsample for that index is the same in every run); the rule required an identity check in which 5 latents gave identical values in all 19 metric columns at both sites (`docs/SCORER_EQUIVALENCE.json`, branch `sweep-paper`). Identity was shown on those latents only.

Table A. Where fits were made and scored, per experiment: tasks, scored fits of all fits, the graphics card or processor of the fits, the processor and site of the scores (with the number of score files where a task was scored at two sites), the fitting and scoring commits, and the dates of the score files. The Scanorama setting that could not be fitted has no row.

Experiment	Tasks	Scored/fits	Fitted on	Scored on	Commits	Score files dated
Sweep	ATAC large	378/378	L40S	AMD EPYC 7702 (JHPCE)	10a1ca5	2026-10-28 to 2026-10-30
Sweep	human–mouse immune	378/378	L40S	AMD EPYC 7702 (JHPCE): 377; AMD Ryzen 5 3600 (workstation): 1	10a1ca5	2026-10-28 to 2026-10-29
Sweep	lung, pancreas, simulation 2	1,134/1,134	L40S	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-10-28 to 2026-10-30
Sweep	ATAC small, immune, simulation 1	1,143/1,143	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5, c88cce3	2026-10-04 to 2026-10-19
Input pilot	ATAC small, immune	108/108	RTX 3080	AMD Ryzen 5 3600 (workstation)	c88cce3	2026-10-10
Standardisation pilot	ATAC small, immune	180/180	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5, c88cce3	2026-10-10 to 2026-10-20
scvi-tools adversary	ATAC large	5/5	L40S	AMD EPYC 7702 (JHPCE)	10a1ca5, c88cce3	2026-11-01
scvi-tools adversary	human–mouse immune	5/5	L40S	AMD EPYC 7702 (JHPCE): 1; AMD Ryzen 5 3600 (workstation): 4	10a1ca5, c88cce3	2026-10-12 to 2026-11-01
scvi-tools adversary	lung, pancreas, simulation 2	15/15	L40S	AMD Ryzen 5 3600 (workstation)	c88cce3	2026-10-06
scvi-tools adversary	ATAC small, immune, simulation 1	15/15	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-03
scANVI	ATAC large, human–mouse immune	10/10	L40S	AMD EPYC 7702 (JHPCE)	10a1ca5, c88cce3	2026-11-01
scANVI	lung, pancreas, simulation 2	15/15	L40S	AMD Ryzen 5 3600 (workstation)	c88cce3	2026-10-06 to 2026-10-07
scANVI	ATAC small, immune, simulation 1	15/15	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-03
sysVI	ATAC large, human–mouse immune	60/60	L40S	AMD EPYC 7702 (JHPCE)	10a1ca5, c88cce3	2026-11-01
sysVI	lung, pancreas, simulation 2	90/90	L40S	AMD Ryzen 5 3600 (workstation)	c88cce3	2026-10-06 to 2026-10-07
sysVI	ATAC small, immune, simulation 1	90/90	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-03
Harmony	all eight	56/56	CPU AMD Ryzen 5 3600	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-01 to 2026-11-03
Scanorama	all eight	55/55	CPU AMD Ryzen 5 3600	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-01 to 2026-11-03
PCA	all eight	16/16	CPU AMD Ryzen 5 3600	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-01 to 2026-11-03
Replication	ATAC small, immune, simulation 1	285/285	RTX 3080	AMD Ryzen 5 3600 (workstation)	10a1ca5	2026-11-03 to 2026-11-05
Replication	ATAC large, human–mouse immune, lung, pancreas, simulation 2	610/610	latent not on the workstation	AMD EPYC 7702 (JHPCE)	10a1ca5	2026-11-03 to 2026-11-05

Table B. Package versions per environment: the fitting environments on the workstation and on the cluster (with the PyTorch build recorded in the latent files), the environment of Harmony, Scanorama and PCA (CPU tools, versions recorded in their latent files), the scoring environment (versions recorded in every score file, Python and R from its export), the sweep analysis and the biology check; – where the package is not in that environment or its record.

Package	Fits, workstation	Fits, JHPCE	CPU tools	Scoring	Sweep analysis	Biology check
Python	3.11.15	–	–	3.11.15	3.11.15	3.11.15
R	–	–	–	4.5.2	–	–
scvi-tools	1.4.2	1.4.2	–	–	1.4.2	–
torch	2.13.0+cu130	2.13.0+cu126	2.4.1+cu121	–	2.13.0	–
lightning	2.6.5	2.6.5	–	–	2.6.5	–
pyro-ppl	1.9.1	1.9.1	–	–	1.9.1	–
anndata	0.12.19	0.12.19	0.11.4	0.11.4	0.12.19	0.12.19
scanpy	1.11.5	1.11.5	1.11.5	1.11.5	1.11.5	1.11.5
numpy	2.4.6	2.4.6	1.26.4	1.26.4	2.4.6	2.4.6
scipy	1.17.1	1.17.1	–	1.14.1	1.17.1	1.17.1
pandas	2.3.3	2.3.3	–	1.5.3	2.3.3	2.3.3
scikit-learn	1.9.0	1.9.0	1.9.0	1.9.0	1.9.0	1.9.0
numba	0.67.0	0.67.0	–	0.66.0	0.67.0	–
pynndescent	0.6.0	0.6.0	–	0.5.13	0.6.0	–
scib	–	–	1.1.7	1.1.7	–	–
kBET (R)	–	–	–	0.99.6	–	–
rpy2	–	–	–	3.6.7	–	–
harmony-pytorch	–	–	0.1.7	–	–	–
scanorama	–	–	1.7.4	–	–	–

Run counts. A fit fails if its training loss or its latent becomes non-finite; failed fits are counted and left out of seed means. Completed and failed fits per task are in Supplementary Table S5.

- Penalty-strength sweep: 3,024 fits, made of 1,098 fits at the ten base strengths and 36 edge-strength fits on the three small tasks and 1,890 fits on the five further tasks (378 per task, 12 of them at edge strengths), plus 9 fits that the matched-strength rule added on the immune task; 3,033 fits analysed, 0 failed.

- Input pilots (small chromatin-accessibility and immune tasks, both decoders, seeds 100–102): 108 fits with a posterior sample as the adversary's input and 180 fits with standardised input, 0 failed; the other half of each comparison is the sweep's own fits.
- Baselines: 448 rows of the committed run list (the scvi-tools adversary, scANVI and sysVI with seeds 0–4; Harmony, Scanorama and PCA one run per setting); 447 scored, 0 failed, and one setting (Scanorama with 80 neighbours on the large chromatin-accessibility task) could not be fitted.
- Planned fixed-strength benchmark: about 572 fits made before it was stopped on 21 October 2026, 167 on the workstation and 405 on the cluster (the cluster count is the record's and was not re-counted); not used.
- Replication across training seeds: 855 fits (775 with a penalty and 80 of plain scVI) and 40 fits of the extension step, 895 in all, 0 failed.
- Direct biology check: 783 saved fits read again, no new fits (Supplementary Note 12).
- Scaling checks (Supplementary Figure S5): no further fits; they re-score the sweep and baseline fits by other scalings of the same metrics.

Checksums. Task files: the figshare download of the eight task files is checked against the byte sizes and md5 sums recorded in the repository, and the prepared files against `docs/prepped_fingerprints_scib.json`; the SHA-256 of each prepared file is in the record of the biology check (Supplementary Table S1 gives the file checksums). Fits and scores: every score file records the SHA-256 of the latent it scored, and the sweep analysis and the biology check refuse a latent whose SHA-256 differs; the run lists carry their SHA-256 (for the replication, `manifests/*.sha256`). Results: `MANIFEST.sha256` in the output folder of the sweep analysis on the data root, `docs/biology_check/MANIFEST.sha256` (branch `biology-check`), `experiments/sweep_replication/agreement_result/MANIFEST.sha256` and `agreement_result_with_extension/MANIFEST.sha256` (branch `sweep-replication`), and, on the paper branch, `paper/figure_data/MANIFEST.sha256`, `paper/figures/MANIFEST.sha256` and `paper/MANIFEST.sha256` (text sources, figures, supplement, number ledgers and PDFs). `paper/REPRODUCE.md` gives the commands that re-run each result and check these files. Repository: github.com/kreid415/wasserstein-critic-deconfounding; [*to be decided: release tag and commit of the submitted version*].

Supplementary Note 12. Direct biology check.

Status. The direct biology check is exploratory and descriptive. Its rules (`docs/BIOLOGY_CHECK.md`, branch `biology-check`) were written on 5 November 2026 (commit 639c302), after the sweep scores had been read (the batch and biology scores of every sweep fit, b^*, the matched strengths and the label-transfer macro F1 at the matched strength) and before any value of this check was computed. It makes no new fits, runs no test and makes no claim of significance; every value is reported per method with its spread over seeds. The code is `scripts/biology_check.py` at commit c4a793f, which made the run of 5 November; commit 697cc6c changed only the one-page summary and the script that writes it (the summary's seed spread had been averaged over all eight tasks instead of the six real tasks).

Fits. For each penalty, decoder and task, the label-transfer strength of Supplementary Note 9 (the failure-free sweep strength whose seed-mean batch score is nearest b^*, ties to the smaller strength; the choice uses batch scores only), with seeds 100–102; plain scVI with both decoders, seeds 100–102; and every scored baseline row at each of its settings (the scvi-tools adversary, scANVI and sysVI with seeds 0–4; Harmony, Scanorama and PCA one run per setting). In all 783 saved fits on the eight tasks; low-signal pairs are included and flagged. The fits, latent files and their SHA-256 checks are those of the sweep analysis, read with its code, and the expression and labels are the prepared task files.

Neighbours. For each cell, its 15 nearest other cells by Euclidean distance in the saved latent, the cell itself excluded. The same lists serve the purity and the smoothing below.

Readouts, per fit and cell type:

- *Neighbourhood purity:* for each cell, the share of its 15 neighbours that carry its own cell-type label; per type, the mean over the type's cells; per fit, the unweighted mean over types. It measures whether cells of one type stay together in the latent space.
- *Rare types:* types with fewer than 1% of a task's cells, with their purity and their label-transfer recall. They show whether batch alignment merges small populations into large ones. Tasks without a rare type report none.
- *Marker-gene preservation,* on the four RNA tasks (pancreas, lung, immune and human–mouse immune): the expression as distributed (log-normalised), restricted to its 2,000 highly variable genes; for each type the top 20 marker genes by the Wilcoxon test against all other cells, computed once on that expression and once on each fit's expression averaged over each cell's 15 neighbours; the readout is the Jaccard overlap of the two sets of 20. It measures whether the neighbourhoods a fit forms keep the genes that define each type. Types with fewer than 10 cells get no marker readout.
- *Label-transfer confusion,* from the classifier of Supplementary Note 9: types present in one batch only are dropped, each batch is held out in turn and its cells are predicted by a 15-nearest-neighbour classifier trained on the other batches; per type, the recall (the share of the type's cells predicted as that type) and the row-normalised confusion table. It shows which types are mistaken for which.

Differences and summaries. Each per-type readout is reported as its difference from plain scVI's value for the same type: for the penalties and plain scVI, plain scVI with the same decoder and the same seed; for the baselines, whose seeds do not match the sweep's, the mean of plain scVI with the batch-conditioned decoder over seeds 100–102. A fit's difference is the mean of its per-type differences over the types where both values exist. Per task, method, setting and decoder the check reports the mean and standard deviation over seeds, then the mean over tasks of the per-task means, separately for the six real and the two simulated tasks. Baselines with several settings are averaged over their settings. scANVI is trained with the cell-type labels, so

its readouts are not label-free.

Checks that stop the run. The latent's SHA-256 equals the one its score file records; the cell order and the labels match the task file; the number of fits per task matches the list above; every penalty and plain-scVI fit has its seed-matched reference; purity and Jaccard overlap lie in $[0, 1]$; no value is missing where a readout is defined; and the macro F1 recomputed from the confusion predictions equals the label-transfer F1 of Supplementary Note 9.

Results. Supplementary Table S10 gives, per method and decoder, the mean over the six real tasks of the purity difference with its seed spread, the rare-type purity and recall differences, and the marker-gene overlap with its difference from plain scVI. The per-task, per-fit, per-type and confusion tables, the record (with the SHA-256 of every input) and the one-page summary are in `docs/biology_check/` on branch `biology-check`, with their checksums in `docs/biology_check/MANIFEST.sha256`.

Supplementary Tables

Supplementary Table S1: Tasks: the eight integration tasks of the single-cell integration benchmark scIB (main-text reference; figshare article 12420968, version 8), as prepared for every fit and score. "Batches": number of batches of the batch variable used (scIB's name in parentheses); for simulation 2 these are sub-batches, nested in the batches whose number follows "in". "Features": training features, scIB's highly variable genes for RNA and all gene-activity features for the chromatin-accessibility tasks, of the genes in the file. "Non-integer": batches whose counts, used as distributed, hold non-integer values (Pancreas: celseq, celseq2, fluidigmc1, smarter; Lung: 1, 2, 3, 4, 5, 6, A1, A2, A3, A4, A5, A6; Immune: Villani; Human–mouse immune: Villani). "Family": tasks of one family never choose a strength or setting for each other in the single-strength placement; the simulations belong to no family and are ranked separately. Shared batches are counted by batch name, shared cells by batch and barcode (the cell name up to its first hyphen). Part B: the downloaded file, its figshare file id and MD5 checksum (checked at download), and the SHA-256 checksum of the prepared file; the prepared files also match the content fingerprints committed in `docs/prepped_fingerprints_scib.json` at commit 10a1ca5.

Task	Kind	Cells	Batches	Cell types	Features	Non-integer	Family; shared batches
Part A: task contents							
Pancreas	real, RNA	16,382	9 (tech)	14	2,000 of 19,093	4 of 9	pancreas
Lung	real, RNA	32,472	16 (batch)	17	2,000 of 15,148	12 of 16	lung
Immune	real, RNA	33,506	10 (batch)	16	2,000 of 12,303	1 of 10	immune; all cells and batches also in Human–mouse immune
ATAC small	real, gene activity	11,270	3 (batch-name_all)	7	all 3,429	0 of 3	atac; all cells and batches also in ATAC large
ATAC large	real, gene activity	84,813	11 (batch-name_all)	7	all 3,580	0 of 11	atac; holds all cells and batches of ATAC small
Human–mouse immune	real, RNA	97,861	23 (batch)	19	2,000 of 8,135	1 of 23	immune; holds all cells and batches of Immune
Simulation 1	simulated, RNA	12,097	6 (Batch)	7	2,000 of 9,979	0 of 6	–
Simulation 2	simulated, RNA	19,318	16 (SubBatch) in 4	4	2,000 of 10,000	0 of 16	–

Part B: files and checksums

Task	File	figshare id	MD5 of download	SHA-256 of prepared file
Pancreas	`human_pancreas_norm_complexBatch.h5ad`	24539828	`5aa155652b998c90395a0c9bd731b4b3`	`d172898be6d237f6698d0723813644b2afc283276bb429e84debf4256bdeeaa9`
Lung	`Lung_atlas_public.h5ad`	24539942	`aac0832c3e3413ae279343f5de02a2c6`	`1b7ad36e535be6628bd3c9dc072552415cc0dac1ae698092a779119d1c31cf64`
Immune	`Immune_ALL_human.h5ad`	25717328	`482a6a3117432d41a46effd8376b8ee0`	`3d8596256dc6cf9d17b91e974aa27842711c4b1817d53b772d324914b74834e3`
ATAC small	`small_atac_gene_activity.h5ad`	25721789	`8eb4af0e5e2d07842e516d5b8047b449`	`6ddc6fc25645bb9ab4dd2e4a5538731aaaa33f2232729baef8300a715a2d814a`
ATAC large	`large_atac_gene_activity.h5ad`	25721786	`bab0bd97840692f938f4361bd07fac8e`	`7bfddbe8e1d7c09637c61ed4974ce35b66463e115fe2bb9fc87552c52ac33269`
Human–mouse immune	`Immune_ALL_hum_mou.h5ad`	25717166	`3cc1ba9f35b21551dd7a81162998b727`	`60fa7788342ebcdd551e63e42017b0b398453151b640af5412074ee4c75ccdaa`
Simulation 1	`sim1_1_norm.h5ad`	33798263	`ea2589604bec1fd184ee489d1491e54f`	`5c25226fd56187f3826cd661dcabe9ffc00cefb14d5fd92cfd8a65932c9523f0`
Simulation 2	`sim2_norm.h5ad`	33798764	`4d22e971f1e199726f99748cf282e689`	`6e5533ccd18558eb645339e6488e08d20f5cd0d2bb7eb48ade98d0fbf78576ea`

Supplementary Table S2: Hyperparameters and training settings (Supplementary Note 2). Every value is the one recorded in the fits' own latent files (the resolved training plan of each fit), and every fit of a method has the same value. Part A: settings shared by every scVI-backbone fit (plain scVI, the six penalties and the scvi-tools adversary); Part B: settings that differ by method, each method's standard setting; Part C: settings by task; Part D: seeds. The loss of each penalty is in Supplementary Note 1, the baselines' settings in Supplementary Table S4.

Part A: shared settings	
Latent dimensions	10
Encoder and decoder	1 hidden layer of 128 units each
Likelihood	zero-inflated negative binomial
Training split	90% of cells for training, the rest for validation; no early stopping: every fit trains for its task's epochs (Part C)
Minibatch	128 cells

Encoder and decoder optimiser	Adam, learning rate 10^{-3}, betas (0.9, 0.999), epsilon 0.01, weight decay 10^{-6}
KL warm-up	linear from 0 to 1 over 400 epochs (scvi-tools default); the penalty is not scaled by it
Adversary input	posterior mean, no per-dimension standardisation
Scored embedding	posterior mean of every cell (training and validation cells)

Part B: settings by method

Method	Adversary steps per encoder step	Adversary optimiser	Gradient penalty weight	Other
Plain scVI	–	–	–	strength $\lambda = 0$
JS discriminator	1	Adam, learning rate 10^{-3}, betas (0.9, 0.999), epsilon 0.01, weight decay 10^{-6}	–	head: 2 hidden layers of 128 units
Reference critic	5	Adam, learning rate 10^{-4}, betas (0, 0.9), epsilon 10^{-8}, no weight decay	10	head: 2 hidden layers of 128 units; target: the reference batch
Pooled critic	5	Adam, learning rate 10^{-4}, betas (0, 0.9), epsilon 10^{-8}, no weight decay	10	head: 2 hidden layers of 128 units; target: the other batches
Barycenter critic	5	Adam, learning rate 10^{-4}, betas (0, 0.9), epsilon 10^{-8}, no weight decay	10	head: 2 hidden layers of 128 units; target: barycenter, at most 10 iterations per step
MMD	–	–	–	Gaussian kernels, $\sigma = \sqrt{10} \times \{0.25, 0.5, 1, 2, 4\}$
Sinkhorn	–	–	–	$\epsilon = 0.05\sqrt{2 \times 10}$, Euclidean cost
scvi-tools adversary	as scvi-tools	Adam, learning rate 10^{-3}, betas (0.9, 0.999), epsilon 0.01, weight decay 10^{-6}	–	scvi-tools' adversarial classifier (Supplementary Note 1); batch-conditioned decoder only; weight $1 - \beta$

Part C: settings by task

Task	Cells	Epochs	KL weight in the last epoch
Pancreas	16,382	400	0.9975
Lung	32,472	246	0.6125
Immune	33,506	239	0.5950
ATAC small	11,270	400	0.9975
ATAC large	84,813	94	0.2325
Human–mouse immune	97,861	82	0.2025
Simulation 1	12,097	400	0.9975
Simulation 2	19,318	400	0.9975

Epochs: scvi-tools' rule $\min(400, \text{round}(20000 \times 400/n))$ for n cells, the same for every method of a task. KL weight in the last of E epochs: $(E-1)/400$, as the warm-up counts epochs from 0.

Part D: seeds

Experiment	Seeds	Fits
Penalty-strength sweep (plain scVI and the six penalties)	100–102	3,033
Replication across training seeds	0–4	895
Input pilots (two tasks, both decoders)	100–102	–
Baselines: scvi-tools adversary, scANVI, sysVI	0–4	320
Baselines: Harmony, Scanorama, PCA (one run each)	0	127

Supplementary Table S4: Baseline grids (Supplementary Note 8), on each of the eight tasks. "Grid": the settings of the varied knob at the main embedding size ("Dims", latent or principal-component dimensions), each run on every task; "Sensitivity": a further setting at the tool's default embedding size, not used to choose a setting. "Planned" and "Scored" count fits over the eight tasks and all seeds; not fitted: 1 of Scanorama on ATAC large. The neural baselines use the scVI-backbone settings of Supplementary Table S2 (epochs, minibatch, training split, hidden units and latent size of each task); the other tools read the same training features.

Tool	Knob	Grid	Dims	Sensitivity	Seeds	Planned	Scored	Input	Version
scvi-tools adversary	none (weight $1 - \beta$)	–	10	–	0–4	40	40	counts; batch-conditioned decoder	scvi-tools 1.4.2
scANVI (uses cell-type labels)	none	–	10	–	0–4	40	40	counts and cell-type labels; scVI pre-training	scvi-tools 1.4.2
sysVI	cycle-consistency weight	1, 2, 5, 10, 20, 50	10	–	0–4	240	240	scIB's normalised values; Gaussian likelihood	scvi-tools 1.4.2
Harmony	diversity penalty θ	0, 0.5, 1, 2, 4, 8	10	2 at 50 dimensions	0	56	56	PCA of scIB's normalised values	harmony-pytorch 0.1.7

Scanorama	neighbours (knn)	2, 5, 10, 20, 40, 80	10	20 at 100 dimensions	0	56	55	scIB's normalised values	scanorama 1.7.4
PCA (uncorrected)	none	–	10	50 dimensions	0	16	16	scIB's normalised values	scanpy 1.11.5
All						448	447		

Supplementary Table S5: Run counts (Supplementary Note 11): fits per method, decoder and task. Sweep: the fits read by the sweep analysis (the ten strengths, the edge strengths and the strength-extension fits on the immune task, Supplementary Note 4); each was fitted and scored, and none failed (Supplementary Note 11). Baselines: fits scored, with the planned number where it differs ("55 of 56"); – for tools without a decoder; not fitted: 1 of Scanorama on ATAC large (Supplementary Note 8; Supplementary Table S4 gives the settings). The input and standardisation pilots, the replication across training seeds and the stopped fixed-strength benchmark are counted in Supplementary Note 11, the replication per penalty in Supplementary Table S9.

Method	Decoder	Pancreas	Lung	Immune	ATAC small	ATAC large	Human–mouse immune	Simulation 1	Simulation 2	All
Plain scVI	batch-conditioned	3	3	3	3	3	3	3	3	24
Plain scVI	unconditioned	3	3	3	3	3	3	3	3	24
JS discriminator	batch-conditioned	30	30	33	30	30	30	30	30	243
JS discriminator	unconditioned	36	36	36	36	36	36	36	36	288
Reference critic	batch-conditioned	30	30	33	30	30	30	30	30	243
Reference critic	unconditioned	30	30	30	30	30	30	30	30	240
Pooled critic	batch-conditioned	30	30	33	30	30	30	30	30	243
Pooled critic	unconditioned	30	30	30	30	30	30	30	30	240
Barycenter critic	batch-conditioned	30	30	30	30	30	30	30	30	240
Barycenter critic	unconditioned	30	30	30	30	30	30	30	30	240
MMD	batch-conditioned	30	30	30	30	30	30	30	30	240
MMD	unconditioned	30	30	30	30	30	30	30	30	240
Sinkhorn	batch-conditioned	36	36	36	36	36	36	36	36	288
Sinkhorn	unconditioned	30	30	30	30	30	30	30	30	240
Sweep, all		378	378	387	378	378	378	378	378	3,033
scvi-tools adversary	batch-conditioned	5	5	5	5	5	5	5	5	40
scANVI (uses cell-type labels)	batch-conditioned	5	5	5	5	5	5	5	5	40
sysVI	batch-conditioned	30	30	30	30	30	30	30	30	240
Harmony	–	7	7	7	7	7	7	7	7	56
Scanorama	–	7	7	7	7	6 of 7	7	7	7	55 of 56
PCA (uncorrected)	–	2	2	2	2	2	2	2	2	16
Baselines, all		56	56	56	56	55 of 56	56	56	56	447 of 448

- Table S6, ranking detail: biology at b^* per penalty, task and decoder with leave-one-seed-out standard errors, ranks and the range of each rank over the three repeats, a separate table of low-signal pairs, and the pairs where a penalty's curve does not reach b^* marked.
- Table S7, single-strength placement for the unconditioned decoder (the counterpart of Table 3): selection score per method and real task at the strength (for Harmony, Scanorama and sysVI, the setting) chosen on other task families, with the task's own best strength or setting in parentheses.

Supplementary Table S8: Secondary results per penalty, task and decoder (definitions in Supplementary Note 7; all descriptive). HV: hypervolume, the area of the batch-score by biology-score plane reached by the penalty's sweep points beyond the reference point (plain scVI's batch score, plain scVI's biology score minus 0.10), divided by the batch gap that sets b^*; n/a in low-signal pairs. Non-dom.: strengths that no other strength of the penalty beats on both scores, of all its strengths. Collapse: number of fits n (30, 33 or 36: all strengths times three seeds) and the percentage that failed or lost 0.10 or more biology against the same seed's plain scVI, with the exact (Clopper–Pearson) 95% interval, which is nominal because the fits of one seed share one threshold. Failed: diverged and non-finite-latent fits (0 in all). Best: the strength with the highest mean selection score; Width: decades spanned by the strengths whose mean selection score is at least the lower end of the best strength's 95% t interval; * a strength inside that span falls below the bound (19 windows); Good: number of such strengths. Local and global: mean of the three neighbourhood metrics (iLISI, kBET, graph connectivity) and of the two whole-embedding metrics (principal-component regression, batch ASW) read at the first crossing of b^* with the same interpolation as the biology reading; since $3\,\text{local} + 2\,\text{global} = 5b^*$ there, $\text{local} = b^* + 0.4\,(\text{L} - \text{G})$. **Part B**: low-signal pairs, shown for completeness and left out of every summary that uses b^* (Supplementary Note 5).

Penalty	HV	Non-dom.	n	Collapse % (95% CI)	Failed	Best	Width	Good	Local	Global	L−G
Part A: ranked pairs											

Supplementary Table S8, continued

Penalty	HV	Non-dom.	n	Collapse % (95% CI)	Failed	Best	Width	Good	Local	Global	L−G
Pancreas, batch-conditioned decoder; $b^* = 0.684$											
JS discriminator	0.255	4/10	30	30 (15–49)	0/0	3	0.5	2	0.550	0.885	−0.335
Reference critic	0.269	6/10	30	10 (2–27)	0/0	3	2.0	5	0.553	0.880	−0.327
Pooled critic	0.344	5/10	30	13 (4–31)	0/0	10	0.0	1	0.549	0.887	−0.338
Barycenter critic	0.108	2/10	30	10 (2–27)	0/0	30	3.5	8	0.549	0.887	−0.338
MMD	0.178	4/10	30	0 (0–12)	0/0	3,000	2.0*	4	0.556	0.876	−0.320
Sinkhorn	0.155	3/12	36	19 (8–36)	0/0	100	2.0	5	0.547	0.889	−0.342
Lung, batch-conditioned decoder; $b^* = 0.648$											
JS discriminator	0.050	4/10	30	67 (47–83)	0/0	0.1	0.5	2	0.505	0.861	−0.356
Reference critic	0.133	5/10	30	13 (4–31)	0/0	1	1.0	3	0.495	0.877	−0.382
Pooled critic	0.154	5/10	30	33 (17–53)	0/0	1	1.5	4	0.494	0.878	−0.384
Barycenter critic	0.093	6/10	30	10 (2–27)	0/0	100	3.5	8	0.493	0.879	−0.386
MMD	0.102	5/10	30	7 (1–22)	0/0	30	4.0	9	0.494	0.878	−0.385
Sinkhorn	0.110	5/12	36	22 (10–39)	0/0	10	3.0	7	0.492	0.882	−0.390
ATAC small, batch-conditioned decoder; $b^* = 0.746$											
JS discriminator	0.029	5/10	30	47 (28–66)	0/0	100	1.0	3	0.658	0.879	−0.221
Reference critic	0.022	5/10	30	57 (37–75)	0/0	10	0.0	1	0.652	0.887	−0.235
Pooled critic	0.047	6/10	30	47 (28–66)	0/0	10	2.0	5	0.649	0.891	−0.242
Barycenter critic	0.055	4/10	30	33 (17–53)	0/0	3	0.0	1	0.649	0.891	−0.242
MMD	0.039	6/10	30	27 (12–46)	0/0	10	4.5	10	0.639	0.907	−0.268
Sinkhorn	0.060	6/12	36	42 (26–59)	0/0	0.3	3.0	7	0.645	0.898	−0.253
ATAC large, batch-conditioned decoder; $b^* = 0.632$											
JS discriminator	0.101	5/10	30	63 (44–80)	0/0	0.3	4.5*	4	0.649	0.606	0.043
Reference critic	0.204	5/10	30	17 (6–35)	0/0	1	3.5*	6	0.648	0.607	0.041
Pooled critic	0.180	6/10	30	37 (20–56)	0/0	1	0.5	2	0.649	0.606	0.043
Barycenter critic	0.107	4/10	30	27 (12–46)	0/0	1	3.0*	4	0.647	0.608	0.039
MMD	0.163	3/10	30	0 (0–12)	0/0	1,000	4.5	10	0.646	0.610	0.036
Sinkhorn	0.122	5/12	36	11 (3–26)	0/0	100	5.5	12	0.647	0.608	0.039
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$											
JS discriminator	0.063	5/10	30	73 (54–88)	0/0	0.1	0.0	1	0.358	0.847	−0.489
Reference critic	0.103	5/10	30	27 (12–46)	0/0	0.1	0.5	2	0.359	0.845	−0.486
Pooled critic	0.112	6/10	30	37 (20–56)	0/0	0.1	0.5	2	0.361	0.843	−0.482
Barycenter critic	0.082	4/10	30	30 (15–49)	0/0	3	2.5*	5	0.359	0.846	−0.487
MMD	0.119	6/10	30	3 (0–17)	0/0	100	0.0	1	0.359	0.846	−0.487
Sinkhorn	0.118	6/12	36	25 (12–42)	0/0	10	3.0	7	0.357	0.848	−0.491
Pancreas, unconditioned decoder; $b^* = 0.492$											
JS discriminator	0.386	1/12	36	0 (0–10)	0/0	1	0.0	1	0.407	0.621	−0.214
Reference critic	0.383	1/10	30	0 (0–12)	0/0	100	1.0	3	0.420	0.601	−0.181
Pooled critic	0.440	2/10	30	0 (0–12)	0/0	30	1.0	3	0.412	0.614	−0.202
Barycenter critic	0.128	2/10	30	10 (2–27)	0/0	300	1.0	3	0.413	0.612	−0.199
MMD	0.240	3/10	30	0 (0–12)	0/0	100	2.0	5	0.403	0.627	−0.224
Sinkhorn	0.192	3/10	30	0 (0–12)	0/0	300	1.5	4	0.405	0.623	−0.218
Lung, unconditioned decoder; $b^* = 0.535$											
JS discriminator	0.103	6/12	36	17 (6–33)	0/0	0.3	3.5	8	0.418	0.711	−0.293
Reference critic	0.124	6/10	30	10 (2–27)	0/0	10	2.0*	4	0.415	0.714	−0.298
Pooled critic	0.137	6/10	30	0 (0–12)	0/0	100	4.5	10	0.416	0.713	−0.297
Barycenter critic	0.080	6/10	30	20 (8–39)	0/0	300	0.0	1	0.416	0.714	−0.298
MMD	0.114	4/10	30	3 (0–17)	0/0	300	2.0	5	0.413	0.717	−0.304
Sinkhorn	0.096	5/10	30	0 (0–12)	0/0	100	1.0	3	0.419	0.708	−0.289
Immune, unconditioned decoder; $b^* = 0.531$											
JS discriminator	0.256	7/12	36	6 (1–19)	0/0	0.3	0.0	1	0.359	0.788	−0.429
Reference critic	0.230	4/10	30	0 (0–12)	0/0	100	3.5	8	0.357	0.791	−0.434
Pooled critic	0.235	5/10	30	10 (2–27)	0/0	10	2.0*	4	0.355	0.794	−0.438
Barycenter critic	0.108	2/10	30	17 (6–35)	0/0	300	1.5	4	0.354	0.795	−0.441
MMD	0.200	4/10	30	0 (0–12)	0/0	1,000	0.0	1	0.354	0.795	−0.441
Sinkhorn	0.161	4/10	30	0 (0–12)	0/0	1,000	4.5	10	0.351	0.800	−0.449
ATAC small, unconditioned decoder; $b^* = 0.522$											
JS discriminator	0.060	7/12	36	61 (43–77)	0/0	0.3	2.0	5	0.466	0.606	−0.140
Reference critic	0.038	6/10	30	57 (37–75)	0/0	30	1.0	3	0.459	0.617	−0.159
Pooled critic	0.074	5/10	30	47 (28–66)	0/0	10	0.5	2	0.456	0.621	−0.164
Barycenter critic	0.099	3/10	30	20 (8–39)	0/0	30	1.5	4	0.457	0.620	−0.163
MMD	0.089	5/10	30	13 (4–31)	0/0	100	1.5*	3	0.449	0.631	−0.182
Sinkhorn	0.087	5/10	30	30 (15–49)	0/0	100	1.0	3	0.449	0.632	−0.182
ATAC large, unconditioned decoder; $b^* = 0.550$											
JS discriminator	0.108	5/12	36	58 (41–74)	0/0	0.03	0.0	1	0.592	0.486	0.105
Reference critic	0.295	9/10	30	53 (34–72)	0/0	1	4.0	9	0.593	0.484	0.109
Pooled critic	0.374	7/10	30	57 (37–75)	0/0	1	1.5*	3	0.594	0.483	0.111
Barycenter critic	0.099	2/10	30	30 (15–49)	0/0	3	2.5	6	0.595	0.482	0.113
MMD	0.390	5/10	30	13 (4–31)	0/0	30	3.5	8	0.592	0.486	0.106
Sinkhorn	0.085	5/10	30	20 (8–39)	0/0	0.1	3.5	8	0.586	0.495	0.091
Human–mouse immune, unconditioned decoder; $b^* = 0.450$											
JS discriminator	0.201	3/12	36	22 (10–39)	0/0	0.1	3.0	7	0.306	0.666	−0.360
Reference critic	0.261	4/10	30	3 (0–17)	0/0	300	4.0*	8	0.307	0.665	−0.357
Pooled critic	0.249	7/10	30	7 (1–22)	0/0	300	4.0	9	0.304	0.670	−0.367
Barycenter critic	0.072	5/10	30	20 (8–39)	0/0	10	1.5*	3	0.302	0.673	−0.370
MMD	0.179	7/10	30	0 (0–12)	0/0	30	2.5	6	0.301	0.675	−0.374
Sinkhorn	0.092	6/10	30	0 (0–12)	0/0	3	1.0	3	0.301	0.674	−0.372

Supplementary Table S8, continued

Penalty	HV	Non-dom.	n	Collapse % (95% CI)	Failed	Best	Width	Good	Local	Global	L−G
Simulation 1, unconditioned decoder; b^* = 0.369											
JS discriminator	0.472	1/12	36	0 (0–10)	0/0	1	0.0	1	0.221	0.591	−0.370
Reference critic	0.451	2/10	30	10 (2–27)	0/0	10	0.5	2	0.227	0.583	−0.356
Pooled critic	0.508	2/10	30	7 (1–22)	0/0	30	0.5	2	0.237	0.567	−0.330
Barycenter critic	0.138	1/10	30	20 (8–39)	0/0	30	0.5	2	0.238	0.566	−0.328
MMD	0.448	1/10	30	0 (0–12)	0/0	300	0.5	2	0.225	0.586	−0.361
Sinkhorn	0.172	2/10	30	0 (0–12)	0/0	100	2.5	6	0.224	0.587	−0.363
Simulation 2, unconditioned decoder; b^* = 0.318											
JS discriminator	0.134	3/12	36	0 (0–10)	0/0	1	0.0	1	0.209	0.482	−0.273
Reference critic	0.146	4/10	30	10 (2–27)	0/0	30	0.0	1	0.170	0.541	−0.371
Pooled critic	0.148	6/10	30	13 (4–31)	0/0	100	1.5	4	0.161	0.555	−0.394
Barycenter critic	0.072	5/10	30	13 (4–31)	0/0	10	4.0	9	0.187	0.516	−0.329
MMD	0.163	2/10	30	0 (0–12)	0/0	100	0.0	1	0.164	0.550	−0.386
Sinkhorn	0.104	4/10	30	0 (0–12)	0/0	300	2.0	5	0.187	0.515	−0.328
Part B: low-signal pairs (not ranked)											
Immune, batch-conditioned decoder; b^* = 0.648											
JS discriminator	n/a	6/11	33	21 (9–39)	0/0	0.03	1.0	3	0.478	0.902	−0.424
Reference critic	n/a	4/11	33	6 (1–20)	0/0	0.3	0.0	1	0.477	0.903	−0.426
Pooled critic	n/a	4/11	33	6 (1–20)	0/0	1	2.0*	4	0.477	0.904	−0.427
Barycenter critic	n/a	2/10	30	20 (8–39)	0/0	0.3	1.0*	2	0.475	0.907	−0.432
MMD	n/a	5/10	30	0 (0–12)	0/0	10	2.5*	4	0.474	0.908	−0.433
Sinkhorn	n/a	3/12	36	22 (10–39)	0/0	10	2.5*	3	0.472	0.911	−0.440
Simulation 1, batch-conditioned decoder; b^* = 0.815											
JS discriminator	n/a	4/10	30	50 (31–69)	0/0	1	1.0	3	0.711	0.972	−0.260
Reference critic	n/a	3/10	30	50 (31–69)	0/0	0.3	1.0	3	0.711	0.972	−0.261
Pooled critic	n/a	3/10	30	43 (25–63)	0/0	0.3	1.5	4	0.711	0.972	−0.260
Barycenter critic	n/a	3/10	30	30 (15–49)	0/0	0.3	1.5	4	0.711	0.972	−0.260
MMD	n/a	4/10	30	3 (0–17)	0/0	3	2.0*	4	0.711	0.971	−0.260
Sinkhorn	n/a	5/12	36	28 (14–45)	0/0	3	2.0	5	0.711	0.972	−0.261
Simulation 2, batch-conditioned decoder; b^* = 0.660											
JS discriminator	n/a	8/10	30	53 (34–72)	0/0	3,000	0.0	1	0.460	0.960	−0.500
Reference critic	n/a	5/10	30	27 (12–46)	0/0	0.1	2.5*	4	0.460	0.961	−0.501
Pooled critic	n/a	4/10	30	40 (23–59)	0/0	0.1	2.5*	3	0.458	0.963	−0.506
Barycenter critic	n/a	4/10	30	37 (20–56)	0/0	0.1	0.0	1	0.455	0.968	−0.512
MMD	n/a	4/10	30	20 (8–39)	0/0	0.3	0.5	2	0.457	0.964	−0.507
Sinkhorn	n/a	5/12	36	25 (12–42)	0/0	0.1	3.5*	5	0.458	0.964	−0.506

- Table S9, replication across training seeds: per penalty, task and decoder, the two bracketing strengths, the batch and biology means over seeds 0–4 at each, the batch score reached, biology at b^* on the new seeds beside the sweep reading, the replication rank beside the sweep rank, and cells that needed the extra outward strength or did not replicate the crossing; Kendall's τ_b per pair; the four large sweep gaps with their sign on the new seeds; plain scVI at seeds 0–4 beside the sweep's plain scVI per task and decoder.
- Table S10, direct biology check: per method and decoder, the mean over the six real tasks of neighbourhood purity minus plain scVI's with its seed spread, rare-type purity and label-transfer recall minus plain scVI's, and marker-gene Jaccard overlap with its difference from plain scVI (typeset after Table 3); per-task, per-cell-type and confusion tables in the repository.

Supplementary Figures

- Figure S1, simulation tasks: trade-off curves and per-task ranking at b^*, each simulation on its own.
- Figure S2, sensitivity: ranking under alternative readings (per-run reading, biology without trajectory conservation on the two immune tasks, points with a failed fit removed).
- Figure S3, label transfer: F1 per cell type, method and task at matched strength, with the gap to b^*.
- Figure S4, per-metric strength curves: each batch and biology metric against strength, per method and decoder.
- Figure S5, scaling checks.
- Figure S6, best of k settings per task: each method's expected best selection score among k settings drawn at random, $k = 1$ to 6, one panel per real task; the mean of these curves over the six real tasks is Figure 6a.

Figures and tables

400 words of captions and notes

- Figure 1: (a) model schematic: scVI encoder and decoder with or without the batch label, swappable alignment term; (b) how biology is read at matched batch removal b^*. Second cut, after Figure 5.
- Figure 2: ranking at matched batch removal: (a) heatmap of biology score at b^* (colour) and rank (number) for the six penalties in each real task, conditioned and unconditioned decoder, not-ranked pairs hatched; (b) mean rank over the real tasks that are not low-signal per penalty and decoder with the range across tasks.
- Figure 3: biology score against batch score trade-off curves, decoders by real tasks, line per method, plain scVI point, b^* line, standard tools as points.

- Figure 4: per-penalty dot plots, one point per real task and the mean: (a) hypervolume; (b) collapse rate with 95% intervals; (c) local minus global batch removal at b^*.
- Figure 5: good-strength width per penalty and decoder, one point per task. First to move to the supplement if pages run short.
- Figure 6: (a) expected best selection score against number of tries, mean over the six real tasks of each method's per-task curve; (b) batch against biology score at single strength.
- Table 1: methods: alignment term, network, update steps, sweep strengths.
- Table 2: benchmark ranking: mean rank per penalty and decoder over the real tasks that are not low-signal, range over tasks, mean biology at b^*, simulation ranks, tasks not ranked, tasks ranked.
- Table 3: single-strength placement: selection score per method, decoder, task.

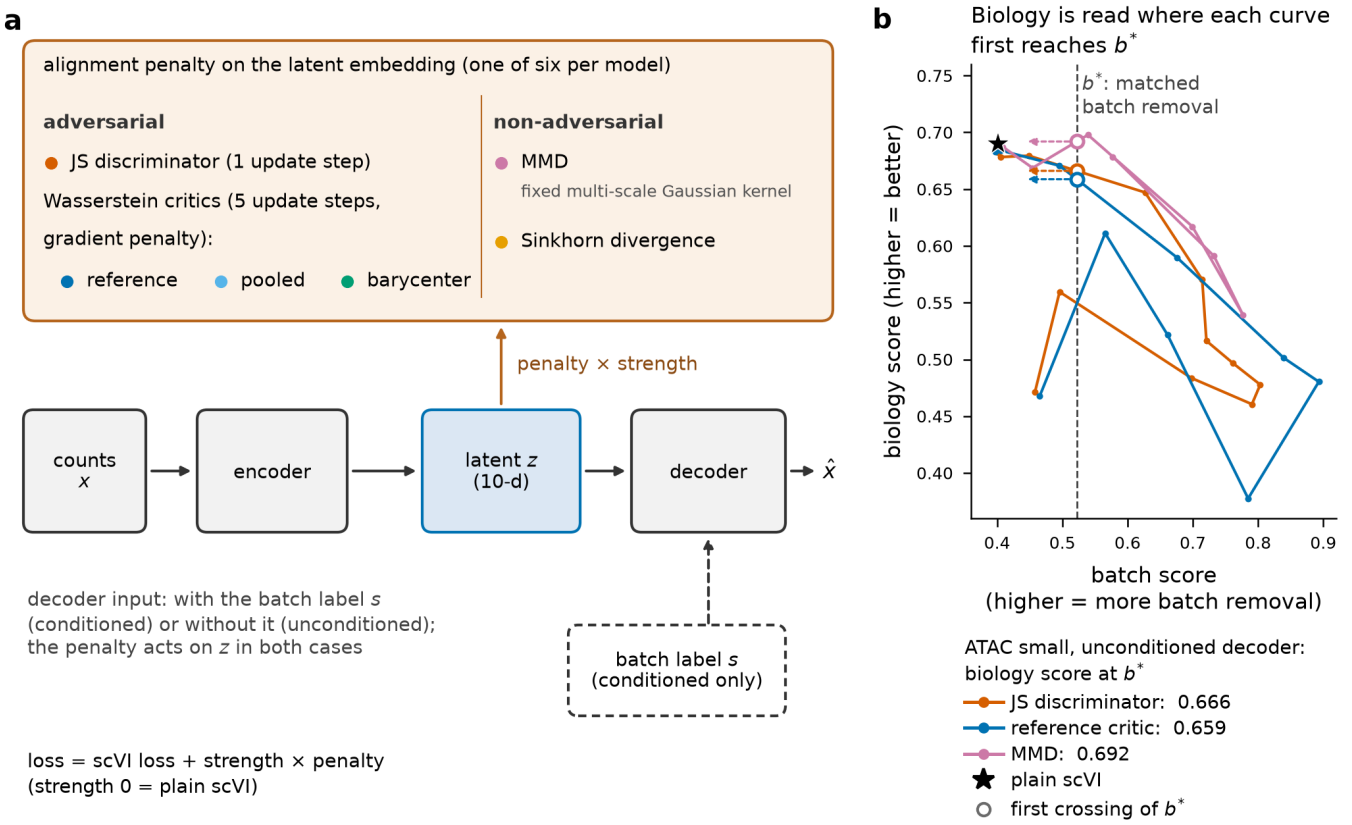


Figure 1: Model and read-out. (a) One scVI encoder–decoder is shared by all methods; the decoder receives the batch label (conditioned) or not (unconditioned), and only the alignment penalty on the 10-dimensional latent embedding is swapped. Four adversarial penalties (a Jensen–Shannon discriminator, one update step; Wasserstein critics, reference, pooled and barycenter, five update steps with a gradient penalty) and two non-adversarial penalties (MMD with a fixed multi-scale Gaussian kernel; the Sinkhorn divergence) are compared. The loss is the scVI loss plus the strength times the penalty; strength 0 is plain scVI. (b) One real task and decoder from the sweep, ATAC small with the unconditioned decoder, for three of the six penalties (colours as in Figures 2–6). Each curve joins plain scVI (star) and the sweep strengths in increasing order (10 strengths for the reference critic and MMD, 12 for the discriminator, which has two further low strengths); each point is a mean over the three seeds. Biology is read where the curve first reaches the matched batch removal b^* (dashed line, 0.522), interpolating between neighbouring strengths (open circles; dotted arrows read the biology score): JS discriminator 0.666, reference critic 0.659, MMD 0.692. The discriminator and reference-critic curves cross b^* a second time at higher strengths; the reading uses the first crossing.

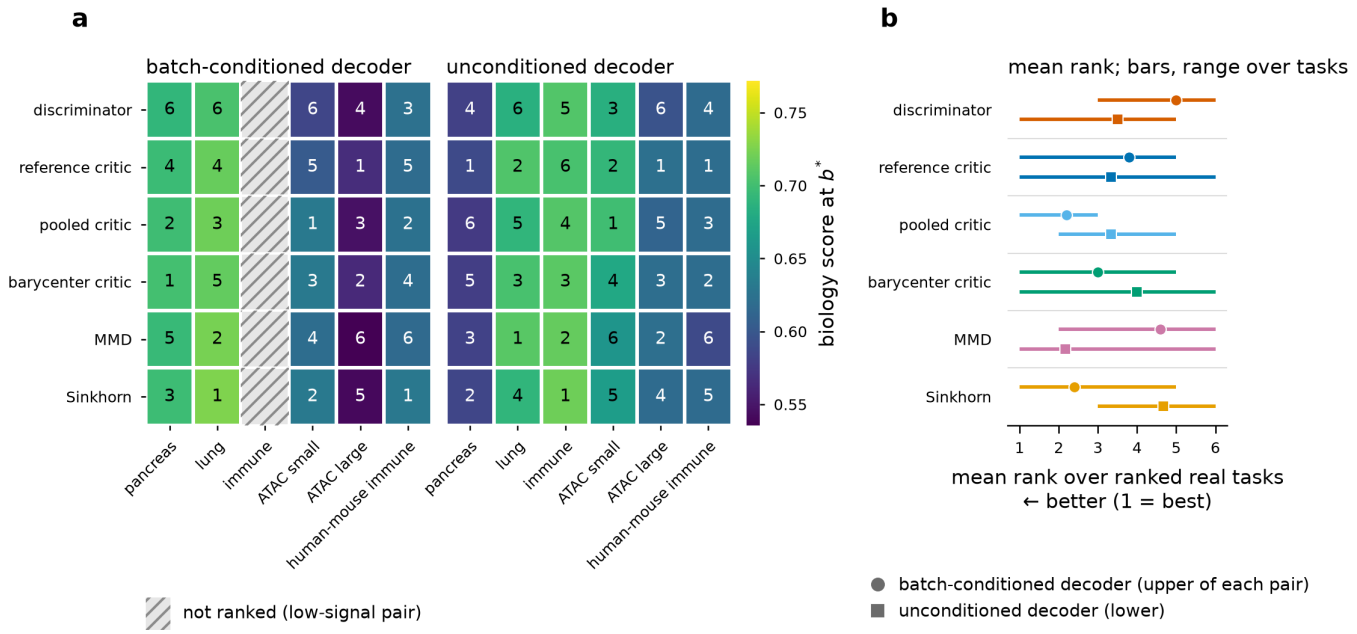


Figure 2: Ranking of the six penalties by biology kept at matched batch removal b^*. (a) Biology score at b^* (colour) and rank among the six penalties within each task column (number; tied penalties share the average rank) for each real task, with the batch-conditioned and the unconditioned decoder. Hatched cells: the immune task with the batch-conditioned decoder, a low-signal pair that is not ranked. A penalty whose curve does not reach b^* would take the bottom rank; every curve reached it here. (b) Mean rank over the ranked real tasks (five with the batch-conditioned decoder, six with the unconditioned decoder) per penalty and decoder; rank 1 is best; bars span the lowest and highest rank over those tasks. The pooled critic had the best mean rank with the batch-conditioned decoder (2.2) and MMD with the unconditioned decoder (2.17). The leave-one-seed-out standard error of each reading in (a) and the range of each rank over the three leave-one-seed-out repeats are in Supplementary Table S6. Simulations are ranked separately (Supplementary Figure S1).

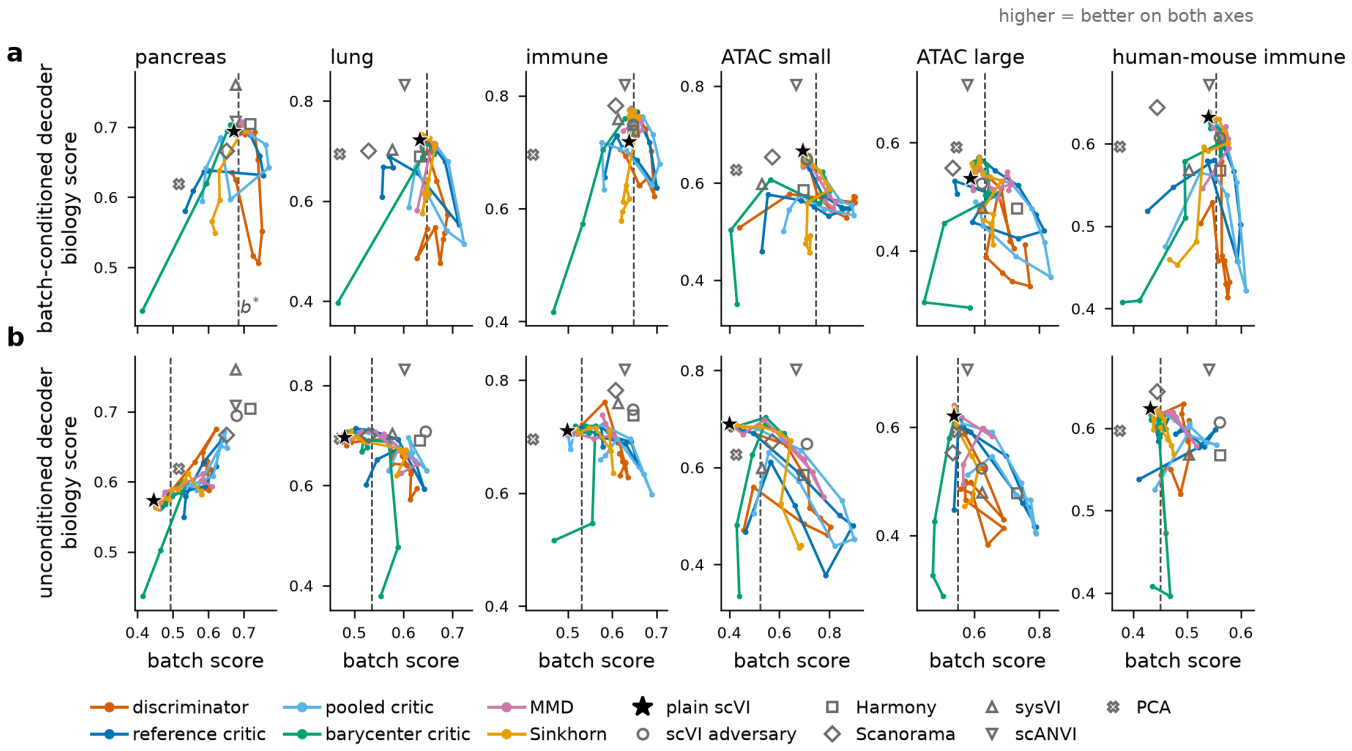


Figure 3: Batch score against biology score for the six real tasks (columns) with the batch-conditioned decoder (a) and the unconditioned decoder (b). Each line joins plain scVI (star) and one penalty's fitted strengths in increasing order, by mean scores over three seeds; no real-task curve ended early because of failed fits. Dashed line, matched batch removal b^*. Grey open symbols, each standard tool at its best setting by selection score (0.4 batch plus 0.6 biology) among the settings run; scANVI uses cell-type labels. Panels in a column share axes.

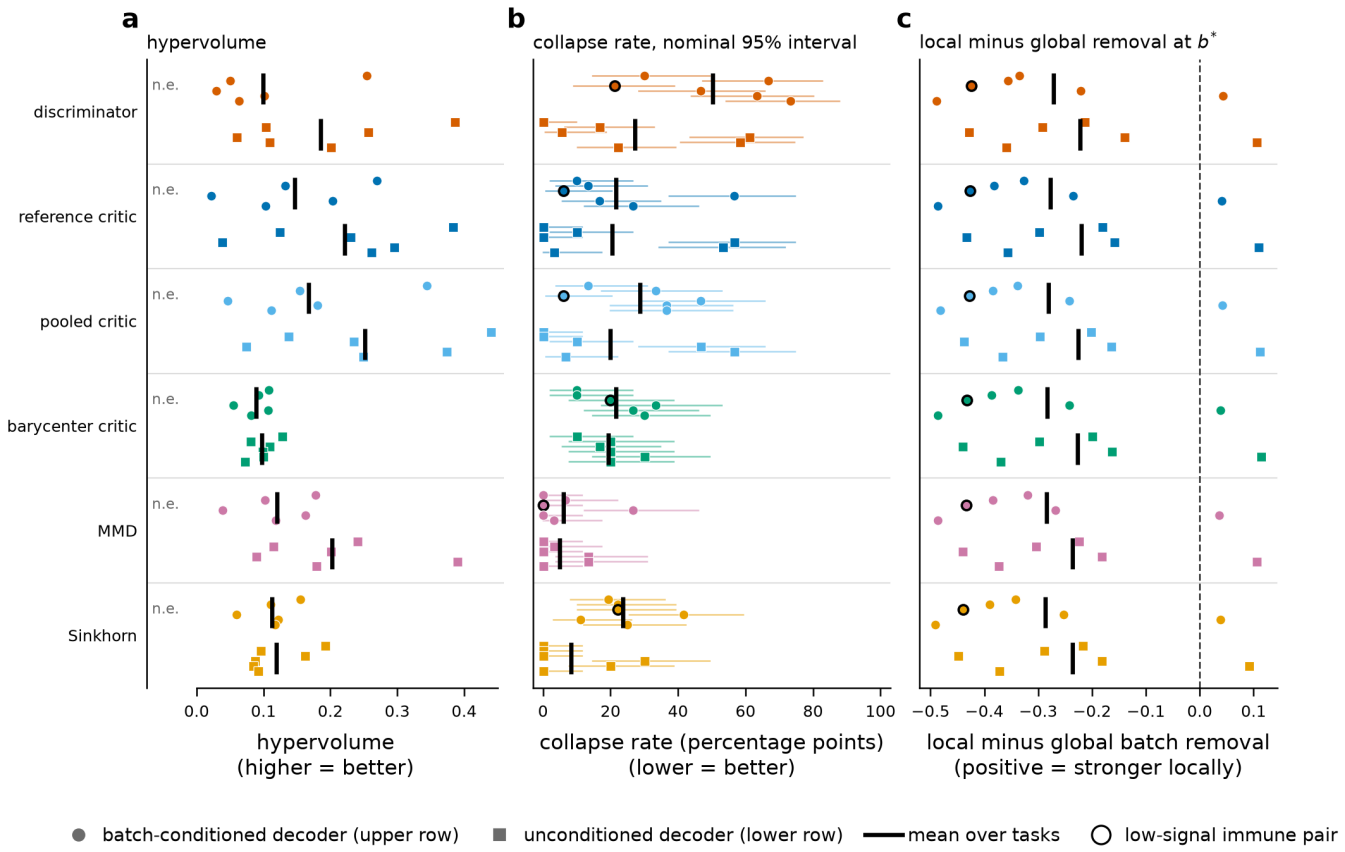


Figure 4: Secondary summaries per penalty with the batch-conditioned (circles, upper row of each band) and unconditioned decoder (squares, lower row). One point per real task, in the order of Figure 3 from top to bottom within each row; black tick, mean over tasks. (a) Hypervolume of the strength curve (higher is better); it is not evaluable (n.e.) for the low-signal immune pair with the batch-conditioned decoder, so that point is absent and the mean is over five tasks. (b) Collapse rate in percentage points with nominal 95% intervals (lower is better); the mean includes the low-signal pair. (c) Local minus global batch removal at b^*; positive means stronger local removal; the mean excludes the low-signal pair. Ringed point, the low-signal immune pair with the batch-conditioned decoder.

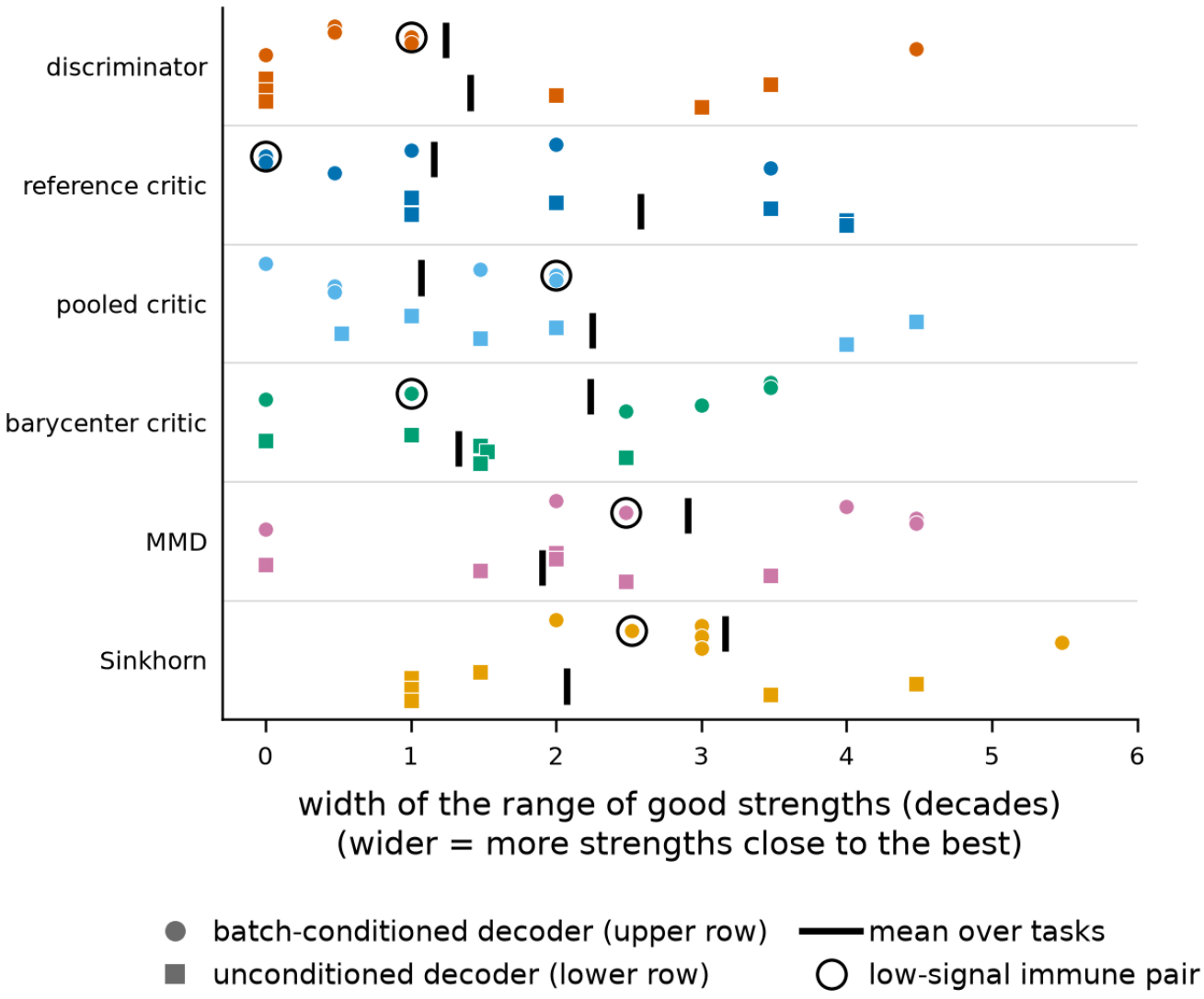


Figure 5: Width of each penalty's range of good strengths, in decades, one point per real task and decoder (batch-conditioned decoder, circles, upper row of each band; unconditioned decoder, squares, lower row; tasks in the order of Figure 3 from top to bottom; black tick, mean over tasks). A good strength has a mean selection score at least the lower 95% confidence bound of the best strength; a width of zero means a single qualifying strength. Ringed point, the low-signal immune pair with the batch-conditioned decoder, included as in the Results text.

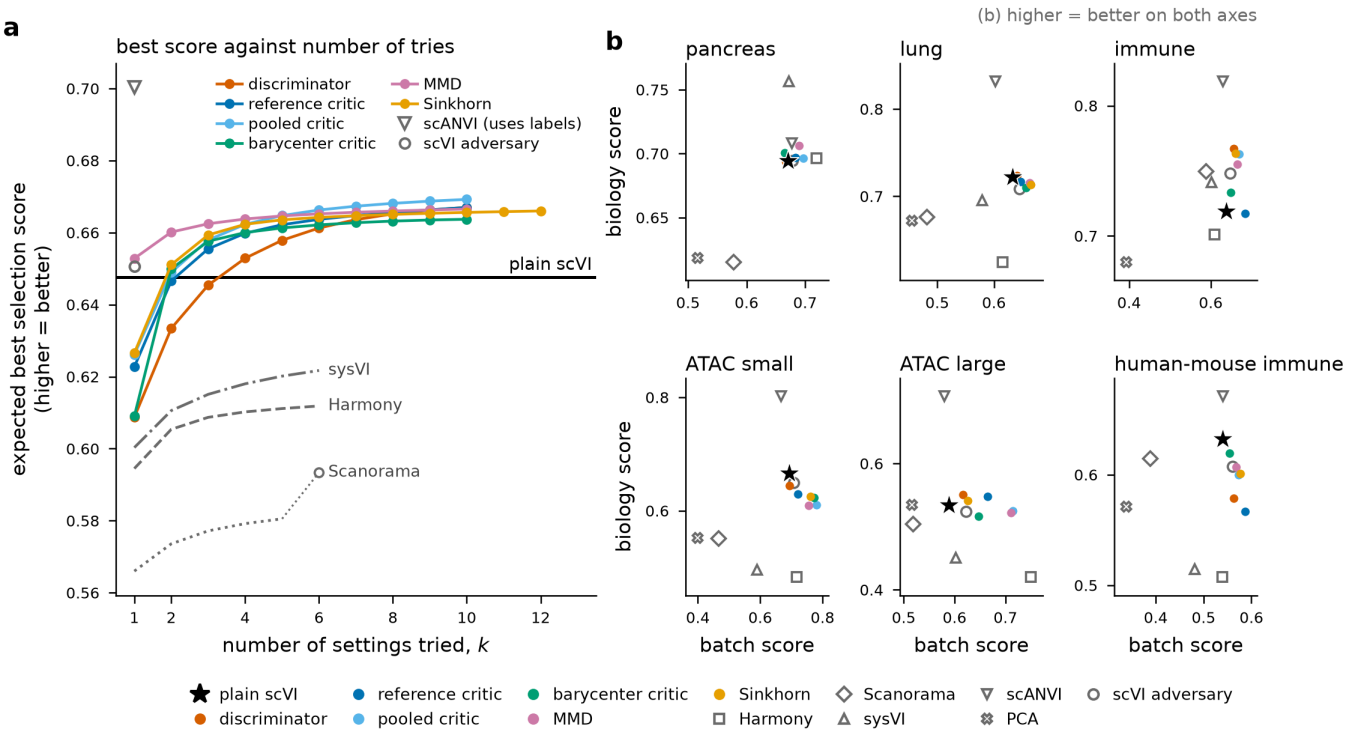


Figure 6: Placement against plain scVI and standard tools with the batch-conditioned decoder. (a) Expected best selection score among k settings drawn at random, mean over the six real tasks of each method's per-task curve (Supplementary Figure S6; selection score: 0.4 batch plus 0.6 biology; optimistic, because the best setting is chosen and scored on the same data). Each curve is drawn up to the largest k for which all six tasks have k settings: $k = 10$ for five penalties (the discriminator, reference critic and pooled critic have an eleventh setting on the immune task only) and $k = 12$ for the Sinkhorn divergence, and $k = 6$ for Harmony and sysVI. Scanorama has six settings on five tasks and five on the large chromatin-accessibility task (80 neighbours could not be fitted); it is drawn to $k = 5$, and its $k = 6$ value (open marker) is the mean of five tasks. The Results compare methods at $k = 1$ and $k = 6$, the number of settings of Harmony, Scanorama and sysVI; the penalty curves beyond $k = 6$ are shown for completeness. Horizontal line, plain scVI (0.648); scANVI (uses cell-type labels) and the scVI adversary have one setting. At $k = 1$ every penalty except MMD scored below plain scVI; the other penalties first exceeded plain scVI at $k = 2$ (pooled critic, barycenter critic, Sinkhorn divergence), 3 (reference critic) and 4 (discriminator). (b) Batch against biology score for each real task at the single strength (penalties, Harmony, Scanorama, sysVI) chosen on other task families, or at the fixed setting (scANVI, scVI adversary, PCA with 10 dimensions); star, plain scVI. The unconditioned decoder is in Supplementary Table S7.

Table 1: Alignment methods compared on the shared scVI backbone. Each penalty is multiplied by a fixed strength and added to the scVI loss. W1, Wasserstein-1; JS, Jensen–Shannon; MMD, maximum mean discrepancy; OT, optimal transport; KL, Kullback–Leibler. Equations in Supplementary Note 1; training settings in Supplementary Note 2 and Table S2.

Method	Alignment term	Network	Adversary steps per encoder step	Gradient penalty or regularisation	Sweep strengths (Methods)
JS discriminator	scvi-tools fool loss (JS-type): encoder trained to make the batch classifier wrong	batch classifier, two hidden layers of 128 units, one output per batch	1	none (plain)	0.1–3,000; also 0.01 and 0.03 with the unconditioned decoder
Reference critic	W1 distance of each batch to one reference batch (highest cell-type entropy)	critic, two hidden layers of 128 units, one output per batch	5	gradient penalty, weight 10	0.1–3,000
Pooled critic	W1 distance of each batch to all other batches pooled	critic, as above	5	gradient penalty, weight 10	0.1–3,000
Barycenter critic	W1 distance of each batch to the minibatch Wasserstein barycenter (an average distribution)	critic, as above	5	gradient penalty, weight 10	0.1–3,000
MMD	kernel two-sample distance between batches, multi-scale Gaussian kernel	none	–	none	0.1–3,000
Sinkhorn	debiased Sinkhorn divergence (entropy-regularised OT); not W1 at this smoothing	none	–	entropic smoothing (Supplementary Table S2)	0.1–3,000; also 10,000 and 30,000 with the conditioned decoder
scvi-tools adversary	scvi-tools adversarial batch classifier (reference method, conditioned decoder only)	batch classifier (scvi-tools default, two hidden layers of 32 units)	1	none; its weight is one minus the KL weight	no strength (weight fixed)

Table 2: Benchmark ranking of the six penalties by biology score at matched batch removal b^*. Mean rank over the real tasks that are not low-signal (rank 1 is best; tied penalties share the average rank, and a penalty whose curve does not reach b^* takes the bottom rank) and its range over tasks; mean biology score at b^*; ranks in simulation 1 and simulation 2, which are ranked separately; real tasks not ranked (low-signal) and number of real tasks ranked, of six. Per-task values are in Supplementary Table S6. Range, lowest and highest rank over the ranked real tasks; biology, mean over those tasks of the biology score at b^*; the simulations are low-signal with the conditioned decoder (–); ranked, real tasks ranked of six.

Penalty	Decoder	Mean rank	Range	Biology	Sim. 1 rank	Sim. 2 rank	Not ranked	Ranked
JS discriminator	conditioned	5.00	3–6	0.629	–	–	immune	5/6
	unconditioned	3.50	1–5	0.649	1	1	none	6/6
Reference critic	conditioned	3.80	1–5	0.640	–	–	immune	5/6
	unconditioned	3.33	1–6	0.646	2	4	none	6/6
Pooled critic	conditioned	2.20	1–3	0.644	–	–	immune	5/6
	unconditioned	3.33	2–5	0.651	5	2	none	6/6
Barycenter critic	conditioned	3.00	1–5	0.646	–	–	immune	5/6
	unconditioned	4.00	1–6	0.650	6	5	none	6/6
MMD	conditioned	4.60	2–6	0.638	–	–	immune	5/6
	unconditioned	2.17	1–6	0.655	3	3	none	6/6
Sinkhorn	conditioned	2.40	1–5	0.645	–	–	immune	5/6
	unconditioned	4.67	3–6	0.645	4	6	none	6/6

Table 3: Single-strength placement: mean selection score over seeds (three for plain scVI and the penalties; five for the scvi-tools adversary, scANVI and sysVI; one run for Harmony, Scanorama and PCA) (0.4×batch + 0.6×biology) per method and real task, conditioned decoder, at the strength with the best mean selection score on real tasks of other families (related tasks never choose for each other). Values in parentheses use the task's own best strength (optimistic bound). Harmony, Scanorama and sysVI are placed by the same rule, at the setting chosen on real tasks of other families, with the task's own best setting in parentheses; scANVI and principal-component analysis (PCA) run at one setting, so no choice is made; scANVI uses cell-type labels. Unconditioned decoder in Supplementary Table S7. Scanorama on the large chromatin-accessibility task has five of its six settings (80 neighbours could not be fitted). The immune task is low-signal with this decoder. Values are rounded to three decimals: the Sinkhorn divergence on pancreas (0.6846) is 0.0004 below plain scVI (0.6850) and counts as not exceeding it.

Method	Pancreas	Lung	Immune	ATAC small	ATAC large	Human–mouse immune
Plain scVI	0.685	0.686	0.686	0.677	0.556	0.595
scvi-tools adversary	0.688	0.683	0.708	0.673	0.564	0.588
JS discriminator	0.683 (0.708)	0.690 (0.690)	0.723 (0.724)	0.664 (0.703)	0.577 (0.582)	0.573 (0.595)
Reference critic	0.692 (0.700)	0.689 (0.690)	0.704 (0.725)	0.666 (0.682)	0.595 (0.606)	0.575 (0.598)
Pooled critic	0.697 (0.709)	0.694 (0.694)	0.725 (0.725)	0.679 (0.689)	0.601 (0.601)	0.589 (0.597)
Barycenter critic	0.687 (0.700)	0.689 (0.693)	0.700 (0.721)	0.683 (0.683)	0.569 (0.591)	0.593 (0.594)
MMD	0.700 (0.702)	0.694 (0.694)	0.719 (0.721)	0.668 (0.672)	0.598 (0.608)	0.591 (0.601)
Sinkhorn	0.685 (0.701)	0.694 (0.697)	0.722 (0.724)	0.679 (0.683)	0.575 (0.591)	0.591 (0.602)
Harmony	0.706 (0.710)	0.621 (0.621)	0.664 (0.664)	0.576 (0.587)	0.552 (0.558)	0.520 (0.531)
Scanorama	0.600 (0.613)	0.598 (0.600)	0.685 (0.692)	0.517 (0.538)	0.510 (0.522)	0.524 (0.525)
sysVI	0.723 (0.726)	0.649 (0.653)	0.685 (0.700)	0.533 (0.571)	0.511 (0.537)	0.501 (0.542)
scANVI (labels)	0.696	0.740	0.743	0.748	0.656	0.619
PCA (uncorrected)	0.577	0.585	0.565	0.490	0.527	0.478

Supplementary Table S10: Direct biology check (exploratory; rules written after the sweep scores had been read; Supplementary Note 12). Means over the six real tasks of per-task seed means. Purity and rare-type columns are differences from plain scVI (penalties and plain scVI: same decoder and seed; baselines: conditioned plain scVI, mean of seeds 100–102). Seed SD, mean over the six real tasks of the SD of the purity difference over seeds (n/a for one-run tools). Rare types on the four real tasks that have them; marker-gene Jaccard on the four RNA tasks. Baselines with several settings: mean over settings (number of settings in parentheses; Scanorama has six on the large chromatin-accessibility task). scANVI uses cell-type labels.

Method	Decoder	Purity diff.	Seed SD	Rare purity diff.	Rare recall diff.	Marker Jaccard	Jaccard diff.
JS discriminator	conditioned	−0.052	0.030	−0.107	−0.121	0.398	0.000
	unconditioned	−0.077	0.028	−0.076	−0.129	0.392	+0.001
Reference critic	conditioned	−0.004	0.008	−0.002	+0.005	0.399	0.000
	unconditioned	−0.011	0.011	+0.002	+0.008	0.394	+0.003
Pooled critic	conditioned	−0.019	0.008	−0.024	−0.036	0.398	−0.001
	unconditioned	+0.001	0.005	+0.010	+0.011	0.390	−0.001
Barycenter critic	conditioned	−0.007	0.007	−0.004	+0.001	0.395	−0.003
	unconditioned	+0.001	0.007	+0.010	+0.005	0.391	0.000
MMD	conditioned	−0.001	0.006	+0.003	+0.007	0.397	−0.001
	unconditioned	−0.004	0.005	+0.001	+0.003	0.394	+0.003
Sinkhorn	conditioned	−0.022	0.008	+0.006	+0.002	0.394	−0.005
	unconditioned	−0.033	0.008	−0.045	−0.079	0.397	+0.006
Plain scVI	conditioned	0.000	0.000	0.000	0.000	0.399	0.000
	unconditioned	0.000	0.000	0.000	0.000	0.391	0.000
scvi-tools adversary (1)	conditioned	−0.007	0.009	−0.001	+0.013	0.398	−0.001
scANVI (labels) (1)	–	+0.115	0.003	+0.069	+0.061	0.390	−0.008
sysVI (6)	–	−0.040	0.023	−0.045	−0.078	0.392	−0.007
Harmony (7)	–	−0.141	n/a	−0.234	−0.330	0.367	−0.031
Scanorama (7)	–	−0.049	n/a	−0.076	−0.087	0.382	−0.017
PCA (2)	–	−0.021	n/a	−0.062	−0.088	0.371	−0.028

References

Matthew Amodio, David van Dijk, Krishnan Srinivasan, William S. Chen, Hussein Mohsen, Kevin R. Moon, Allison Campbell, Yujiao Zhao, Xiaomei Wang, Manjunatha Venkataswamy, Anita Desai, V. Ravi, Priti Kumar, Ruth Montgomery, Guy Wolf, and Smita Krishnaswamy. Exploring single-cell data with deep multitasking neural networks. *Nature Methods*, 16(11):1139–1145, 2019. doi: 10.1038/s41592-019-0576-7.

Martin Arjovsky, Soumith Chintala, and Léon Bottou. Wasserstein generative adversarial networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 214–223. PMLR, 2017. URL https://proceedings.mlr.press/v70/arjovsky17a.html.

Kai Cao, Qiyu Gong, Yiguang Hong, and Lin Wan. A unified computational framework for single-cell data integration with optimal transport. *Nature Communications*, 13(1):7419, 2022. doi: 10.1038/s41467-022-35094-8.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc., 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/file/af21d0c97db2e27e13572cbf59eb343d-Paper.pdf.

Marco Cuturi and Arnaud Doucet. Fast computation of Wasserstein barycenters. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 685–693. PMLR, 2014. URL https://proceedings.mlr.press/v32/cuturi14.html.

Reut Danino, Iftach Nachman, and Roded Sharan. Batch correction of single-cell sequencing data via an autoencoder architecture. *Bioinformatics Advances*, 4(1):vbad186, 2024. doi: 10.1093/bioadv/vbad186.

Ayse B Dincer, Joseph D Janizek, and Su-In Lee. Adversarial deconfounding autoencoder for learning robust gene expression embeddings. *Bioinformatics*, 36(Supplement 2):i573–i582, 2020. doi: 10.1093/bioinformatics/btaa796.

Jean Feydy, Thibault Séjourné, François-Xavier Vialard, Shun-ichi Amari, Alain Trouve, and Gabriel Peyré. Interpolating between optimal transport and MMD using Sinkhorn divergences. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 2681–2690. PMLR, 2019. URL https://proceedings.mlr.press/v89/feydy19a.html.

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016. URL https://jmlr.org/papers/v17/15-239.html.

Adam Gayoso, Romain Lopez, Galen Xing, Pierre Boyeau, Valeh Valiollah Pour Amiri, Justin Hong, Katherine Wu, Michael Jayasuriya, Edouard Mehlman, Maxime Langevin, Yining Liu, Jules Samaran, Gabriel Misrachi, Achille Nazaret, Oscar Clivio, Chenling Xu, Tal Ashuach, Mariano Gabitto, Mohammad Lotfollahi, Valentine Svensson, Eduardo da Veiga Beltrame, Vitalii Kleshchevnikov, Carlos Talavera-López, Lior Pachter, Fabian J. Theis, Aaron Streets, Michael I. Jordan, Jeffrey Regier, and Nir Yosef. A Python library for probabilistic analysis of single-cell omics data. *Nature Biotechnology*, 40(2):163–166, 2022. doi: 10.1038/s41587-021-01206-w.

Aude Genevay, Gabriel Peyré, and Marco Cuturi. Learning generative models with Sinkhorn divergences. In *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 1608–1617. PMLR, 2018. URL https://proceedings.mlr.press/v84/genevay18a.html.

Wilson Wen Bin Goh, Chern Han Yong, and Limsoon Wong. Are batch effects still relevant in the age of big data? *Trends in Biotechnology*, 40(9):1029–1040, 2022. doi: 10.1016/j.tibtech.2022.02.005.

Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Paper.pdf.

Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012. URL https://jmlr.org/papers/v13/gretton12a.html.

Weizhen Gu, Han Zhang, and Mingjing Han. scStarCorrect: a StarGAN-based adversarial model for reference-guided batch correction in single-cell RNA-Seq data. In *2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 921–926. IEEE, 2025. doi: 10.1109/bibm66473.2025.11356167.

Ishaan Gulrajani, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron Courville. Improved training of Wasserstein GANs. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/892c3b1c6dccd52936e27cbd0ff683d6-Paper.pdf.

Brian Hie, Bryan Bryson, and Bonnie Berger. Efficient integration of heterogeneous single-cell transcriptomes using Scanorama. *Nature Biotechnology*, 37(6):685–691, 2019. doi: 10.1038/s41587-019-0113-3.

Karin Hrovatin, Amir Ali Moinfar, Luke Zappia, Shrey Parikh, Alejandro Tejada Lapuerta, Ben Lengerich, Manolis Kellis, and Fabian J. Theis. Integrating single-cell RNA-seq datasets with substantial batch effects. *BMC Genomics*, 26(1):974, 2025. doi: 10.1186/s12864-025-12126-3.

Chenghan Jiang, Ren Zheng, Yongle Ji, Shubin Cao, Yanhua Fang, Zhe Wang, Ruoyu Wang, Shanshan Liang, and Shuai Tao. Geometric-aware and interpretable deep learning for single-cell batch correction via explicit disentanglement and optimal transport. *bioRxiv*, 2026. doi: 10.64898/2026.02.17.706490. Preprint, posted 2026-02-20.

Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

Jeffrey T. Leek, Robert B. Scharpf, Héctor Corrada Bravo, David Simcha, Benjamin Langmead, W. Evan Johnson, Donald Geman, Keith Baggerly, and Rafael A. Irizarry. Tackling the widespread and critical impact of batch effects in high-throughput data. *Nature Reviews Genetics*, 11(10):733–739, 2010. doi: 10.1038/nrg2825.

Hui Li, Davis J. McCarthy, Heejung Shim, and Susan Wei. Trade-off between conservation of biological variation and batch effect removal in deep generative modeling for single-cell transcriptomics. *BMC Bioinformatics*, 23(1):460, 2022. doi: 10.1186/s12859-022-05003-3.

Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

Mohammad Lotfollahi, Mohsen Naghipourfar, Fabian J Theis, and F Alexander Wolf. Conditional out-of-distribution generation for unpaired data using transfer VAE. *Bioinformatics*, 36(Supplement 2):i610–i617, 2020. doi: 10.1093/bioinformatics/btaa800.

Malte D. Luecken, M. Büttner, K. Chaichoompu, A. Danese, M. Interlandi, M. F. Mueller, D. C. Strobl, L. Zappia, M. Dugas, M. Colomé-Tatché, and Fabian J. Theis. Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50, 2022. doi: 10.1038/s41592-021-01336-8.

Malte D. Luecken, Scott Gigante, Daniel B. Burkhardt, Robrecht Cannoodt, Daniel C. Strobl, Nikolay S. Markov, Luke Zappia, Giovanni Palla, Wesley Lewis, Daniel Dimitrov, Michael E. Vinyard, D. S. Magruder, Michaela F. Mueller, Alma Andersson, Emma Dann, Qian Qin, Dominik J. Otto, Michal Klein, Olga Borisovna Botvinnik, Louise Deconinck, Kai Waldrant, Sai Nirmayi Yasa, Artur Szałata, Andrew Benz, Zhijian Li, Open Problems Jamboree Members, Bastian Rieck, Constantin Ahlmann-Eltze, Eduardo da Veiga Beltrame, Carmen Bravo González-Blas, Ann T. Chen, Benjamin DeMeo, Can Ergen, Swann Floc'hlay, Adam Gayoso, Stephanie Hicks, Yuge Ji, Vitalii Kleshchevnikov, Gioele La Manno, Maximilian G. Lombardo, Romain Lopez, Dario Righelli, Hirak Sarkar, Valentine Svensson, Alexander Tong, Galen Xing, Chenling Xu, Jonathan M. Bloom, Angela Oliveira Pisco, Julio Saez-Rodriguez, Drausin Wulsin, Luca Pinello, Yvan Saeys, Fabian J. Theis, and Smita Krishnaswamy. Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology*, 43(7):1035–1040, 2025. doi: 10.1038/s41587-025-02694-w.

Hassaan Maan, Lin Zhang, Chengxin Yu, Michael J. Geuenich, Kieran R. Campbell, and Bo Wang. Characterizing the impacts of dataset imbalance on single-cell data integration. *Nature Biotechnology*, 42(12):1899–1908, 2024. doi: 10.1038/s41587-023-02097-9.

Lily Monnier and Paul-Henry Cournède. A novel batch-effect correction method for scRNA-seq data based on Adversarial Information Factorization. *PLOS Computational Biology*, 20(2):e1011880, 2024. doi: 10.1371/journal.pcbi.1011880.

Uri Shaham, Kelly P Stanton, Jun Zhao, Huamin Li, Khadir Raddassi, Ruth Montgomery, and Yuval Kluger. Removal of batch effects using distribution-matching residual networks. *Bioinformatics*, 33(16):2539–2546, 2017. doi: 10.1093/bioinformatics/btx196.

Hoa Thi Nhu Tran, Kok Siong Ang, Marion Chevrier, Xiaomeng Zhang, Nicole Yee Shin Lee, Michelle Goh, and Jinmiao Chen. A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21(1):12, 2020. doi: 10.1186/s13059-019-1850-9.

Po-Yuan Tung, John D. Blischak, Chiaowen Joyce Hsiao, David A. Knowles, Jonathan E. Burnett, Jonathan K. Pritchard, and Yoav Gilad. Batch effects and the effective design of single-cell gene expression studies. *Scientific Reports*, 7(1):39921, 2017. doi: 10.1038/srep39921.

Dongfang Wang, Siyu Hou, Lei Zhang, Xiliang Wang, Baolin Liu, and Zemin Zhang. iMAP: integration of multiple single-cell datasets by adversarial paired transfer networks. *Genome Biology*, 22(1):63, 2021. doi: 10.1186/s13059-021-02280-8.

Xun Wang, Chaogang Zhang, Ying Zhang, Xiangyu Meng, Zhiyuan Zhang, Xin Shi, and Tao Song. IMGG: integrating multiple single-cell datasets through connected graphs and generative adversarial networks. *International Journal of Molecular Sciences*, 23(4):2082, 2022a. doi: 10.3390/ijms23042082.

Xun Wang, Chaogang Zhang, Lulu Wang, and Pan Zheng. Integrating multiple single-cell RNA sequencing datasets using adversarial autoencoders. *International Journal of Molecular Sciences*, 24(6):5502, 2023. doi: 10.3390/ijms24065502.

Yuge Wang, Tianyu Liu, and Hongyu Zhao. ResPAN: a powerful batch correction model for scRNA-seq data through residual adversarial networks. *Bioinformatics*, 38(16):3942–3949, 2022b. doi: 10.1093/bioinformatics/btac427.

Yifan Wu, Ezra Winston, Divyansh Kaushik, and Zachary Lipton. Domain adaptation with asymmetrically-relaxed distribution alignment. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6872–6881. PMLR, 2019. URL `https://proceedings.mlr.press/v97/wu19f.html`.

Chenling Xu, Romain Lopez, Edouard Mehlman, Jeffrey Regier, Michael I Jordan, and Nir Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17(1):MSB20209620, 2021. doi: 10.15252/msb.20209620.

Han Xu, Yusen Ye, Ran Duan, Yong Gao, Yuxuan Hu, and Lin Gao. Beaconet: a reference-free method for integrating multiple batches of single-cell transcriptomic data in original molecular space. *Advanced Science*, 11(26):2306770, 2024. doi: 10.1002/advs.202306770.

Chenxin Yi, Jinyu Cheng, Jiajun Chen, Wanquan Liu, Junwei Liu, and Yixue Li. Benchmarking deep learning methods for biologically conserved single-cell integration. *Genome Biology*, 26(1):398, 2025. doi: 10.1186/s13059-025-03869-z.

Ying Yu, Yuanbang Mai, Yuanting Zheng, and Leming Shi. Assessing and mitigating batch effects in large-scale omics studies. *Genome Biology*, 25(1):254, 2024. doi: 10.1186/s13059-024-03401-9.

Han Zhao, Remi Tachet Des Combes, Kun Zhang, and Geoffrey Gordon. On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 7523–7532. PMLR, 2019. URL `https://proceedings.mlr.press/v97/zhao19a.html`.

Jia Zhao, Gefei Wang, Jingsi Ming, Zhixiang Lin, Yang Wang, The Tabula Microcebus Consortium, Angela Ruohao Wu, and Can Yang. Adversarial domain translation networks for integrating large-scale atlas-level single-cell datasets. *Nature Computational Science*, 2(5):317–330, 2022. doi: 10.1038/s43588-022-00251-y.

Eckart Zitzler and Lothar Thiele. Multiobjective optimization using evolutionary algorithms – a comparative case study. In *Parallel Problem Solving from Nature – PPSN V*, volume 1498 of *Lecture Notes in Computer Science*, pages 292–301, Berlin, Heidelberg, 1998. Springer. doi: 10.1007/BFb0056872.

Supplementary content (Note 1, Tables S6, S7 and S9, Figures S1–S6)

Supplementary Note 1. Model and loss equations.

Notation. A cell i has counts x_{ig} for genes $g = 1, \dots, G$ (2,000 highly variable genes, or all features for the chromatin-accessibility tasks), a batch label $s_i \in \{1, \dots, K\}$ and a latent vector $z_i \in \mathbb{R}^d$ with $d = 10$. The encoder gives $q(z_i \mid x_i) = \mathcal{N}(\mu_i, \operatorname{diag} \sigma_i^2)$. $\mathcal{B}$ is a minibatch of 128 cells, $\mathcal{B}_k$ its cells from batch k and $\mathcal{B}_{-k}$ its cells from the other batches. $K_{\mathcal{B}}$ is the set of batches for which both $\mathcal{B}_k$ and $\mathcal{B}_{-k}$ are non-empty; a minibatch with no such batch adds zero to the penalty. Every penalty is applied to the posterior mean, $u_i = \mu_i$ (no per-dimension standardisation); $\hat{P}_k$ is the empirical distribution of $\{u_i : i \in \mathcal{B}_k\}$ and $\hat{P}_{-k}$ that of $\{u_i : i \in \mathcal{B}_{-k}\}$. The strength is $\lambda \geq 0$; plain scVI is $\lambda = 0$.

Encoder objective. For all methods the encoder and decoder minimise, over each minibatch,

$$\mathcal{L}_G \;=\; \mathcal{L}_{\text{scVI}} \;+\; \lambda\, \mathcal{A}(u_{\mathcal{B}}), \qquad \mathcal{L}_{\text{scVI}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \Big[- \mathbb{E}_q \sum_g \log p_{\text{ZINB}}(x_{ig} \mid \mu_{ig}, \theta_g, \pi_{ig}) + \beta_e \operatorname{KL}\big(q(z_i \mid x_i) \,\|\, \mathcal{N}(0, I_d)\big) \Big], \tag{4}$$

where $\mathcal{A}$ is the alignment term of the penalty (below), the expectation is over the posterior of the latent, and the likelihood is the zero-inflated negative binomial

$$p_{\text{ZINB}}(x \mid \mu, \theta, \pi) = \pi\, \mathbf{1}[x = 0] + (1 - \pi)\, \frac{\Gamma(x + \theta)}{\Gamma(\theta)\, x!} \Big(\frac{\theta}{\theta + \mu} \Big)^{\theta} \Big(\frac{\mu}{\theta + \mu} \Big)^{x}, \qquad \mu_{ig} = \ell_i\, \rho_g(z_i, s_i), \quad \ell_i = \textstyle\sum_{g'} x_{ig'}, \tag{5}$$

with ℓ_i the observed library size (scvi-tools' default), $\rho(z_i, s_i)$ the decoder's softmax over genes, θ_g a per-gene dispersion and π_{ig} the dropout probability produced by the decoder. The encoder reads $\log(1 + x_i)$. In the batch-conditioned setting the decoder receives the one-hot batch label s_i; in the unconditioned setting no batch label reaches the encoder or the decoder. The weight of the latent KL term follows scvi-tools' linear warm-up, $\beta_e = e/400$ for epochs $e < 400$ and 1 afterwards (the final value for each task is in Table S2). The penalty acts from the first step and is not scaled by the warm-up. The adversary heads below are trained on detached $u_{\mathcal{B}}$, and only the encoder gradient of $\mathcal{A}$ enters $\mathcal{L}_G$.

Jensen–Shannon-type discriminator. A classifier head $h : \mathbb{R}^d \to \mathbb{R}^K$ (two hidden layers of 128 units) is trained with one step per encoder step to predict the batch,

$$\min_h \; \mathcal{L}_D = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \mathcal{L}_{\text{CE}}\big(h(u_i), s_i\big), \qquad \mathcal{L}_{\text{CE}}(h(u), s) = -\log \operatorname{softmax}(h(u))_s, \tag{6}$$

and the encoder minimises the bounded fool loss of scvi-tools, the cross-entropy between the prediction and the uniform distribution over the other $K - 1$ batches,

$$\mathcal{A}_{\text{JS}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \frac{1}{K - 1} \sum_{k \neq s_i} \log \operatorname{softmax}\big(h(u_i)\big)_k. \tag{7}$$

Wasserstein-1 critics. A critic head $C = (C_1, \dots, C_K)$ (two hidden layers of 128 units, one output per batch) estimates, for each active batch k with target distribution T_k, the Kantorovich–Rubinstein dual

$$\hat{W}_k = \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} C_k(u_i) - \frac{1}{|T_k|} \sum_{t \in T_k} C_k(t) \;\approx\; W_1(\hat{P}_k, \hat{T}_k), \tag{8}$$

where the right-hand side is the distance between the empirical distributions; the dual is a lower bound of it for a 1-Lipschitz C_k, which the gradient penalty below enforces softly, so $\hat{W}_k$ is an estimate and not a bound. The targets are: reference critic, $T_k = \{u_j : s_j = r\}$, the cells of the reference batch r, the batch with the highest cell-type entropy $-\sum_c p_{r,c} \log p_{r,c}$ (head r is inactive; this rule uses the cell-type labels); pooled critic, $T_k = \{u_j : j \in \mathcal{B}_{-k}\}$; barycenter critic, $T_k = \hat{B}$, the support of the Wasserstein-2 barycenter of the batches present in the minibatch,

$$\hat{B} = \arg\min_B \; \frac{1}{K_{\text{pres}}} \sum_{k \text{ present}} W_2^2(\hat{P}_k, B), \tag{9}$$

with $|\mathcal{B}|$ atoms of equal weight, solved by the free-support fixed-point algorithm (at most 10 iterations, stopping threshold 10^{-7}), started from the minibatch cells and recomputed at every step; $\hat{B}$ is detached, so no gradient passes through it. A head is active when its batch is present in the minibatch and its target is non-empty; $\mathcal{H}$ is the set of active heads. The critic takes five steps per encoder step on

$$\min_C \; \mathcal{L}_C = -\frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \hat{W}_k \;+\; \lambda_{\text{GP}}\, \frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} \Big(\big\| \nabla_x C_k(\hat{x}_i) \big\|_2 - 1 \Big)^2, \qquad \hat{x}_i = \varepsilon_i u_i + (1 - \varepsilon_i) t_i, \tag{10}$$

with $\lambda_{\text{GP}} = 10$, $\varepsilon_i \sim \text{U}(0, 1)$ and t_i drawn uniformly from T_k for each cell i of batch k. In the encoder step the gradient penalty is dropped and the encoder minimises the critic's estimate,

$$\mathcal{A}_W = \frac{1}{|\mathcal{H}|} \sum_{k \in \mathcal{H}} \hat{W}_k, \tag{11}$$

where the reference cells (reference critic) and the cells of the other batches (pooled critic) are not detached and so receive gradient, while the barycenter support is a constant. No spectral normalisation is used.

Maximum mean discrepancy. Each batch is compared with the cells of all other batches in the minibatch, with the squared MMD as a biased (V-statistic) estimate,

$$\mathcal{A}_{\mathrm{MMD}} = \frac{1}{|K_{\mathcal{B}}|} \sum_{k \in K_{\mathcal{B}}} \widehat{\mathrm{MMD}}^2(\hat{P}_k, \hat{P}_{-k}), \qquad \widehat{\mathrm{MMD}}^2 = \frac{1}{n_k^2} \sum_{i,j \in \mathcal{B}_k} \kappa(u_i, u_j) + \frac{1}{n_{-k}^2} \sum_{i,j \in \mathcal{B}_{-k}} \kappa(u_i, u_j) - \frac{2}{n_k n_{-k}} \sum_{i \in \mathcal{B}_k} \sum_{j \in \mathcal{B}_{-k}} \kappa(u_i, u_j), \tag{12}$$

with $n_k = |\mathcal{B}_k|$, $n_{-k} = |\mathcal{B}_{-k}|$ and the multi-scale Gaussian kernel

$$\kappa(u, u') = \frac{1}{5} \sum_{m \in \{0.25,\, 0.5,\, 1,\, 2,\, 4\}} \exp\Big(- \frac{\|u - u'\|^2}{2\sigma_m^2} \Big), \qquad \sigma_m = \sqrt{d}\, m, \tag{13}$$

so that for $d = 10$ the bandwidths are $\sigma = 0.79$, 1.58, 3.16, 6.32 and 12.65. They are fixed and not tuned.

Sinkhorn divergence. With the Euclidean ground cost $c(u, u') = \|u - u'\|_2$ and entropic smoothing $\epsilon = 0.05\sqrt{2d}$,

$$\mathcal{A}_{\mathrm{SD}} = \frac{1}{|K_{\mathcal{B}}|} \sum_{k \in K_{\mathcal{B}}} S_\epsilon(\hat{P}_k, \hat{P}_{-k}), \qquad S_\epsilon(\alpha, \beta) = \mathrm{OT}_\epsilon(\alpha, \beta) - \tfrac{1}{2}\mathrm{OT}_\epsilon(\alpha, \alpha) - \tfrac{1}{2}\mathrm{OT}_\epsilon(\beta, \beta), \tag{14}$$

$$\mathrm{OT}_\epsilon(\alpha, \beta) = \max_{f, g} \langle \alpha, f \rangle + \langle \beta, g \rangle - \epsilon \left\langle \alpha \otimes \beta,\ e^{(f \oplus g - c)/\epsilon} - 1 \right\rangle, \tag{15}$$

which at the optimum equals $\langle \alpha, f \rangle + \langle \beta, g \rangle$. The potentials are found by Sinkhorn updates, annealed in ϵ from the diameter of the cost matrix with ratio 0.8 per step, then iterated at the final ϵ in double precision until the largest L_1 marginal error of the plans is below 10^{-4} (a cap of 500,000 iterations raises an error). The gradient is the envelope-theorem gradient: the potentials are held fixed and the cost carries gradient through its row point only. At this smoothing S_ϵ is below W_1 and is not an estimate of it.

scvi-tools adversarial classifier (batch-conditioned decoder only). A classifier with 32 hidden units, applied to a sample z_i from the posterior, is trained in one step per encoder step on $\kappa\, \mathcal{L}_{\mathrm{CE}}$ (as in Eq. 6, detached input), and the encoder minimises $\mathcal{L}_{\mathrm{scVI}} + \kappa\, \mathcal{A}_{\mathrm{JS}}$ with $\mathcal{A}_{\mathrm{JS}}$ as in Eq. 7, where $\kappa = 1 - \beta_e$ is fixed by the KL weight and there is no strength to vary.

Supplementary Table S6: Ranking detail: biology score at the matched batch removal b^* per penalty, task and decoder, with the leave-one-seed-out standard error (SE) of the reading (the same seed left out of plain scVI, b^* held fixed), the number m of the three leave-one-seed-out repeats that gave a reading, the rank (1 is the highest biology; ties share the average rank) and the lowest to highest rank over the three repeats. The SE and the rank range are given only where $m \geq 2$ (otherwise n/a; 0 of 78 ranked readings). "Cross." is the number of times the penalty's seed-mean curve crosses b^*; the reading is taken at the first crossing. In every pair every penalty's curve reaches b^*, so no reading is marked as not reached. 13 task and decoder pairs are ranked; the immune task and both simulations with the batch-conditioned decoder are low-signal and are in Part B. Simulations are ranked separately from the real tasks (Supplementary Notes 5 and 6; Table 2 gives the mean ranks). **Part B (last block)**: Low-signal task and decoder pairs (3; the gap between plain scVI's batch score and the highest batch score every penalty reaches is under 0.02): biology score at b^* read in the same way, not ranked and without standard errors (Supplementary Note 5).

Penalty	Biology at b^*	SE	m	Rank	Rank range	Cross.
Part A: ranked pairs						
Pancreas, batch-conditioned decoder; $b^* = 0.684$						
JS discriminator	0.691	0.001	3	6	6	2
Reference critic	0.697	0.005	3	4	3–5	2
Pooled critic	0.699	0.006	3	2	2–4	2
Barycenter critic	0.701	0.006	3	1	1–2	2
MMD	0.694	0.002	3	5	3–5	3
Sinkhorn	0.697	0.001	3	3	1–4	4
Lung, batch-conditioned decoder; $b^* = 0.648$						
JS discriminator	0.704	0.011	3	6	5–6	3
Reference critic	0.717	0.008	3	4	2–6	2
Pooled critic	0.719	0.003	3	3	3–4	2
Barycenter critic	0.715	0.005	3	5	3–5	2
MMD	0.723	0.003	3	2	1–3	2
Sinkhorn	0.726	0.003	3	1	1–2	4
ATAC small, batch-conditioned decoder; $b^* = 0.746$						
JS discriminator	0.585	0.003	3	6	6	2
Reference critic	0.603	0.003	3	5	5	2
Pooled critic	0.632	0.007	3	1	1–3	2
Barycenter critic	0.631	0.001	3	3	1–2	2
MMD	0.619	0.007	3	4	4	1
Sinkhorn	0.631	0.004	3	2	1–3	2
ATAC large, batch-conditioned decoder; $b^* = 0.632$						
JS discriminator	0.542	0.010	3	4	4–5	1
Reference critic	0.565	0.009	3	1	1–2	4
Pooled critic	0.546	0.014	3	3	2–5	2
Barycenter critic	0.563	0.004	3	2	1–3	2
MMD	0.536	0.017	3	6	3–6	1
Sinkhorn	0.540	0.017	3	5	3–6	4
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$						
JS discriminator	0.622	0.005	3	3	3–6	2
Reference critic	0.618	0.003	3	5	3–5	2
Pooled critic	0.624	0.002	3	2	2–3	2
Barycenter critic	0.619	0.004	3	4	4	2
MMD	0.617	0.009	3	6	2–6	2
Sinkhorn	0.630	0.002	3	1	1	2
Pancreas, unconditioned decoder; $b^* = 0.492$						
JS discriminator	0.585	0.008	3	2	1–5	1
Reference critic	0.582	0.004	3	3	2–4	1
Pooled critic	0.579	0.003	3	5	4–5	1
Barycenter critic	0.576	0.005	3	6	6	2
MMD	0.588	0.005	3	1	1–3	1
Sinkhorn	0.581	0.005	3	4	1–3	1
Lung, unconditioned decoder; $b^* = 0.535$						
JS discriminator	0.690	0.006	3	4	4–5	1
Reference critic	0.710	0.004	3	1	1	2
Pooled critic	0.703	0.007	3	3	2–3	1
Barycenter critic	0.690	0.005	3	5	4–5	1
MMD	0.707	0.001	3	2	2–3	1
Sinkhorn	0.685	0.004	3	6	6	1
Immune, unconditioned decoder; $b^* = 0.531$						
JS discriminator	0.720	0.012	3	1	1–3	1
Reference critic	0.716	0.002	3	2	1–3	1
Pooled critic	0.711	0.006	3	3	2–5	1
Barycenter critic	0.708	0.003	3	4	2–6	2
MMD	0.703	0.006	3	6	4–6	1
Sinkhorn	0.708	0.003	3	5	3–5	1
ATAC small, unconditioned decoder; $b^* = 0.522$						
JS discriminator	0.666	0.008	3	5	5–6	2
Reference critic	0.659	0.003	3	6	5–6	2
Pooled critic	0.677	0.004	3	4	3–4	2
Barycenter critic	0.700	0.015	3	1	1–2	2
MMD	0.692	0.015	3	2	1–4	1

Sinkhorn	0.684	0.013	3	3	1–3	1
ATAC large, unconditioned decoder; $b^* = 0.550$						
JS discriminator	0.617	0.009	3	4	2–5	1
Reference critic	0.621	0.006	3	2	2–4	2
Pooled critic	0.620	0.015	3	3	1–5	1
Barycenter critic	0.611	0.010	3	5	3–6	2
MMD	0.622	0.010	3	1	1–3	1
Sinkhorn	0.596	0.021	3	6	4–6	1
Human–mouse immune, unconditioned decoder; $b^* = 0.450$						
JS discriminator	0.617	0.001	3	5	2–4	1
Reference critic	0.589	0.030	3	6	6	2
Pooled critic	0.619	0.002	3	2	1–5	2
Barycenter critic	0.618	0.022	3	3	3–5	4
MMD	0.619	0.001	3	1	1–2	1
Sinkhorn	0.617	0.002	3	4	2–4	1
Simulation 1, unconditioned decoder; $b^* = 0.369$						
JS discriminator	0.709	0.002	3	1	1	1
Reference critic	0.691	0.003	3	2	2–3	3
Pooled critic	0.681	0.003	3	5	5	1
Barycenter critic	0.677	0.000	3	6	6	3
MMD	0.688	0.008	3	3	2–4	1
Sinkhorn	0.683	0.002	3	4	3–4	1
Simulation 2, unconditioned decoder; $b^* = 0.318$						
JS discriminator	0.584	0.002	3	1	1–3	2
Reference critic	0.579	0.002	3	4	3–5	3
Pooled critic	0.582	0.001	3	2	2	2
Barycenter critic	0.578	0.001	3	5	4–5	3
MMD	0.581	0.001	3	3	1–4	2
Sinkhorn	0.573	0.003	3	6	6	1
Part B: low-signal pairs (not ranked)						
Immune, batch-conditioned decoder; $b^* = 0.648$						
JS discriminator	0.772	n/a	–	not ranked	–	2
Reference critic	0.756	n/a	–	not ranked	–	2
Pooled critic	0.751	n/a	–	not ranked	–	2
Barycenter critic	0.749	n/a	–	not ranked	–	2
MMD	0.769	n/a	–	not ranked	–	2
Sinkhorn	0.766	n/a	–	not ranked	–	2
Simulation 1, batch-conditioned decoder; $b^* = 0.815$						
JS discriminator	0.897	n/a	–	not ranked	–	2
Reference critic	0.896	n/a	–	not ranked	–	2
Pooled critic	0.897	n/a	–	not ranked	–	2
Barycenter critic	0.896	n/a	–	not ranked	–	2
MMD	0.896	n/a	–	not ranked	–	4
Sinkhorn	0.897	n/a	–	not ranked	–	2
Simulation 2, batch-conditioned decoder; $b^* = 0.660$						
JS discriminator	0.493	n/a	–	not ranked	–	2
Reference critic	0.508	n/a	–	not ranked	–	2
Pooled critic	0.483	n/a	–	not ranked	–	2
Barycenter critic	0.458	n/a	–	not ranked	–	2
MMD	0.474	n/a	–	not ranked	–	2
Sinkhorn	0.494	n/a	–	not ranked	–	3

Supplementary Table S7: Placement with the unconditioned decoder (counterpart of Table 3 and of Figure 6). **(a)** Mean selection score ($0.4\times$batch $+ 0.6\times$biology) per method and real task at the strength (Harmony, Scanorama, sysVI: the setting) with the best mean selection score on real tasks of other task families (related tasks never choose for each other); in parentheses, the task's own best strength or setting (optimistic bound). Plain scVI here receives no batch information. The scvi-tools adversary is run with the batch-conditioned decoder only and is not shown. scANVI (uses cell-type labels), PCA and the three tools have no decoder, so their values equal those of Table 3. The last column counts the real tasks (of six) on which the method's score exceeds plain scVI's, at the transferred strength and, in parentheses, at the own best. **(b)** Number of real tasks on which the penalty's expected best selection score among k randomly drawn settings exceeds that of the tool at k settings of its own, $k = 1$ and $k = 6$; Scanorama at $k = 6$ is compared on the five tasks where all six of its settings exist (the 80-neighbour setting could not be fitted on the large chromatin-accessibility task). Counts are descriptive (Supplementary Notes 7 and 8).

(a)

Method	Pancreas	Lung	Immune	ATAC small	ATAC large	Human–mouse immune	Above plain scVI
Plain scVI	0.523	0.610	0.626	0.575	0.588	0.547	
JS discriminator	0.601 (0.654)	0.644 (0.644)	0.691 (0.691)	0.639 (0.639)	0.565 (0.594)	0.567 (0.575)	5 (6)
Reference critic	0.592 (0.650)	0.613 (0.651)	0.653 (0.671)	0.540 (0.645)	0.575 (0.607)	0.556 (0.580)	4 (6)
Pooled critic	0.579 (0.660)	0.644 (0.662)	0.648 (0.674)	0.591 (0.659)	0.589 (0.617)	0.564 (0.569)	6 (6)
Barycenter critic	0.560 (0.575)	0.607 (0.640)	0.636 (0.656)	0.572 (0.650)	0.447 (0.595)	0.533 (0.552)	2 (6)
MMD	0.600 (0.600)	0.636 (0.646)	0.663 (0.674)	0.634 (0.650)	0.609 (0.612)	0.558 (0.561)	6 (6)
Sinkhorn	0.575 (0.584)	0.621 (0.645)	0.655 (0.664)	0.533 (0.652)	0.515 (0.598)	0.531 (0.552)	3 (6)
Harmony	0.706 (0.710)	0.621 (0.621)	0.664 (0.664)	0.576 (0.587)	0.552 (0.558)	0.520 (0.531)	4 (4)
Scanorama	0.600 (0.613)	0.598 (0.600)	0.685 (0.692)	0.517 (0.538)	0.510 (0.522)	0.524 (0.525)	2 (2)
sysVI	0.723 (0.726)	0.649 (0.653)	0.685 (0.700)	0.533 (0.571)	0.511 (0.537)	0.501 (0.542)	3 (3)
scANVI (labels)	0.696	0.740	0.743	0.748	0.656	0.619	6
PCA (uncorrected)	0.577	0.585	0.565	0.490	0.527	0.478	1

(b)

	Plain scVI		Harmony		Scanorama		sysVI		scANVI (labels)		PCA	
Penalty	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6	k=1	k=6
JS discriminator	4/6	5/6	3/6	5/6	5/6	4/5	3/6	3/6	0/6	0/6	6/6	6/6
Reference critic	5/6	6/6	5/6	5/6	4/6	4/5	3/6	3/6	0/6	0/6	5/6	6/6
Pooled critic	5/6	6/6	4/6	5/6	5/6	4/5	3/6	4/6	0/6	0/6	6/6	6/6
Barycenter critic	1/6	6/6	2/6	4/6	3/6	3/5	3/6	3/6	0/6	0/6	5/6	5/6
MMD	5/6	6/6	4/6	5/6	4/6	3/5	3/6	3/6	0/6	0/6	5/6	6/6
Sinkhorn	4/6	6/6	4/6	4/6	4/6	3/5	3/6	3/6	0/6	0/6	5/6	6/6

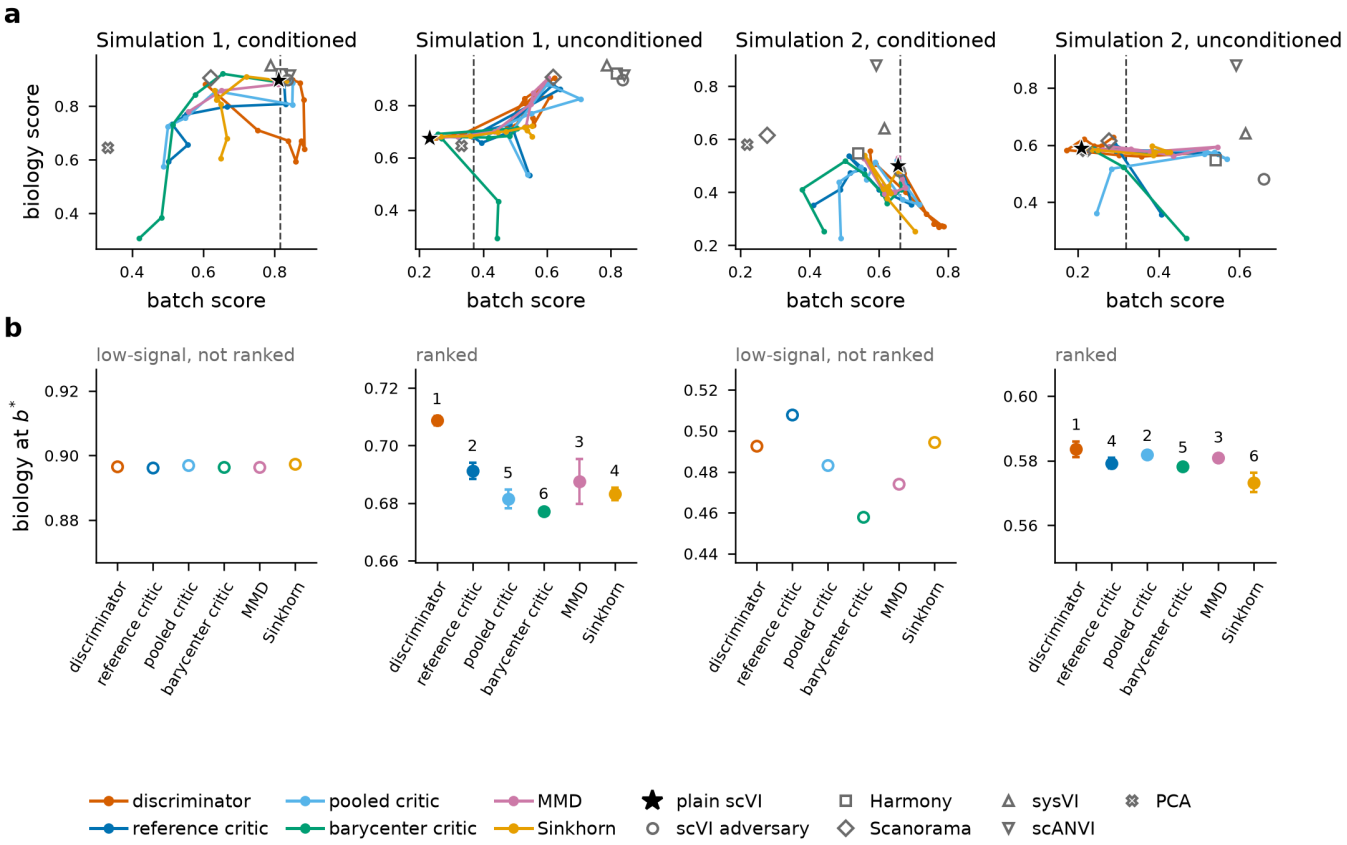
Supplementary Table S9: Replication across training seeds: biology at the matched batch removal b^* (fixed at its sweep value) on new seeds 0–4 beside the sweep reading (seeds 100–102), per penalty, task and decoder, for the 13 ranked pairs. "Strengths" are the two bracketing strengths (0 is plain scVI at seeds 0–4); "Batch" and "Biology" give the seed-mean scores at those two points; the reading interpolates linearly in batch score between them at the first crossing of b^*, walking up from plain scVI at seeds 0–4 (in 7 readings the new-seed mean at the lower refitted strength already reached b^*, so the reading starts at plain scVI). Rank: replication (sweep); 1 is the highest biology; a penalty that does not reach b^* takes the bottom place. "extension": the new-seed means did not bracket b^* at the planned strengths, so the next sweep strength was fitted once with seeds 0–4, as the agreement rule prescribes (8 readings, 40 fits). †Did not replicate the crossing: the highest new-seed mean batch score is shown. Each block header gives b^*, plain scVI's batch score at seeds 0–4 beside the sweep's and Kendall's τ_b between the sweep and replication ranks. All 895 fits were scored and 0 failed. The last block gives the four large sweep gaps (0.04 or more within a pair) and their sign on the new seeds (Supplementary Notes 4 and 10).

Penalty	Strengths	Batch	Biology	At b^*, new	At b^*, sweep	Rank	Note
Pancreas, batch-conditioned decoder; $b^* = 0.684$; plain scVI batch score 0.668 (sweep 0.672); $\tau_b = -0.067$							
JS discriminator	0.3–1	0.681/0.703	0.697/0.688	0.695	0.691	4 (6)	
Reference critic	0–0.3	0.668/0.685	0.696/0.704	0.703	0.697	1 (4)	
Pooled critic	0.3–1	0.677/0.696	0.698/0.694	0.696	0.699	3 (2)	
Barycenter critic	1–3	0.679/0.695	0.697/0.691	0.695	0.701	5 (1)	
MMD	up to 3	0.674 (max)	–	–	0.694	6 (5)	extension; not reached†
Sinkhorn	3–10	0.682/0.696	0.699/0.701	0.699	0.697	2 (3)	extension
Pancreas, unconditioned decoder; $b^* = 0.492$; plain scVI batch score 0.451 (sweep 0.446); $\tau_b = 0.200$							
JS discriminator	0.1–0.3	0.488/0.552	0.587/0.614	0.589	0.585	1 (2)	
Reference critic	1–3	0.481/0.517	0.582/0.594	0.585	0.582	2 (3)	
Pooled critic	1–3	0.492/0.536	0.584/0.599	0.584	0.579	3 (5)	extension
Barycenter critic	1–3	0.481/0.497	0.580/0.586	0.584	0.576	4 (6)	
MMD	10–30	0.480/0.527	0.578/0.589	0.581	0.588	5 (1)	
Sinkhorn	3–10	0.490/0.495	0.580/0.580	0.580	0.581	6 (4)	
Lung, batch-conditioned decoder; $b^* = 0.648$; plain scVI batch score 0.633 (sweep 0.633); $\tau_b = 0.867$							
JS discriminator	0.3–1	0.644/0.671	0.713/0.610	0.700	0.704	6 (6)	
Reference critic	0.3–1	0.647/0.661	0.723/0.707	0.722	0.717	4 (4)	
Pooled critic	0.3–1	0.647/0.663	0.726/0.708	0.725	0.719	2 (3)	
Barycenter critic	0–1	0.633/0.649	0.723/0.720	0.720	0.715	5 (5)	
MMD	3–10	0.640/0.650	0.728/0.722	0.723	0.723	3 (2)	
Sinkhorn	3–10	0.646/0.653	0.730/0.713	0.726	0.726	1 (1)	
Lung, unconditioned decoder; $b^* = 0.535$; plain scVI batch score 0.479 (sweep 0.481); $\tau_b = 0.600$							
JS discriminator	0.1–0.3	0.531/0.565	0.687/0.683	0.687	0.690	5 (4)	
Reference critic	1–3	0.505/0.539	0.697/0.705	0.704	0.710	2 (1)	
Pooled critic	1–3	0.519/0.557	0.701/0.698	0.700	0.703	3 (3)	
Barycenter critic	100–300	0.523/0.557	0.689/0.670	0.682	0.690	6 (5)	
MMD	10–30	0.511/0.550	0.707/0.710	0.709	0.707	1 (2)	
Sinkhorn	30–100	0.521/0.612	0.705/0.660	0.698	0.685	4 (6)	
Immune, unconditioned decoder; $b^* = 0.531$; plain scVI batch score 0.499 (sweep 0.499); $\tau_b = 0.200$							
JS discriminator	0.03–0.1	0.505/0.531	0.704/0.720	0.720	0.720	1 (1)	
Reference critic	0.3–1	0.513/0.537	0.716/0.706	0.709	0.716	6 (2)	
Pooled critic	0.3–1	0.510/0.536	0.712/0.718	0.717	0.711	3 (3)	
Barycenter critic	0–3	0.499/0.534	0.711/0.715	0.715	0.708	4 (4)	
MMD	10–30	0.525/0.563	0.711/0.710	0.711	0.703	5 (6)	
Sinkhorn	10–30	0.528/0.543	0.717/0.719	0.718	0.708	2 (5)	
ATAC small, batch-conditioned decoder; $b^* = 0.746$; plain scVI batch score 0.690 (sweep 0.694); $\tau_b = 0.467$							
JS discriminator	0.3–1	0.719/0.753	0.628/0.576	0.587	0.585	6 (6)	
Reference critic	0.3–1	0.720/0.773	0.638/0.593	0.616	0.603	5 (5)	
Pooled critic	0.3–1	0.724/0.779	0.634/0.601	0.621	0.632	4 (1)	
Barycenter critic	1–3	0.738/0.772	0.648/0.615	0.640	0.631	1 (3)	
MMD	10–30	0.725/0.757	0.632/0.623	0.626	0.619	3 (4)	
Sinkhorn	3–10	0.732/0.761	0.639/0.620	0.630	0.631	2 (2)	
ATAC small, unconditioned decoder; $b^* = 0.522$; plain scVI batch score 0.398 (sweep 0.402); $\tau_b = 0.600$							
JS discriminator	0.1–0.3	0.460/0.617	0.682/0.650	0.669	0.666	5 (5)	
Reference critic	1–3	0.496/0.678	0.678/0.589	0.665	0.659	6 (6)	
Pooled critic	1–3	0.509/0.682	0.682/0.631	0.678	0.677	3 (4)	
Barycenter critic	3–10	0.520/0.619	0.677/0.690	0.677	0.700	4 (1)	extension
MMD	10–30	0.438/0.552	0.683/0.692	0.689	0.692	1 (2)	
Sinkhorn	10–30	0.521/0.596	0.685/0.665	0.685	0.684	2 (3)	extension
ATAC large, batch-conditioned decoder; $b^* = 0.632$; plain scVI batch score 0.591 (sweep 0.589); $\tau_b = 0.600$							
JS discriminator	0.1–0.3	0.621/0.661	0.525/0.502	0.518	0.542	6 (4)	
Reference critic	0.1–0.3	0.618/0.663	0.548/0.552	0.550	0.565	1 (1)	
Pooled critic	0.1–0.3	0.614/0.654	0.540/0.550	0.545	0.546	3 (3)	
Barycenter critic	0.3–1	0.620/0.636	0.550/0.544	0.546	0.563	2 (2)	
MMD	3–10	0.612/0.652	0.536/0.543	0.540	0.536	4 (6)	
Sinkhorn	0–10	0.591/0.633	0.553/0.537	0.538	0.540	5 (5)	
ATAC large, unconditioned decoder; $b^* = 0.550$; plain scVI batch score 0.538 (sweep 0.538); $\tau_b = 0.467$							
JS discriminator	0.03–0.1	0.543/0.587	0.601/0.588	0.599	0.617	5 (4)	
Reference critic	0.1–0.3	0.544/0.582	0.628/0.586	0.622	0.621	1 (2)	
Pooled critic	0.1–0.3	0.544/0.558	0.635/0.604	0.622	0.620	2 (3)	
Barycenter critic	0–1	0.538/0.554	0.637/0.612	0.619	0.611	3 (5)	

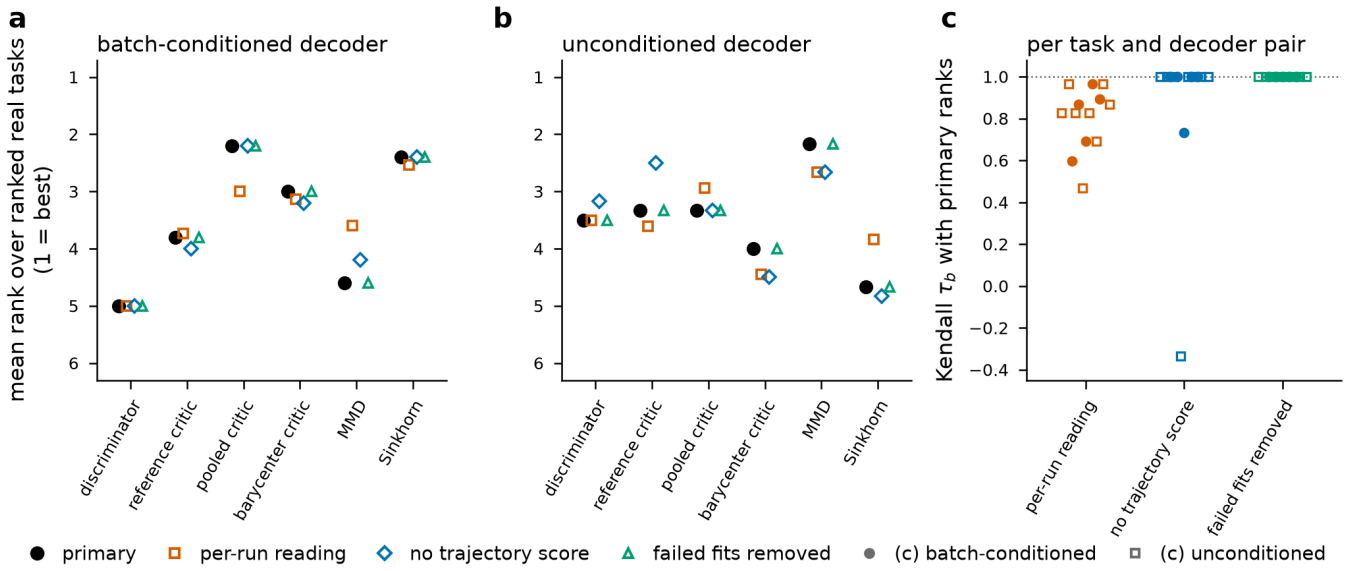
MMD	3–10	0.542/0.553	0.629/0.603	0.611	0.622	4 (1)	
Sinkhorn	30–100	0.543/0.565	0.616/0.522	0.587	0.596	6 (6)	
Human–mouse immune, batch-conditioned decoder; $b^* = 0.554$; plain scVI batch score 0.540 (sweep 0.540); $\tau_b = -0.067$							
JS discriminator	0–0.1	0.540/0.557	0.630/0.613	0.617	0.622	4 (3)	
Reference critic	0.3–1	0.553/0.570	0.620/0.590	0.619	0.618	2 (5)	
Pooled critic	0.1–0.3	0.547/0.556	0.630/0.598	0.606	0.624	6 (2)	
Barycenter critic	1–3	0.552/0.557	0.619/0.606	0.615	0.619	5 (4)	extension
MMD	3–10	0.549/0.562	0.622/0.606	0.617	0.617	3 (6)	
Sinkhorn	3–10	0.553/0.559	0.624/0.616	0.624	0.630	1 (1)	
Human–mouse immune, unconditioned decoder; $b^* = 0.450$; plain scVI batch score 0.432 (sweep 0.432); $\tau_b = 0.333$							
JS discriminator	0.01–0.03	0.436/0.451	0.608/0.617	0.617	0.617	1 (5)	
Reference critic	0–0.3	0.432/0.451	0.610/0.603	0.603	0.589	6 (6)	
Pooled critic	0–0.3	0.432/0.457	0.610/0.619	0.616	0.619	2 (2)	
Barycenter critic	10–30	0.444/0.457	0.621/0.610	0.616	0.618	4 (3)	
MMD	3–10	0.445/0.485	0.619/0.598	0.616	0.619	3 (1)	
Sinkhorn	3–10	0.450/0.453	0.615/0.615	0.615	0.617	5 (4)	
Simulation 1, unconditioned decoder; $b^* = 0.369$; plain scVI batch score 0.232 (sweep 0.232); $\tau_b = 0.867$							
JS discriminator	0.1–0.3	0.338/0.557	0.681/0.832	0.703	0.709	1 (1)	
Reference critic	1–3	0.338/0.484	0.679/0.721	0.688	0.691	3 (2)	
Pooled critic	1–3	0.360/0.487	0.679/0.713	0.682	0.681	5 (5)	extension
Barycenter critic	1–3	0.285/0.403	0.677/0.679	0.678	0.677	6 (6)	
MMD	10–30	0.321/0.449	0.692/0.694	0.693	0.688	2 (3)	
Sinkhorn	3–10	0.336/0.424	0.684/0.695	0.688	0.683	4 (4)	
Simulation 2, unconditioned decoder; $b^* = 0.318$; plain scVI batch score 0.203 (sweep 0.209); $\tau_b = 0.867$							
JS discriminator	0.1–0.3	0.222/0.423	0.586/0.586	0.586	0.584	1 (1)	
Reference critic	3–10	0.284/0.461	0.581/0.573	0.580	0.579	3 (4)	
Pooled critic	3–10	0.305/0.505	0.581/0.574	0.581	0.582	2 (2)	
Barycenter critic	3–10	0.255/0.332	0.582/0.576	0.577	0.578	5 (5)	
MMD	10–30	0.254/0.346	0.584/0.578	0.580	0.581	4 (3)	
Sinkhorn	30–100	0.307/0.432	0.574/0.603	0.576	0.573	6 (6)	extension

Large sweep gaps

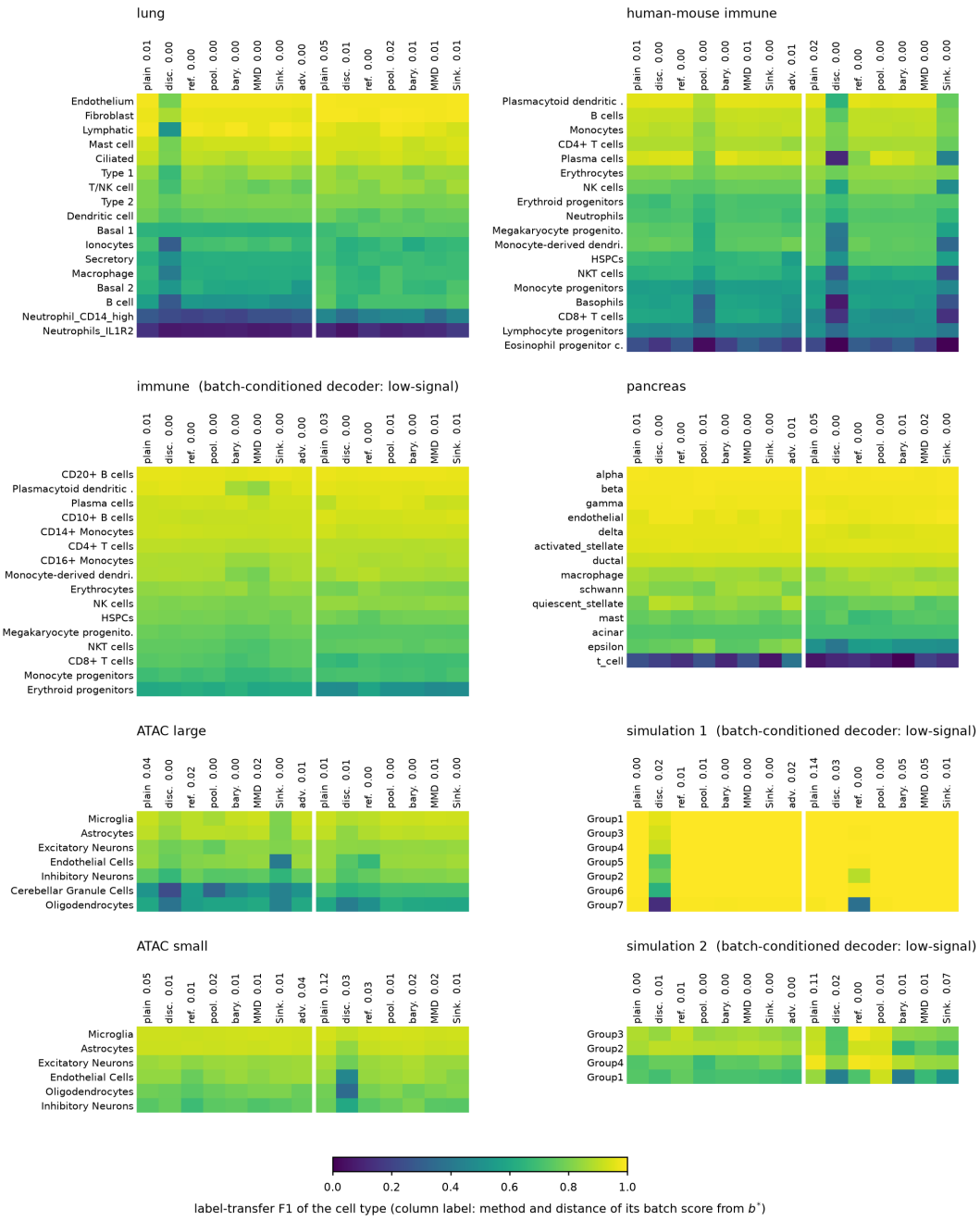
Pair	Difference	Sweep	New	Sign kept
ATAC small, unconditioned decoder	Barycenter critic − Reference critic	+0.041	+0.013	yes
ATAC small, batch-conditioned decoder	Barycenter critic − JS discriminator	+0.046	+0.053	yes
ATAC small, batch-conditioned decoder	JS discriminator − Pooled critic	−0.046	−0.034	yes
ATAC small, batch-conditioned decoder	JS discriminator − Sinkhorn	−0.046	−0.043	yes



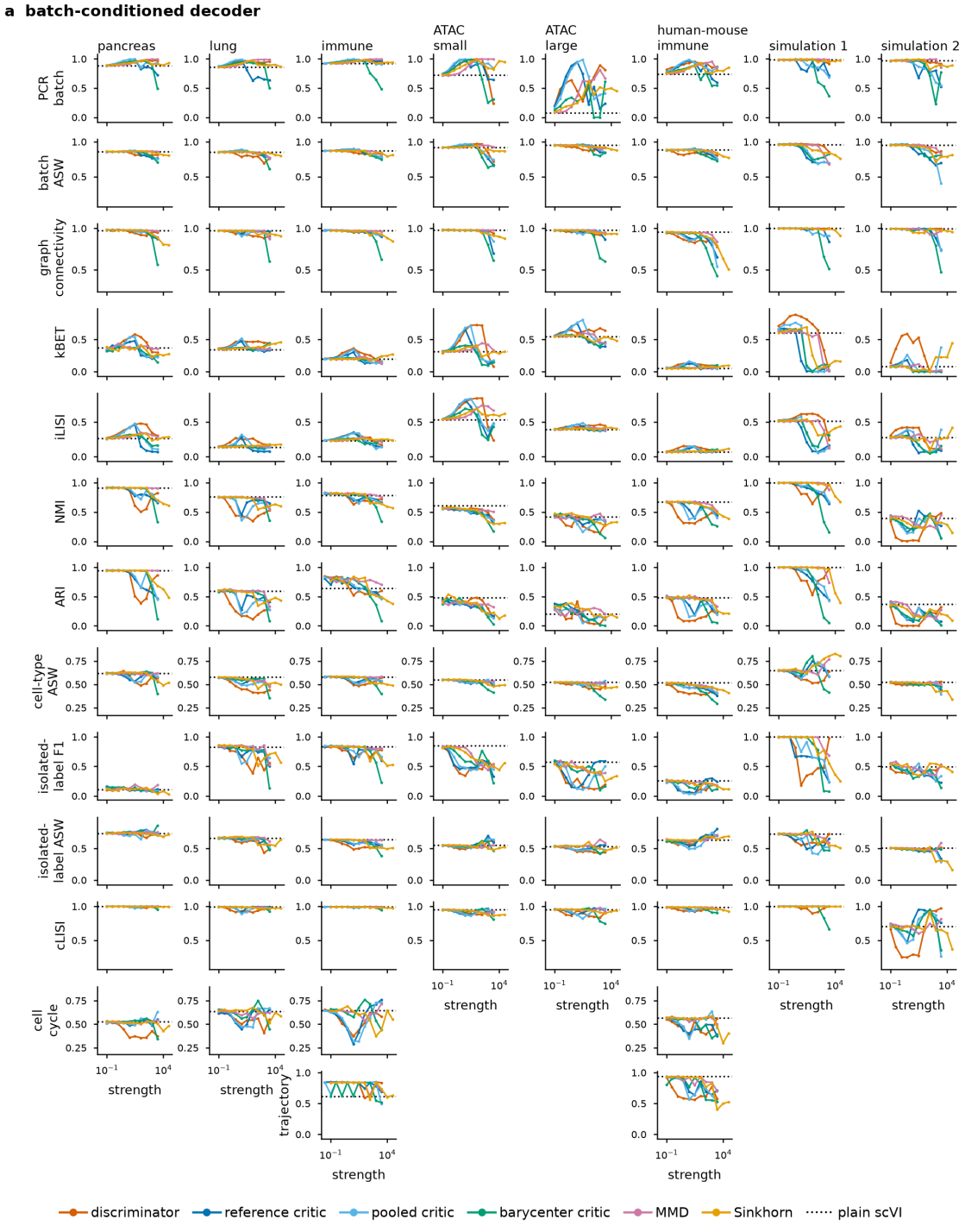
Supplementary Figure S1: Simulation tasks, each on its own, both decoders. **(a)** Batch score against biology score for each penalty's sweep strengths (seed means over seeds 100–102), plain scVI (star), the matched batch removal b^* (dashed line) and the best setting of each standard tool by selection score (open grey markers). **(b)** Biology score at b^* per penalty with its leave-one-seed-out standard error, and the rank above each point. With the batch-conditioned decoder both simulations are low-signal: their readings are shown with open markers, without standard errors and ranks. Simulations are ranked separately from the real tasks (Supplementary Notes 5 and 6; Supplementary Table S6).



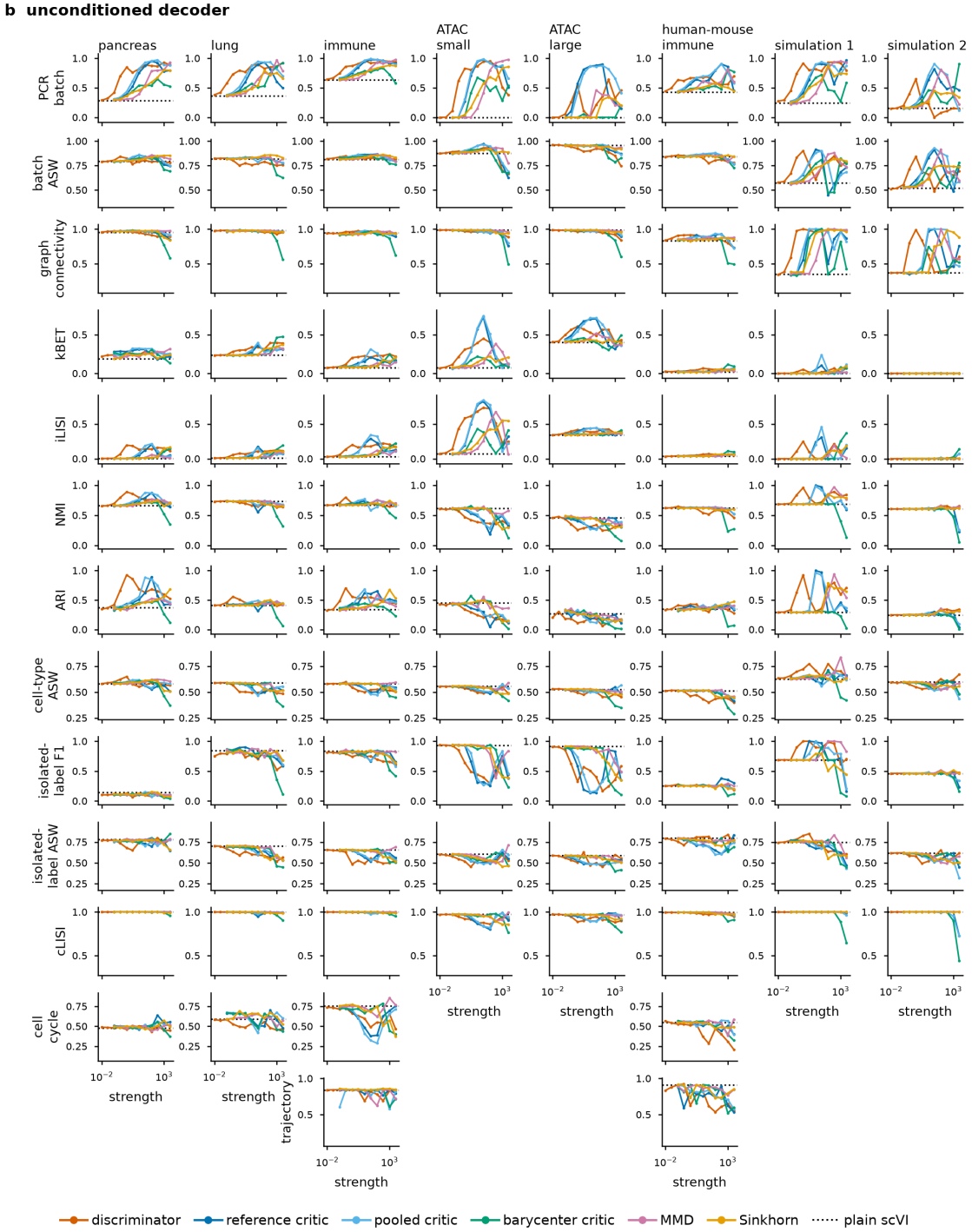
Supplementary Figure S2: Sensitivity of the ranking to the reading rules. Each variant re-reads biology at b^* from the same sweep fits: the per-run reading (each seed read on its own curve and the three per-seed ranks averaged), biology without trajectory conservation on the two immune tasks, and the primary reading with points that contain a failed fit removed (the sweep has no failed fit, so this variant reproduces the primary ranks). **(a, b)** Mean rank over the ranked real tasks per penalty and decoder. **(c)** Kendall's τ_b between the primary ranks of the six penalties and each variant's ranks in each of the 13 ranked task and decoder pairs (simulations included); points at 1 overlap. Descriptive; Supplementary Notes 5 and 6.



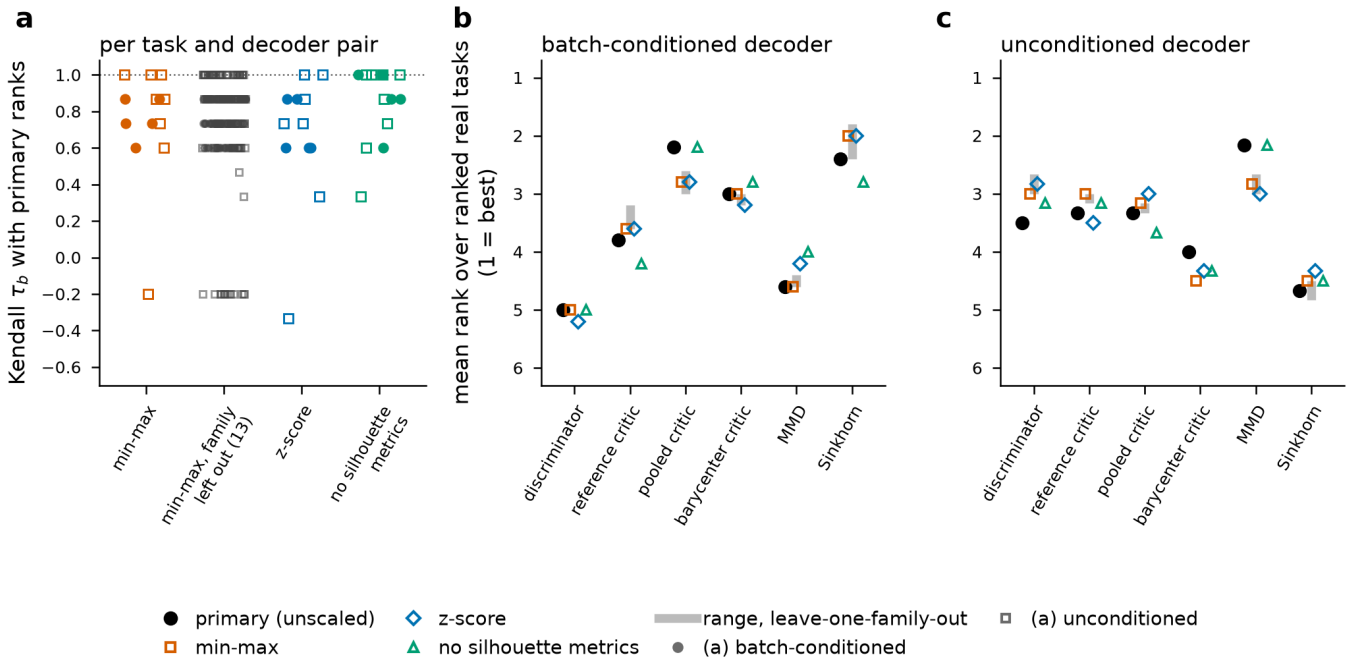
Supplementary Figure S3: Label transfer: F1 per cell type, method and task at the matched strength. Left block of each panel: batch-conditioned decoder (plain scVI, the six penalties and the scvi-tools adversary); right block: unconditioned decoder (plain scVI, which receives no batch information, and the six penalties). Each batch is held out in turn and its cells are predicted from the other batches by a 15-nearest-neighbour classifier; cell types present in one batch only are dropped; a type's F1 is averaged over the held-out batches that contain it and over seeds (three for plain scVI and the penalties, five for the scvi-tools adversary). Penalties are at the strength whose seed-mean batch score is nearest the matched batch removal b^*, never chosen by accuracy; plain scVI and the scvi-tools adversary are unmatched. The number in each column label is the distance between the column's seed-mean batch score and b^*. Cell types are ordered by their mean F1 over the columns of the panel. The immune task and both simulations with the batch-conditioned decoder are low-signal; they are tabulated here and left out of summaries (Supplementary Note 9).



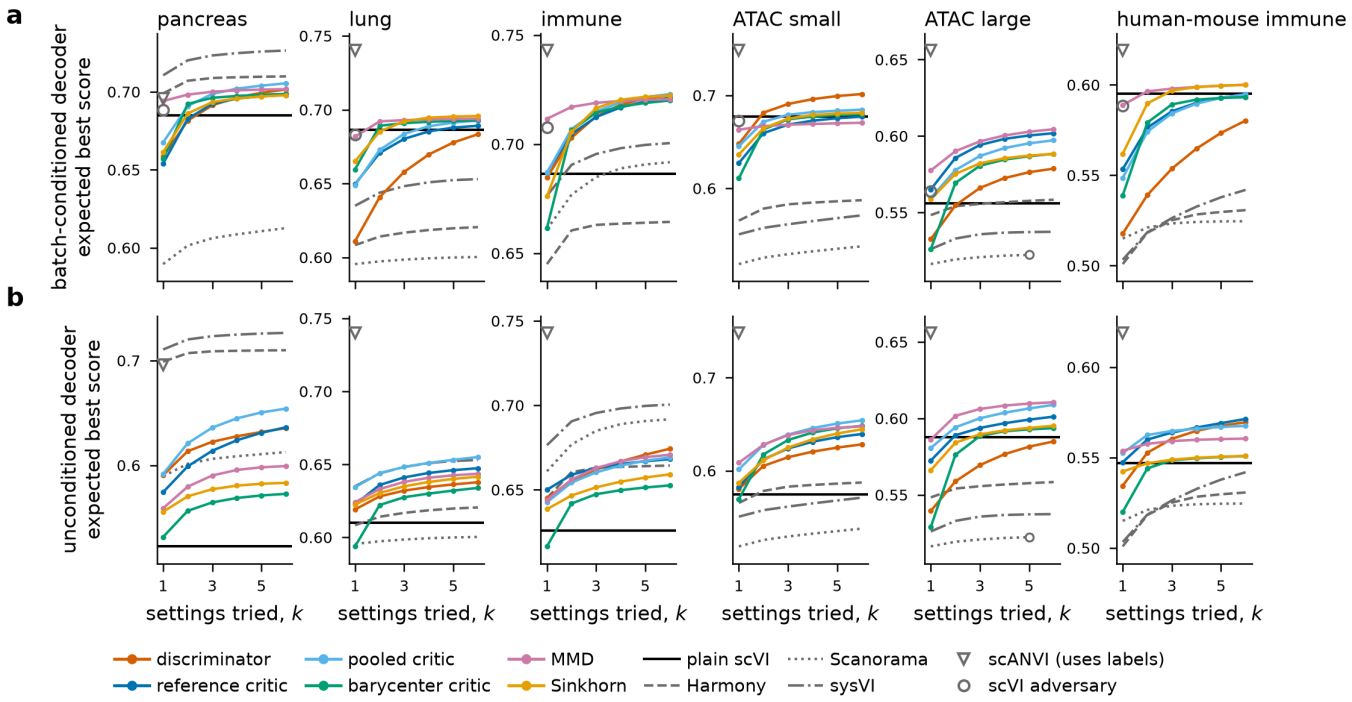
Supplementary Figure S4a: Per-metric strength curves, batch-conditioned decoder. Each batch metric (rows 1–5) and biology metric (rows 6–13) of the scIB set, unscaled, against penalty strength on a log scale, seed means over the three sweep seeds, one line per penalty, per task (columns). The dotted line is plain scVI. Each row has one vertical range for all tasks. Cell-cycle conservation exists for the four RNA real tasks and trajectory conservation for the two immune tasks; other panels are empty (Supplementary Note 3).



Supplementary Figure S4b: Per-metric strength curves, unconditioned decoder. As Supplementary Figure S4a, with the unconditioned decoder.



Supplementary Figure S5: Scaling checks (exploratory, used for no conclusion). The six penalties are re-ranked at matched batch removal after changing how the metrics are scaled or combined: scIB min-max scaling of each metric over one comparison set per task (every scored fit of the task, both decoders, all penalty strengths, plain scVI and the standard tools); the same with each of the 13 method families left out of the set that fixes the minimum and maximum; z-score standardisation over the same set; and a composite without the three silhouette metrics (batch average silhouette width, cell-type average silhouette width, isolated-label silhouette). The primary reading uses unscaled metrics. Batch and biology scores, b^*, the readings and the ranks are recomputed for every variant on the 13 ranked task and decoder pairs. **(a)** Kendall's τ_b between the primary ranks and each variant's ranks, one point per pair (leave-one-family-out: one point per pair and family). **(b, c)** Mean rank over the ranked real tasks per penalty and decoder; the grey bar is the range over the 13 leave-one-family-out runs.



Supplementary Figure S6: Best of k settings per task. Expected best selection score ($0.4\times$batch $+ 0.6\times$biology) among k settings drawn at random, $k = 1$ to 6, for each penalty, Harmony, Scanorama and sysVI, one panel per real task and decoder. Plain scVI (black line), scANVI (uses cell-type labels) and the scvi-tools adversary (batch-conditioned decoder only) have one setting. Scanorama at $k = 6$ is missing on the large chromatin-accessibility task (open marker at $k = 5$), where its 80-neighbour setting could not be fitted. The mean of the batch-conditioned curves over the six real tasks is Figure 6a. Every value is optimistic, because the best setting is chosen and scored on the same data (Supplementary Note 7).